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CM3070 PROJECT

FINAL PROJECT REPORT

Healthcare Worker Burnout-Related Wellbeing Monitoring (Multimodal AI System)

Author : Subathra Sundarbabu

Student Number : 10263704

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Supervisor : Sarita Singh

GitHub Link :

https://github.com/SubathraSIM/CM3020_Solace_Wellbeing_Monitoring_FYP

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CHAPTER 1: INTRODUCTION (968/1000 Words)

1.1 Overview

This chapter describes the project concept, selected template, justification for the healthcare burnout focus, objectives, and data sources.

1.2 Selected Project Template

This project follows the Artificial Intelligence Project Idea 1: Orchestrating AI Models to Achieve a Goal. According to the brief, a system that generates useful results for a problem area must use at least three pre-trained models from various domains, such as text, image, or audio. The project focuses on selecting, testing, rejecting and integrating pre-trained models into a functional system.

1.3 Project Concept

The system, named Solace, analyses speech, transcribed text, and facial-expression signals from a one-minute audio or video check-in using six orchestrated pre-trained models spanning text, audio, vision, speech transcription, translation, and language generation. On a private dashboard, it then displays the user's current indicator, historical trend, and a customised recommendation, alongside an assistant that can answer questions about the user's own check-in history in their chosen language.

Unlike existing burnout tools, it requires only a guided one-minute check-in, is a desktop application that runs on everyday hardware with a camera and microphone, and captures voice, text, and vision from a single input. Instead of diagnosing burnout or replacing professional evaluation, this application is meant to encourage reflection and early awareness. It is developed for the Singapore context, so personal data is handled in accordance with Singapore's Personal Data Protection Act (PDPA) [1]. The check-in and its recommendations are available in English and Singapore's other main languages which are Malay, Chinese and Tamil.

1.4 Project Motivations

1.4.1 Academic Motivation

One of the most well-documented occupational mental health issues of the past decade is burnout among healthcare professionals. A systematic review and meta-analysis of eight studies involving 215,787 public health workers estimated a pooled burnout prevalence of 39% [2]. Among healthcare professionals surveyed in a Singapore healthcare cluster, 43.9% met the study's burnout criterion, with the highest prevalence observed among allied health professionals at 52.5% [3]. During the COVID-19 public health emergency, nurses exposed to COVID-19 patients and those working in emergency departments or intensive care units reported significantly higher burnout levels [4].

1.4.2 Personal Motivation

The personal inspiration came from my father, who works in a factory in Singapore and described his coworkers being emotionally drained after long overtime periods. Emotional exhaustion is one of three recognised dimensions of burnout [5]. Similarly, doctors and nurses frequently experience depersonalisation and a reduced sense of personal accomplishment [3]. Healthcare was selected as the priority due to the strong scientific evidence in this field and serious implications, which affect both patients and employees.

1.4.3 Domain Consultation

Early in the project, informal consultations with two practising surgeons and a former healthcare administrator confirmed the need. They described being frequently exhausted by overtime with no one checking in on them and saw a brief, private check-in tool as potentially useful. This shaped the design toward a low-effort, confidential daily check-in, though their schedules did not permit participation in the later evaluation.

1.5 Project Aims, Objectives, and Research Questions

1.5.1 Project Aim

The main goal is to develop an integrated multimodal AI system using a one-minute daily check-in to assist healthcare professionals in identifying indicators associated with burnout. Using three primary pre-trained AI models for text, audio, and vision, it generates a single burnout-related risk indicator and trend, and a local LLM converts the combined outputs into personalised, non-diagnostic recommendations.

1.5.2 Project Objectives

The project is structured around four main objectives. Each is written as a measurable and observable result.

Objective		Outcome	Justification
1	Single Fused Burnout Indicator	Normalises the text, audio and facial-expression scores from a one-minute check-in and combines the available scores by median fusion, adjusted by supporting signals, into a single internal 0-100 burnout indicator, shown to the user as an inverted wellbeing score. Met when unit tests verify the fusion and a valid indicator is produced for every valid check-in.	The aim requires one coherent indicator from three modalities. This is the measurable end-state of that fusion, independent of how it is built.

2	Evidence Backed Models	For each modality, the highest performing model among at least three benchmarked candidates is deployed, judged by the metric suited to that component. Macro-F1 is used for the text, audio and vision emotion models, CER for speech transcription, and chrF++ [6] for translation, and the comparison evidence is documented. Met when the evidence table is present and the deployed model wins on its component's chosen metric for each modality.	The goal is an evidence-based approach. The intended outcome is the best-performing model on the recorded evidence for each modality, not the process of choosing.
3	Early-Warning from Personal Baseline	An early-warning status is raised for a check-in whose combined burnout indicator is above the rolling baseline mean by at least one standard deviation. Met when in-range cases fail to trigger while seeded above-threshold cases trigger correctly.	The baseline deviation behaviour turns the aim's early warning capacity into a testable result that may be attained without labelled burnout data.
4	Complete Check-In	Each completed check-in yields a stored indicator, an updated trend, and one personalised non-diagnostic recommendation. Met when all three outputs are present and at least 95% of test check-ins are completed end-to-end.	With the dashboard, LLM, and storage serving as means rather than ends, the aim is an integrated working system. The result is a usable end-to-end loop.

Table 1: Objectives and Justifications

Each objective maps to a concrete deliverable, the working desktop application (Objective 1, 4), the documented model-selection evidence (Objective 2), and the personal-baseline early-warning mechanism (Objective 3), with the CBI study providing external validation.

1.5.3 Research Questions

The project is guided by the following questions:

Q1: How to orchestrate pre-trained text, audio and vision models into a single pipeline that produces a coherent burnout-related indicator from a one-minute check-in?

Q2: How should the pipeline behave when one modality is missing, noisy, or low confidence, so that fusion degrades gracefully rather than failing?

Q3: Can a personal-baseline mechanism flag a reading that deviates from a user's established range, acting as an early-warning signal in the absence of labelled burnout data?

Q1 and **Q2** are answered through the implemented pipeline, and **Q3** through demonstrating the baseline mechanism on seeded check-ins.

1.6 Data Sources

Because the project uses pre-trained models, no training dataset is required. Publicly available labelled datasets are used for model evaluation and selection. Once operational, healthcare professionals may provide limited usability input.

1.6.1 Test Data for Model Selection

Publicly available labelled datasets are used to compare the candidate models. The datasets are RAF-DB for vision [7], CancerEmo for text [8], and RAVDESS for audio [9], with FLEURS [10] and OPUS-100 [11] supporting the multilingual transcription and translation. These datasets carry emotion or language labels rather than burnout labels and are therefore used to evaluate task performance during model selection.

The deployed models are RoBERTa-Large for text [12], MERaLiON-SER for audio [13], and DDAMFN++ for facial expression [14], with Whisper-Small selected via CER over Tiny/Base for speech transcription [15] and NLLB-200-distilled-600M selected via chrF++ [6, 16] over M2M100 [17] and SeamlessM4T [18] for translation. Full candidate comparisons are presented in Chapter 5.

1.6.2 User Check-in Data

Each check-in results in a short recording that the pipeline analyses in real time. To protect users' privacy, raw audio, video and captured facial images are not stored. They are disposed of after analysis is completed, and only the resulting information is stored in the local database. This retained information consists of the speech transcript, final numerical scores and supporting signals, and generated summary and recommendation text. Separately, a small user study collects Copenhagen Burnout Inventory (CBI) [19] responses with short recordings, used to check the indicator against a recognised burnout measure.

CHAPTER 2: LITERATURE REVIEW (2198/2500 Words)

2.1 Overview

This chapter examines previous research in three areas which are current burnout and wellness apps, the methodology the project expands upon, and the research findings that support the approach.

2.2 Similar Projects

Source credibility: These commercial products serve as examples of current practices. Although they can be used to describe stated benefits, their official websites and app-store listings do not prove clinical effectiveness on their own. Therefore, the peer-reviewed research examined in Section 2.4 supports scientific assertions.

2.2.1 BurnoutGuard

A commercial app called BurnoutGuard creates a burnout risk assessment and individualised suggestions by combining calendar data, work logs, screen usage, sleep, mood, stress, and energy levels [20].

Evaluation:

One benefit is that it requires little work to get health and calendar data. Monitoring is not entirely passive, though, because some measurements require manual logging. Additionally, the website offers no independent clinical validation of its burnout score, and calendar analysis may be less appropriate for frontline employees on set rosters.

2.2.2 Welltory

Welltory uses physiological data, specifically heart rate variability (HRV), that is gathered by smartphone cameras or wearable technology that is compatible [21]. To evaluate stress and recovery and offer personalised advice, proprietary algorithms analyse the data.

Evaluation:

Its use of physiological data instead of just depending on self-report is its primary strength. However, HRV is not a precise indicator of burnout because it is influenced by things like sleep, exercise, and illness. Cost and accessibility issues could also arise from wearable integration. In contrast to Welltory's physiological method, this project analyses voice, text, and facial expression cues using an everyday device's camera and microphone without the need for additional hardware.

2.2.3 Unburn

Unburn is a mobile application that measures overall, personal, work-related, and client-related burnout using a brief questionnaire based on the Copenhagen Burnout Inventory [19]. Additionally, it offers calming sounds, daily recovery activities, and mood tracking [22].

Evaluation:

Its strength is that it uses an established assessment tool and divides burnout into recognised dimensions. However, results are still sensitive to recall and response bias, and repeated questions may cause response load, especially for busy healthcare staff. Additionally, there is no proof in the app description that the modified questionnaire has undergone independent validation. By analysing speech, text, and facial expression cues from a quick check-in to assist early identification of changes, this project reduces the need for repeated questionnaires.

2.2.4 Wysa

Wysa is a conversational mental health platform driven by AI that offers self-help activities, human-guided care routes, and anonymous, CBT-based wellbeing assistance [23].

Evaluation:

Its strength is that it supports text as a helpful modality for this project by analysing users' language through normal conversation rather than just depending on strict questionnaires. However, text-based communication may overlook non-verbal cues like vocal fatigue or facial expression, and Wysa is a general mental health platform rather than a burnout specific monitoring tool. This technique combines text, audio, and face signals to overcome that limitation.

2.2.5 Comparison Summary

Applications	Data Used	User Effort	Multimodal	Early Warning	Needs Hardware
BurnoutGuard	Calendar, work logs, mood, stress, energy, sleep	Medium	No	Yes	No
Welltory	HRV and physiological data	Low	No	Yes	Optional
Unburn	Questionnaire and mood logs	High	No	Partial	No
Wysa	Chat Text	High	No	Partial	No
My Project	Voice, Text, and Face	Low	Yes	Yes	No

Table 2: Comparison of burnout applications

In this table, multimodal refers specifically to combining voice, text, and facial-expression signals. The comparison shows that no application under review integrates text, speech, and facial expression signals into a single low-effort check-in. By offering multimodal early-warning support without the need for extra hardware, this project fills that need.

2.3 Techniques and Methods

2.3.1 Software Libraries

Software Libraries	Applications
Pandas	Organises and simplifies check-in outputs and model-selection results.
PyTorch	Handles the tensor operations for inference while loading and executing the pre-trained audio, text, and vision models.
Scikit-learn	Provides the evaluation of metrics to compare potential models.
Hugging Face Transformers	Makes it simple to evaluate candidate voice, text, and vision models using the same inputs by offering single-interface access.
Librosa	Performs audio loading and resampling, a standard preprocessing step for spectrogram based emotion models.
MediaPipe	Detects facial landmarks used to derive blink rate and head position as supporting signals.
NLTK	Provides tokenisation used to compute speech rate, disfluency, and lexical variety from the transcript.
OpenCV(cv2)	Detects faces with a Haar cascade and samples video frames for the facial-expression model.
jiwer	Computes character and word error rates (CER, WER) for the transcription comparison.
sacreBLEU	Computes chrF++ [6] and BLEU [24] for the translation comparison.
matplotlib	Produces evaluation figures such as confusion matrices, ROC and precision-recall curves, and comparison bar charts.

Table 3: Software Libraries

2.3.2 Algorithms and Models

1. Speech-to-Text Models

Spoken audio is converted into text using speech-to-text models. Strong transcription performance and robustness in noisy conditions have been shown by Whisper, an open-source encoder-decoder Transformer trained on 680,000 hours of multilingual and multitask audio data [15]. Text-based emotion analysis is then performed on the final transcript.

2. Speech Emotion Recognition Models

Instead of focusing only on spoken words, speech emotion recognition models include vocal traits like tone, intensity, and speaking tempo. To classify emotional states, Wav2Vec 2.0 can be refined using labelled emotion data after learning contextually adapted speech representations directly from raw waveform data [25]. Instead, the Audio Spectrogram Transformer (AST) uses a transformer to classify emotion from sound patterns by treating the audio spectrogram as an image [26].

More recent speech emotion models such as MERaLiON-SER [13] use self-supervised speech representations (WavLM) and are trained for the Southeast Asian context, which fits a Singapore deployment. Whisper's speech translation enables reuse of English-based emotion models for Malay, Chinese and Tamil input, and NLLB translates outputs back, to support users across languages.

3. Text Emotion Models

Text-emotion models are used to classify the emotional content of transcribed speech. For text-classification tasks, RoBERTa improves BERT's pre-training methodology [12]. The chosen RoBERTa-large checkpoint, documented on its official model card, has been optimised to classify emotions in English text, including anger, fear, and sadness [27].

4. Facial Expression Models

Video frames are classified into emotions using facial expression models. The Dual-Direction Attention Mixed Feature Network (DDAMFN++) is a mixed feature backbone combined with a dual-direction attention head for facial-expression recognition with good accuracy on benchmark datasets [14]. It is trained on facial-expression data and classifies observable emotional expressions from face images.

5. Signal Fusion Algorithm

Higher values indicate stronger negative-emotion signals that may be relevant to reduced wellbeing. Each modality score is normalised to a common 0-1 scale. The available modality scores are then combined in a late-fusion step using their median, limiting the effect of one unreliable modality, and adjusted with supporting signals from speech and facial behaviour before comparison with the user's baseline. As there is no labelled multimodal burnout dataset available to learn the modality weights, an unweighted median is used as a transparent and robust combination method [28]. The approach is explained in more detail in Section 2.3.3.

6. Tool-Routing and Response Generation

Beyond emotion detection, the project uses a large language model (Qwen2.5-1.5B) [29] to generate personalised recommendations and to power a conversational assistant. Rather than letting the model answer every query freely, the assistant follows a two-stage architecture in which a routing step first selects a tool or response category, and only general or informational queries reach free form generation, consistent with tool-augmented approaches that separate action selection from free text generation to constrain model behaviour [30, 31]. Safety-critical categories such as crisis disclosure and requests for a diagnosis instead return fixed, non-generated responses, reflecting documented risks of hallucination in generative language models [32], particularly in mental-health adjacent applications where an unconstrained response could cause harm [33].

2.3.3 Research Methodologies

1. Comparative Model Selection

After evaluating candidate models on the same labelled evaluation data, the primary metric appropriate to each component is recorded, namely macro-F1 for the text, audio and vision emotion models, CER for speech transcription, and chrF++ [6] for translation, alongside supporting performance metrics, runtime, and applicability. The brief's need to assess and reject models with clear, evidence-based reasoning is then satisfied by selecting the best-performing model for each modality based on this documented data.

2. Personal Baseline Comparison

The project evaluates each reading against a rolling baseline created from previous check-ins rather than evaluating a single check-in in isolation. When the combined indicator rises above a predetermined level above the user's typical range, an early-warning alert is activated. This allows for customised trend tracking, and the baseline threshold is checked against a recognised burnout questionnaire in the evaluation.

3. Score Combination

Since there is no labelled data available for training a fusion model, the available model results are combined using their median, with supporting signals from speech and facial behaviour used as small adjustments rather than learned weights. The median is not affected by a single noisy modality, and the combination remains interpretable and defensible. A fully learned fusion model remains for future work, pending labelled multimodal burnout data. However, a rule-based text-distress floor, informed by healthcare-style check-ins, refines the median fusion in the deployed pipeline (Section 4.6.2).

4. Evaluation and Validation

There are two primary methods for evaluating the system. The accuracy and consistency of the fundamental pipeline operations, namely model loading, score combination, and baseline comparison, are verified by automated unit testing. To identify the optimal model based on evidence, model selection is assessed through comparative testing on labelled datasets using the metric best suited to each task, namely macro-F1 for classification, CER for transcription, and chrF++ [6] for translation, alongside accuracy, supporting metrics and runtime. In addition to unit testing and comparative model selection, the fused indicator is compared to each individual modality to assess if fusion adds value, fairness is evaluated across languages and speaker groups, and a small user study is performed to collect Copenhagen Burnout Inventory [19] responses with short recordings to check the indicator against a recognised burnout measure.

2.4 Research Studies

2.4.1 Burnout can be Measured

Maslach and Leiter define burnout through three measurable elements, namely decreased personal accomplishment or professional inefficacy, depersonalisation or cynicism, and emotional exhaustion [5]. Additionally, they state that changes in mood, behaviour, and appearance may be signs of burnout. Although these signs cannot replace validated assessments, they motivate a low-effort tool that surfaces early indicators from voice, text, and facial cues before exhaustion deepens. This project analyses negative-emotion and fatigue indicators as indirect proxies and tests them against an established burnout questionnaire rather than claiming to measure burnout itself because no model can detect burnout directly.

2.4.2 Voice as Indicator of Emotional State

Machine-learning techniques can categorise a speaker's emotional state from speech signals, according to a review of speech emotion recognition [34]. Additionally, Bandela et al. showed that stressful emotions can be identified using spectral characteristics and acoustic energy [35]. This supports voice as a low-effort early signal for overburdened staff. However, spontaneous speech and background noise during real-world check-ins may cause performance to suffer because many speech-emotion datasets contain performed emotions collected under controlled conditions.

2.4.3 Facial Expression as an Indicator

Fatigue following night shifts was linked to decreased eye opening and facial movement during smiling, according to research involving nurses [36]. This adds weight to investigating facial expressions as indirect markers of exhaustion. However, the vision modality cannot detect burnout on its own because fatigue is different from burnout.

2.4.4 Language as an Indicator of Psychological State

Language analysis can identify psychological indicators from spontaneous speech, including differences in lexical richness and speech coherence [37]. Although these characteristics do not measure burnout directly, this supports the use of transcribed language as an indirect indicator.

2.4.5 Value of Combining Modalities

According to research of multimodal emotion identification, integrating text, audio, and visual data provides more reliable and accurate emotion analysis than single-modality techniques since the modalities offer additional information [38]. This validates the usage of text, speech, and face cues in the project. However, when one modality is unreliable, noisy, or missing, fusion performance may suffer.

2.4.6 Fusion Strategies

Multimodal systems may combine modalities early, by combining features, or late, by combining outputs of separate models. Late fusion is used here because the three pre-trained models are independent and produce comparable scores, and because it degrades gracefully when a modality is missing. Reviews also note that fusion does not always beat the strongest single modality, particularly when one modality is much weaker [39].

2.5 Gaps in Existing Applications

2.5.1 Reliability on a single signal

The examined apps highlight various information sources which include Welltory uses physiological data [21], Wysa uses written language [23], Unburn uses surveys and mood logs [22], and BurnoutGuard uses behavioural and self-reported data [20]. However, none of the reviewed applications combine voice, transcribed text, and facial-expression signals within one check-in.

2.5.2 Active effort and Self-reporting

Several current tools depend on the user actively answering a survey (Unburn) [22] or having a dialogue (Wysa) [23]. In addition to requiring user effort, this could result in recollection or response bias. This burden might be lessened by a simple, user-initiated check-in that is automatically analysed.

2.5.3 Reactive rather than Early-warning

Although they can track changes over time and detect present burnout symptoms, questionnaires may become more challenging for users if they are completed repeatedly. Therefore, investigating a low-effort system that tracks variations from a personal baseline as an early warning indicator is worthwhile.

2.5.4 Wearables

A smartphone camera can be used by Welltory to gather HRV [21], although wearable integration allows for more continuous monitoring and may present accessibility or cost issues. This project needs only a standard computer with a camera and microphone. No additional hardware is required.

2.5.5 Limited focus on healthcare workers

Most tools currently in use are general purpose wellness programs that were not created with frontline healthcare workers' time constraints and long shifts in mind. A tool that is specifically designed to meet the demands and limitations of this population is lacking.

2.6 Contribution to the Field

The primary contribution is a light-weight user-initiated system that orchestrates six pre-trained models, three primary emotion models for voice, text and vision together with speech transcription, translation, and a local language model, to produce a single burnout-related signal. It analyses a single one-minute check-in, and without requiring the user to complete a burnout questionnaire at each check-in. This fills the identified gap among the reviewed applications. Choosing and combining the models shows useful multimodal AI orchestration and provides a way to track potential changes in healthcare professionals' wellbeing over time, even though it is not a replacement for clinical examination.

CHAPTER 3: PROJECT DESIGN (1967/2000 Words)

3.1 Overview

The project concept based on Artificial Intelligence Project Idea 1: Orchestrating AI Models to Achieve a Goal is presented in this chapter.

3.2 Domain and Users

3.2.1 Project Domain

The project uses multimodal AI to improve the welfare of healthcare workers. AI works well in this situation because rule-based software is less suitable for identifying the language, facial, and vocal indicators of burnout.

Domain Area	Explanation
Healthcare Worker Burnout	Spotting early signs of burnout in nurses, doctors, and allied health staff.
Multimodal AI	Using voice and text signals from an audio check-in, with facial-expression signals added when video is used, for a more complete picture.
Automatic Signal Analysis	Cues from a brief user-recorded check-in are automatically analysed without the need for a formal burnout questionnaire.
Occupational Wellbeing	Tracking changes over time so early identification may support earlier action.

Table 4: Project Domain Areas

3.2.2 Project Users

USER GROUP	EXPLANATION
PRIMARY USERS	
Healthcare Workers	A daily private tool to support early awareness in nurses, physicians, and allied health professionals.
SECONDARY STAKEHOLDERS	
Healthcare Organisations	Hospitals benefit indirectly from safer care and reduced employee turnover, and employers are never given access to data.
Researchers	An example of multimodal model orchestration for reference.

Table 5: Project Primary and Secondary Stakeholders

3.3 Design Justification

1. Simple and Low-Effort check-in

Repeated self-reporting may burden users due to time limits in healthcare settings, particularly while working shifts. As a result, the project uses a quick, user-initiated, one-minute check-in that is automatically analysed.

2. Multimodal Strategy

When voice, text, and face are combined, each can fill in the blind spots of the others, producing a more dependable outcome.

3. No additional hardware

Wearable-based tools raise access and cost limitations. Most healthcare workers can use the system on an everyday computer with a built-in camera and microphone.

4. Early Warning

The system finds trends over time in place of a clinical assessment, which is appropriate for a self-awareness use case and prevents too much dependence on an automated judgment.

5. Personal to each person

Healthcare workers can worry that showing signs of exhaustion could damage their standing at work. The tool feels safe and promotes honest use because all scores are kept private, and no information is shared with employers.

6. Local Desktop Processing

As the system is designed as a desktop application, recordings can be fully analysed on the user's own device. This upholds the privacy promise and is appropriate in a sensitive healthcare situation because no video leaves the system and no server is needed.

7. Multilingual Justification

Singapore's healthcare workforce is linguistically diverse, so the check-in and recommendations are offered in English, Malay, Chinese and Tamil to widen access and reduce language as a barrier to honest use.

3.4 Overall Structure

Input capture, analysis by three primary pre-trained models, score combining, and output display make up the system's four-stage pipeline. It creates a single exploratory burnout-related indicator from a one-minute check-in (video or audio-only). The three deployed models are shown in Figure 1 which are RoBERTa-Large (text) [12], MERaLiON-SER (audio) [13], and DDAMFN++ (vision) [14], with Whisper providing transcription.

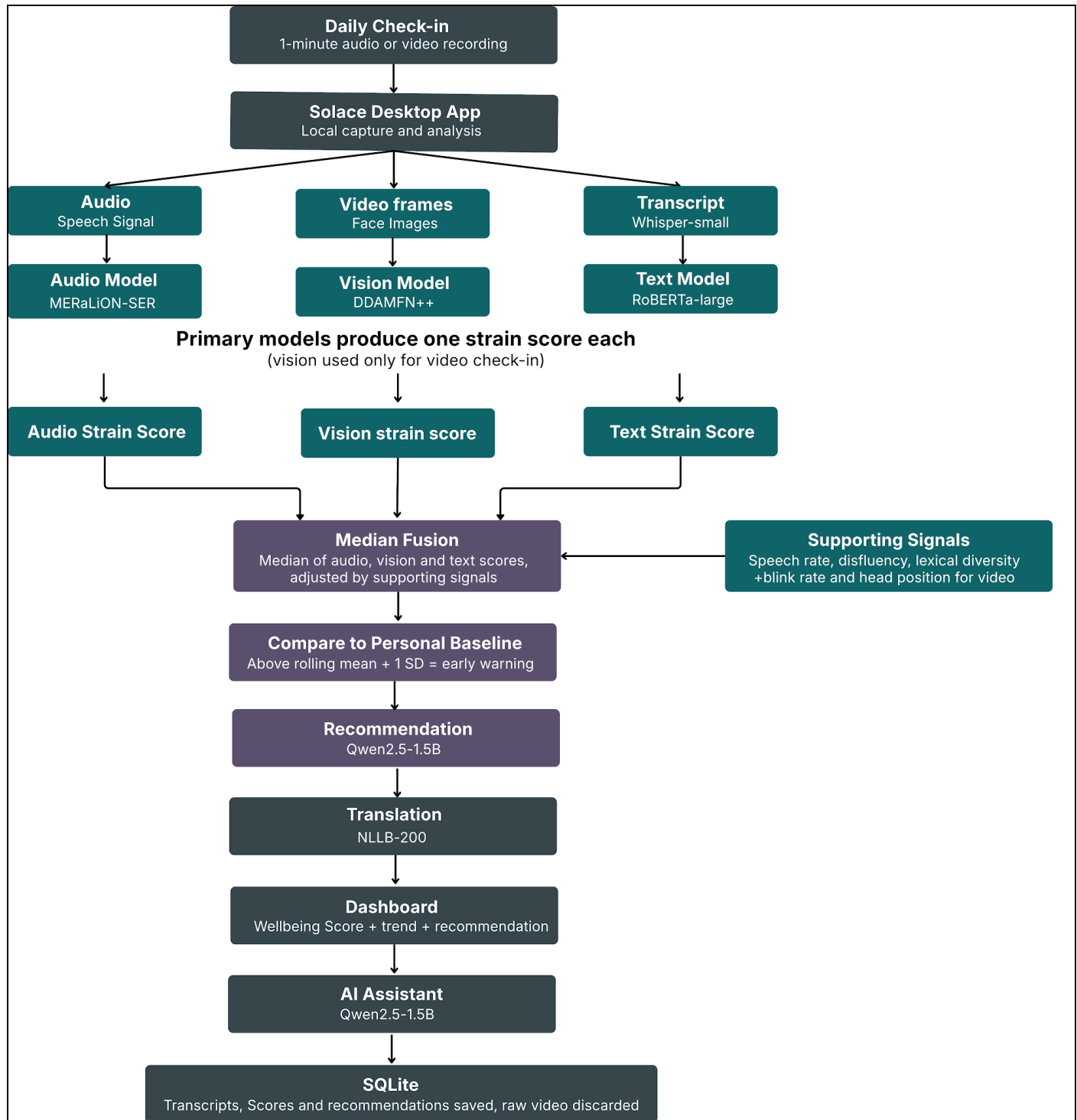


Figure 1: Full Application Pipeline

3.4.1 Processing Pipeline

The desktop application records a one-minute audio or video check-in. Audio is extracted for both check-in types, while video frames are additionally extracted when video is used. The three analysis streams are spoken audio, transcribed text, and, for video check-ins, facial frames. Whisper converts the spoken audio into the transcript. These are scored by the speech-emotion, text-emotion, and facial-expression models respectively. MediaPipe and NLTK are used to derive supporting behavioural signals, such as blink rate, head position, speech rate, disfluency, and lexical variety. The model scores are stored in SQLite and integrated using median fusion with supporting signals to produce the combined strain indicator. This indicator is compared with the user's personal baseline, while its inverse wellbeing score is displayed on the dashboard.

3.4.2 Candidate Models for Respective Modalities

Modalities	Input	Candidate Models	Reason For selecting Models	Output
Text	Transcript	1) DistilBERT 2) BERT 3) RoBERTa 4) DistilRoBERTa 5) RoBERTa-large 6) BERT (nateraw)	To evaluate the prediction performance and runtime of many transformer emotion classifiers, ranging from lightweight (DistilBERT) to huge (RoBERTa-Large).	Text-emotion score
Audio	Voice Audio	1) Wav2Vec2-CREMA-D 2) AST-CREMA-D 3) AST-CREMA-D-Alternative 4) MERaLiON-SER 5) HuBERT-CREMA-D 6) DistilHuBERT-CREMA-D	Compare speech-emotion checkpoints between waveform (Wav2Vec2), spectrogram-transformer (AST), WavLM-based (MERaLiON-SER) and HuBERT approaches in terms of performance and runtime.	Voice score (Tone, energy)
Vision	Video frames	1) ViT 2) ResNet-50 3) ViT-Mixed-Datasets 4) ViT-FER2013 5) ConvNeXt-Tiny 6) Swin-Tiny 7) DDAMFN++	Pre-trained PyTorch checkpoints compare transformer architectures (ViT variants and Swin) with convolutional ones (ResNet-50 and ConvNeXt).	Facial expression score

Speech-to-text	Voice audio	1) Whisper-tiny 2) Whisper-base 3) Whisper-small	To compare Whisper model sizes while weighing runtime for local use vs transcription accuracy.	Transcript
Translation	Text	1) NLLB-200 2) M2M100 3) SeamlessM4T (medium)	To compare leading multilingual translation models on accuracy and runtime across the required languages.	Translated text
Recommendation (LLM)	Fused Result	1) Qwen2.5-1.5B 2) Qwen3-1.7B 3) Malaysian-Qwen2.5-3B 4) Malaysian-Qwen2.5-1.5B 5) Malaysian-Llama3.2-1B 6) Malaysian-Llama3.2-3B	To evaluate the effectiveness of local instruction-tuned LLMs including variations tuned for the Malaysian language in generating reliable, secure, and non-diagnostic suggestions in all supported languages.	Recommendation text

Table 6: Candidate Models by Modality

3.4.3 Combining the Score

Voice, text, and face outputs can be compared on a common scale by converting each model's negative-emotion probability into a documented modality score between 0 and 1.

1. Median fusion with supporting signals

The negative-emotion likelihood of each model is translated into a modality score ranging from 0 to 1. The median of the available modality scores is used to integrate them, preventing distortion from a single unreliable modality. Supporting behavioural signals then apply a small adjustment.

Formula for Video Check-in:

$$\text{Combined strain} = \text{median}(\text{text, audio, vision}) \times 0.98 + \left(\frac{\text{speech-rate} + \text{disfluency} + \text{lexical-variety} + \text{blink-rate} + \text{head-position}}{5} \right) \times 0.02$$

In this case, the five supporting signals are already scaled to 0-1 before being averaged, text, audio, and vision are all scored from 0 to 1, and median () selects the midpoint value to prevent one faulty modality from dominating. The three main signals drive the outcome, with the behavioural signals very slightly influencing it, as 0.98 and 0.02 split the influence.

Formula for Audio Check-in:

$$\text{Combined Strain} = \text{median}(\text{text}, \text{audio}) \times 0.988 + \left(\frac{\text{speech rate} + \text{disfluency} + \text{lexical variety}}{3} \right) \times 0.012$$

For an audio-only check-in there are no vision scores or facial signals, so the median is taken over the text and audio scores with three supporting signals, giving a 0.988 primary weight.

Formula for Wellbeing Score:

$$\text{Wellbeing score} = 100 \times (1 - \text{combined strain})$$

A calmer day produces a higher number, since the wellbeing score is the inverse of the strain expressed as a percentage. An unweighted median is used as a clear and reliable combination method because there is not a labelled multimodal burnout dataset to obtain modality weights.

Worked Example:

For a video check-in with text = 0.30, audio = 0.40, vision = 0.50, and supporting signals averaging 0.50.

$$\text{median}(0.30, 0.40, 0.50) = 0.40$$

$$\text{So, strain} = (0.40 \times 0.98) + (0.50 \times 0.02) = 0.392 + 0.01 = 0.40.$$

$$\text{The wellbeing score shown to the user is } 100 \times (1 - 0.40) = \mathbf{60}.$$

2. Comparison to a personal baseline

An initial personal baseline is created using the first seven valid check-ins. After this period, each indicator is compared with the mean and standard deviation of the previous seven valid readings. A provisional early-warning signal is generated when the new indicator exceeds the rolling baseline mean by one standard deviation. Due to a lack of historical data, the first seven check-ins just show the current reading. The baseline period and alarm threshold are design parameters, and their validity is checked in the evaluation against the Copenhagen Burnout Inventory [19]. An in-app questionnaire page also lets users complete the Copenhagen Burnout Inventory directly within the application, a feature added after a study participant suggested integrating the self-assessment into the tool.

3.4.4 Output

The current indicator, a trend over time, and a personalised wellness recommendation from a local language model, Qwen2.5-1.5B [29], run locally through Hugging Face Transformers, appear on a private dashboard (Figure 4). The home page surfaces a set of static wellness resources for users to browse between check-ins (Figure 2). The navigation bar displays Singapore's national mental health helpline (Mindline) [40] number at all times, not only in response to a flagged interaction. A settings page allows users to permanently delete their account, which removes all stored data from the database, select their preferred interface language (English, Malay, Chinese, or Tamil) and review the privacy and consent disclaimer at any time (Figure 7). On the Trends page, a highlighted banner appears when a reading exceeds the personal baseline, paired with a short supportive tip drawn from the recommendation model, consistent with the considerate presentation described in section 3.9 (Figure 4).

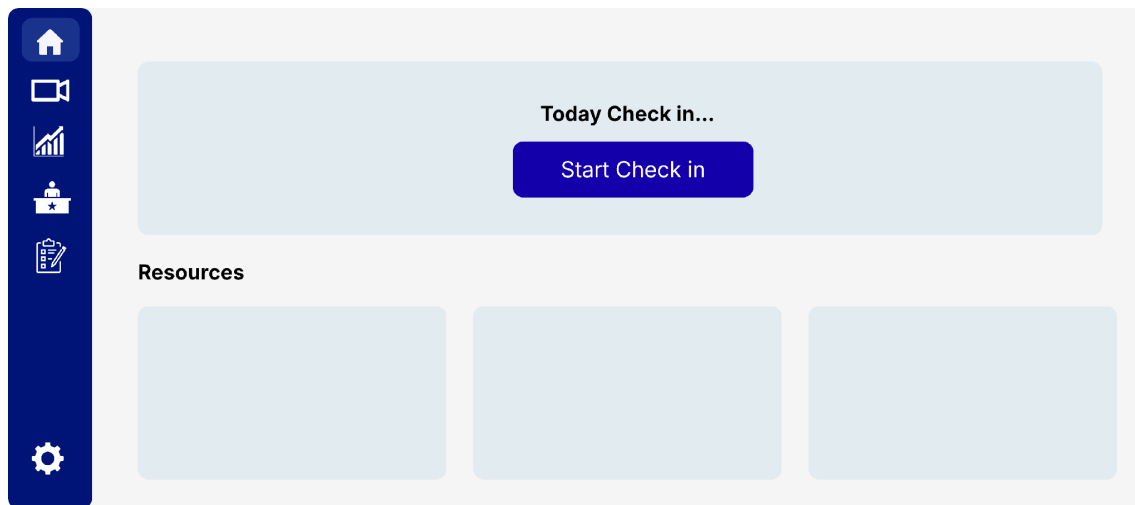


Figure 2: Wireframe for home page

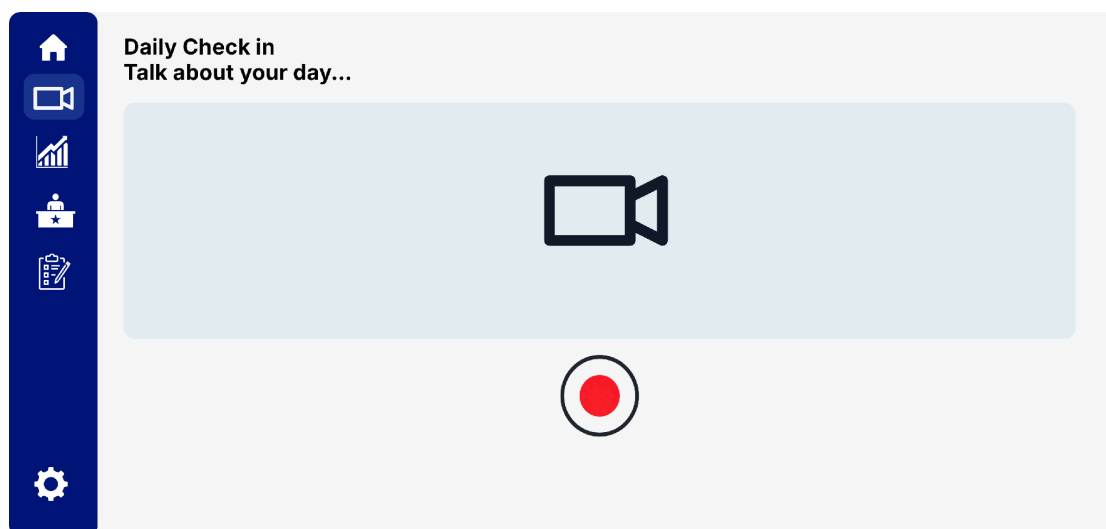


Figure 3: Wireframe for record page

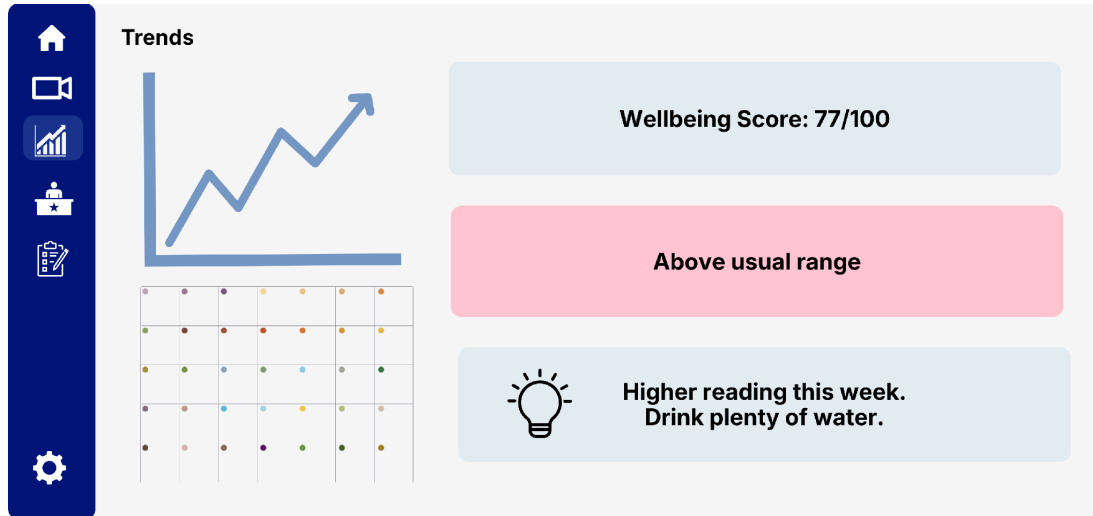


Figure 4: Wireframe for trend page

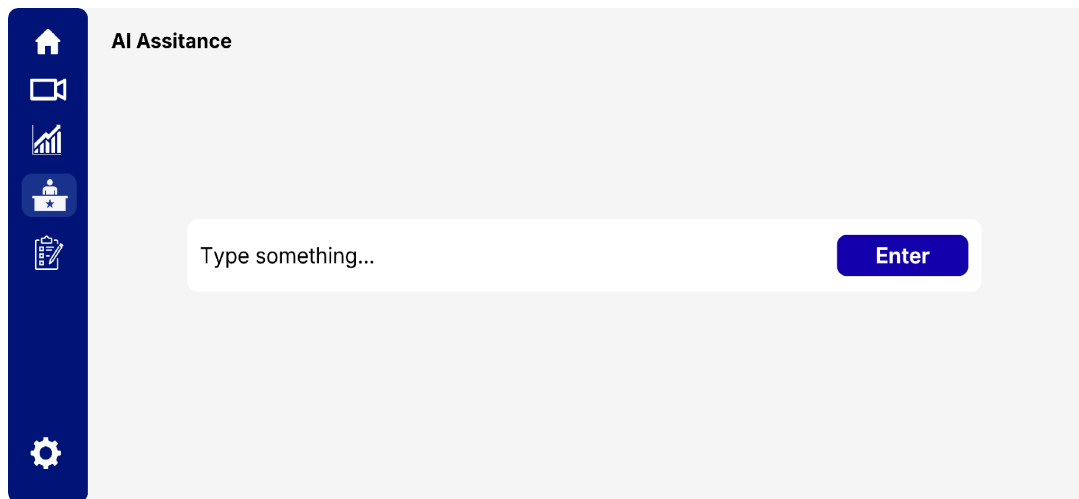


Figure 5: Wireframe for Assistance page

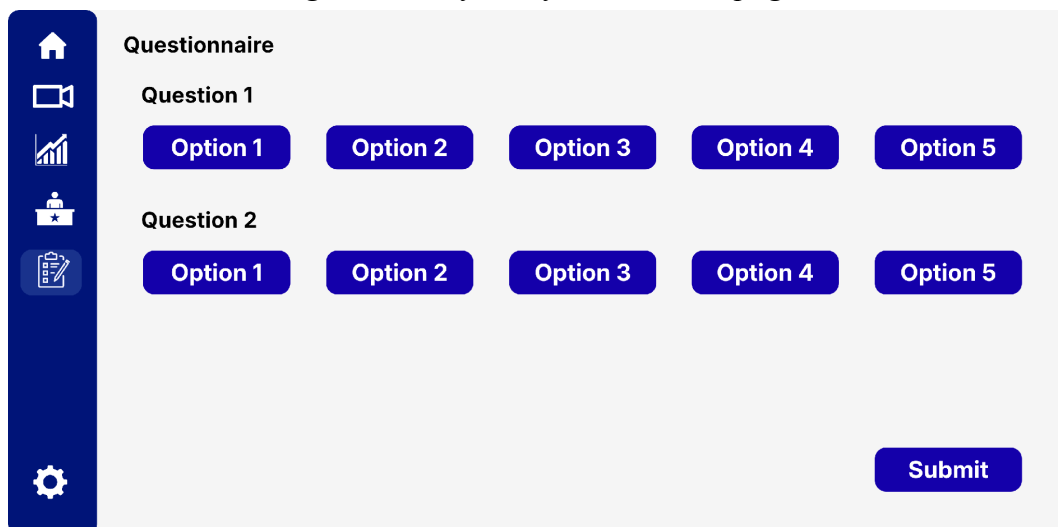


Figure 6: Wireframe for questionnaire page

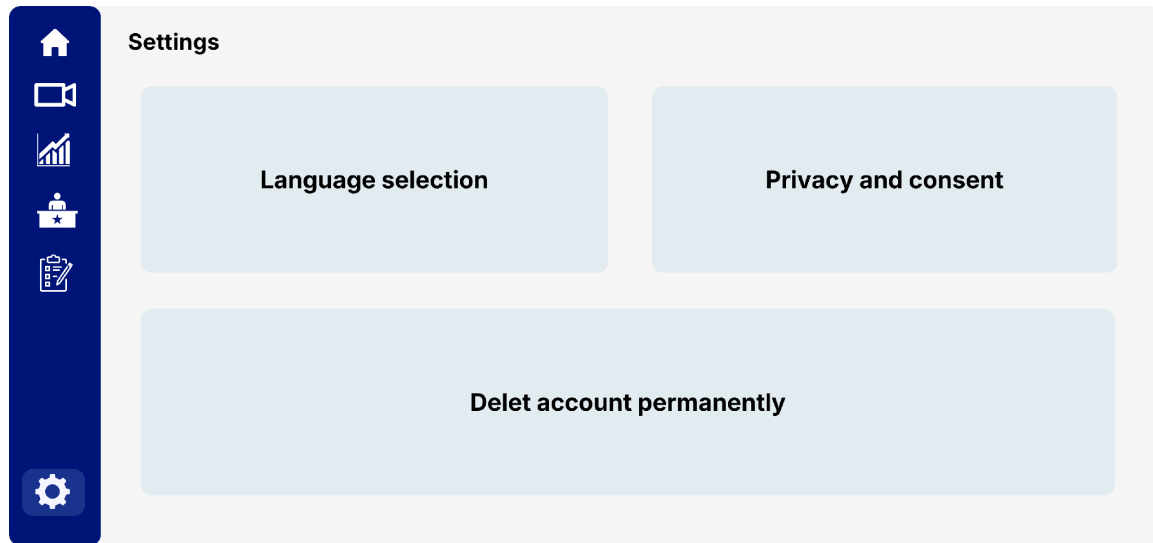


Figure 7: Wireframe for settings page

3.4.5 Assistant Design

A local assistant is also included in the application (Figure 5). It answers questions about the user's own check-in history using a small set of defined data-access tools and refuses out-of-scope or unsafe requests with a fixed safety response (Figure 8).

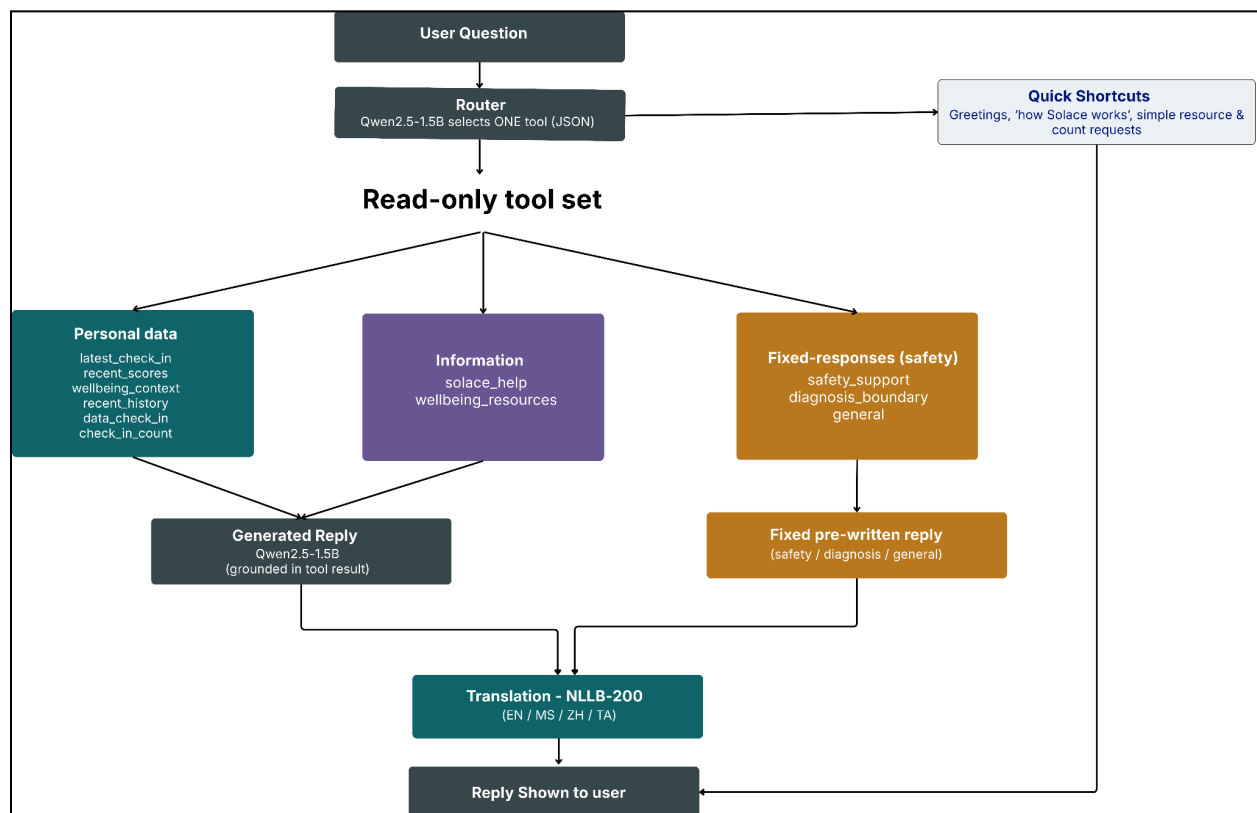


Figure 8: tool-routing pipeline

3.4.6 Future Enhancements

The main planned enhancement is a learned fusion model to replace the median combination, which becomes feasible once enough check-in data with validated burnout ratings is available. Instead of depending on translation, future work could expand the English-based emotion models with natively multilingual emotion models.

3.5 Important Technologies and Methods

CATEGORY	TECHNOLOGY	ROLE IN THE PROJECT
Language	Python	The primary language used for all AI model handling and backend.
AI/ML	Hugging Face Transformers	The candidate pre-trained models are loaded and run.
	PyTorch	Running the pre-trained deep-learning models for inference. No model training is performed.
	Scikit-learn	Calculating macro-F1, precision, recall, accuracy, ROC-AUC, and PR-AUC for model comparison.
Speech-to-text	Whisper	Text transcription of the audio for the text-emotion model
Language Model	Qwen2.5-1.5B (local, via Hugging Face Transformers)	Generates the personalised recommendation and powers the dashboard assistant.
Application/UI	PySide6 (Qt)	Desktop interface for check-in, dashboard, trends, and settings.
Database	SQLite	Keep track of scores and timestamps for the baseline and trends.
Development	VS Code, GitHub	Code editing and version control
Audio Processing	Librosa	Check-in audio is loaded and resampled to 16 kHz mono for the audio-emotion model.
Facial Signals	MediaPipe	Blink rate and head position
Text Signals	NLTK	Speech rate, disfluency, lexical variety

Translation	NLLB	Translating English recommendations, assistant replies and interface text into Malay, Chinese, or Tamil.
Computer Vision	OpenCV (cv2)	Detects faces with a Haar cascade and samples video frames for the facial-expression model.
Testing	PyTest	Runs unit and integration tests that validate the pipeline (model loading, fusion, database persistence).
Evaluation	Jiwer, sacreBLEU	jiwer computes CER/WER for the transcription comparison. sacreBLEU computes chrF++ and BLEU for the translation comparison.
Data Handling	Pandas	Organises check-in outputs and model-selection results.
Visualisation	matplotlib	Produces evaluation figures such as confusion matrices, ROC/PR curves, comparison charts.

Table 7: Technologies and Methods

3.6 Workplan

S/N	Tasks	Estimated Start Date	Estimated End Date	Actual Start Date	Actual End Date	Dependency
1	Project Planning Defining objectives, tools, models, datasets, and deadlines.	27 Apr 2026	3 May 2026	27 Apr 2026	1 May 2026	-
2	Literature Review Research burnout, multimodal AI, existing applications.	4 May 2026	17 May 2026	4 May 2026	17 May 2026	1
3	Project Design Complete the fusion approach, evaluation plan, model selections, and system architecture.	18 May 2026	31 May 2026	18 May 2026	29 May 2026	2
4	Setup Set up environments, libraries, and Git repository	1 Jun 2026	7 Jun 2026	31 May 2026	3 Jun 2026	3
5	First model (text) Construct and evaluate the most basic modality, the text-emotion model	8 Jun 2026	14 Jun 2026	6 Jun 2026	11 Jun 2026	4
6	Audio models Add and compare audio-emotion candidates.	15 Jun 2026	21 Jun 2026	12 Jun 2026	19 Jun 2026	5
7	Preliminary Report Writing & Prototype Submission	8 Jun 2026	29 Jun 2026	8 Jun 2026	29 Jun 2026	5
8	Vision Models Add and compare facial expression candidates.	6 Jul 2026	19 Jul 2026	1 Jul 2026	17 Jul 2026	6
9	Pipeline and Fusion Link models, include baseline comparison, median fusion, and Whisper transcription.	20 Jul 2026	2 Aug 2026	20 Jul 2026	2 Aug 2026	8
10	Application UI Construct the dashboard, Desktop interface (PySide6), SQLite storage, and recommendation model.	3 Aug 2026	16 Aug 2026	3 Aug 2026	14 Aug 2026	9

11	Enhancements blink rate, head position, speech rate, disfluency, and lexical variety, AI Assistant, Multilingual Application	17 Aug 2026	21 Aug 2026	15 Aug 2026	18 Aug 2026	10
12	Report Write-up Draft	19 Aug 2026	19 Aug 2026	19 Aug 2026	19 Aug 2026	11
13	Evaluation and iteration Unit tests and model testing	20 Aug 2026	27 Aug 2026	20 Aug 2026	27 Aug 2026	12
14	Evaluation and iteration fusion comparison, fairness testing and function evaluation	28 Aug 2026	4 Sep 2026	28 Aug 2026	3 Sep 2026	13
15	Evaluation and iteration CBI user study, healthcare speech evaluation and robustness evaluation	5 Sep 2026	13 Sep 2026	4 Sep 2026	12 Sep 2026	14
16	Final Report Editing	14 Sep 2026	27 Sep 2026	14 Sep 2026	26 Sep 2026	15
17	Final Submission	28 Sep 2026	28 Sep 2026	28 Sep 2026	28 Sep 2026	16

Table 8: Workplan

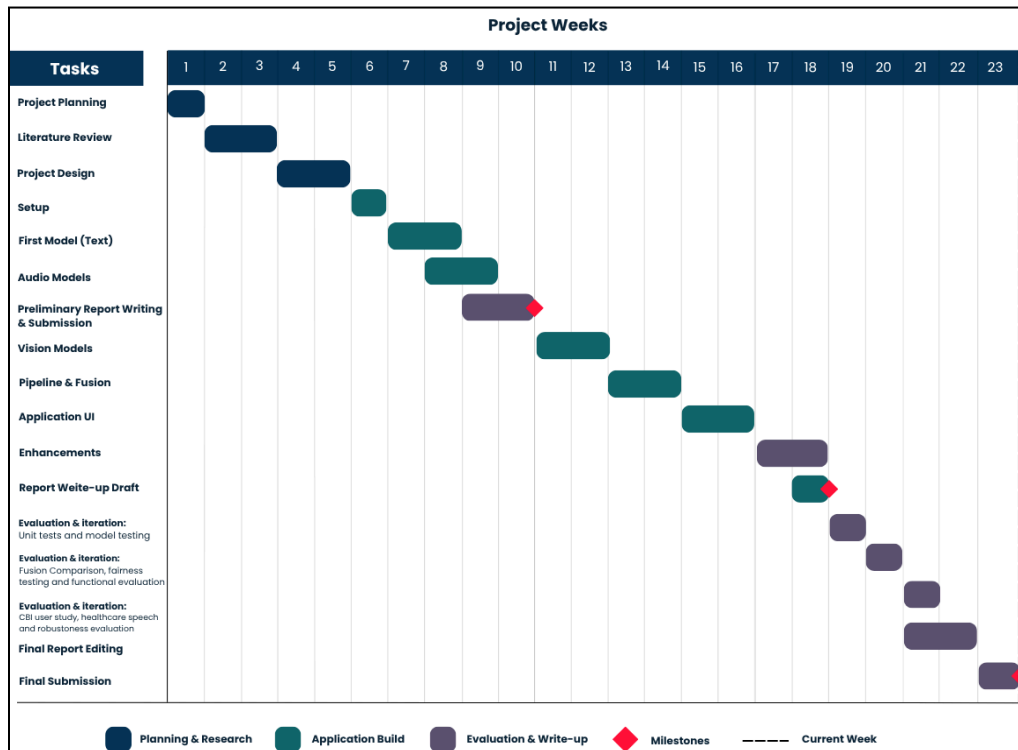


Figure 9: Gantt Chart

3.7 Contingency Plans

Risk	Contingency Plan
Bias or accuracy issues	Tested using a variety of data, documented limitations, and presented as an early warning rather than a diagnostic.
Modality proves unreliable	A clearly labelled partial score is computed from the remaining valid modalities. If one modality fails quality checks, the user is prompted to try again.
Environment Problems	Well-documented, open-source tools with regular Git commits.
Model Instability	Unstable models will be replaced by the next best tested candidate.

Table 9: Risks and Contingency Plan

3.8 Evaluation Strategy

1. Model Selection

RAVDESS, CancerEmo and RAF-DB are used for audio, text, and visual evaluation, where checkpoint documentation shows no training data overlap, with each candidate's fine-tuning corpus disjoint from its evaluation set ([Appendix A](#)). Models are evaluated using macro-F1, precision, recall, accuracy, ROC-AUC, PR-AUC, and runtime to compare performance and efficiency. Based on this data, the most suitable model is selected, and weaker ones are rejected with reasonable justification. These datasets test each model's task-specific reliability rather than exhaustion itself since they contain emotion labels instead of burnout labels.

2. System testing

End-to-end pipeline integrity, including model loading, inference execution, score fusion, and database persistence, is validated using PyTest-based unit and integration tests.

3. Usability testing

After completing a check-in and dashboard activity, 10 to 15 representative users will test the system's usability and output clarity on a five-point scale and offer brief suggestions for improvement.

4. Fusion Comparison

To determine whether combining models outperforms the strongest single modality, the fused indication is compared to each individual modality as well as to alternative fusion techniques.

5. Fairness Testing

Transcription and emotion performance are checked across languages and across speaker groups for example speaker gender.

6. Validation against a Burnout Measure

In a separate study, short audio and video recordings were collected from consenting participants alongside their Copenhagen Burnout Inventory responses, and the combined strain indicator is compared against the CBI score by correlation and band agreement.

7. Functional Evaluation (Assistant)

The dashboard assistant is tested to confirm it correctly routes queries to the defined data-access tools, returns accurate answers about the user's own check-in history, and returns the fixed safety response for out-of-scope, crisis, or diagnosis requests rather than a generated one.

8. Functional Evaluation (Multilingual)

The end-to-end check-in and recommendation flow is tested across English, Malay, Chinese, and Tamil to confirm transcription, translation, and recommendation generation behave correctly in each supported language.

9. Robustness Evaluation

System behaviour is tested under degraded audio, missing modalities, and noisy input to confirm the pipeline degrades gracefully to a clearly labelled partial score rather than failing.

10. Healthcare Speech Evaluation

Beyond acted datasets, the audio-emotion model is evaluated on natural, healthcare-style speech to check that performance holds on spontaneous, domain-relevant input.

11. Defining Success

Success will be determined by technological performance and usability rather than clinical accuracy since there are no ground-truth burnout labels. If all required models for the selected check-in type generate valid scores, candidate models are compared using macro-F1 and runtime, automated tests pass, at least 95% of test check-ins are completed successfully, and users are able to complete the check-in and understand the dashboard without help, the project will be considered successful.

3.9 Ethical Concerns

1. Privacy and data handling

The captured voice, face, and text signals are sensitive to personal data, so collection is kept minimal. In line with Singapore's PDPA, the raw video and audio are processed and immediately deleted, and only the transcript, numerical scores and generated text are stored [1]. A disclaimer appears before first use, which the user must accept before any check-in. Users can review their check-in history on the Trends page and permanently delete their account from the settings page at any time, which removes all associated records from the database giving them direct control.

2. Bias and fairness

Model performance can be impacted by accents, lighting, skin tones, and camera quality. Instead of an authoritative decision, the result is a supportive signal where models are assessed on a variety of data, limitations are noted, and the system makes no claims about being equally reliable for all users.

3. Consent

The use of the app is completely voluntary. The user is presented with a disclaimer regarding data handling, immediate deletion of videos, and the non-diagnostic nature of the tool, which they must agree to before the first check-in. Users can choose to record all check-ins and can stop at any time. Separately, participants in the CBI validation study gave written informed consent before their audio and video were recorded and used. A sample consent form is provided in [Appendix B](#).

4. Not a clinical diagnosis

The system is not a medical diagnosis, it is an early warning system. Users are encouraged to seek expert advice if they have any concerns. If a check-in or assistant interaction suggests acute distress, the interface displays Singapore's national mental health helpline (Mindline) [40] rather than a generated response.

5. Considerate Presentation

The interface begins with a qualitative state compared to the person's personal baseline and minimizes the underlying number, combining any heavier than usual reading with a supportive prompt rather than a bare score, because a raw burnout figure might worsen anxiety in an already stressed user.

6. Safe Response Generation

Because a local language model generates the recommendations and assistant replies, safety-critical requests such as crisis disclosure or requests for a diagnosis return fixed, pre-written responses instead of generated text, reducing the risk of harmful or hallucinated output in a mental-health adjacent setting.

CHAPTER 4: IMPLEMENTATION (2052/2500 Words)

4.1 Overview

This chapter details Solace's architecture, model selection, core pipeline, recommendation and assistant agents, multilingual output, the in-app questionnaire, and data storage.

4.2 System Architecture

With main.py serving as the entry point, Solace is a single desktop application divided into three layers under a UI package which are a user interface layer (UI/ui), an AI pipeline layer (UI/ai), and a data layer (UI/database).

Layer	Location	Key Files	Responsibility
Interface	UI/ui	main_window.py, check_in_page.py, assistant_page.py	Screens, navigation, language switching (PySide6)
AI pipeline	UI/ai	multimodal_pipeline.py, solace_agent.py	Multimodal analysis, fusion, recommendation, assistant
Data	UI/database	database.py, schema.sql	Authentication, consent, check-in storage (SQLite)

Table 10: Application Layers

Account-level actions such as logout, delete, profile save, closing the app are guarded by account_busy(), which blocks them while a recording, analysis or assistant reply is still running, preventing a check-in being corrupted mid-analysis.

```
# Check whether recording or background work is active
def account_busy(self):
    p=self.check_in_page
    # Check recording and background worker before account changes
    return (p.video_recording or p.audio_recording
            or (p.analysis_worker is not None and p.analysis_worker.isRunning())
            or (p.transcription_worker is not None and p.transcription_worker.isRunning())
            or self.assistant_page.worker_running())
```

Code Snippet 1: account_busy() function

4.3 Model Selection Method

Because the project uses pre-trained models rather than training its own, the main engineering task was to select the best model for each modality on documented evidence. For each modality, the same approach was used. Several public checkpoints were gathered for each, ran over the same labelled evaluation set, and rated using the same metrics. To keep the comparison fair, each candidate's fine-tuning corpus was checked against its evaluation set so that no model was tested on data it had been trained on, and every candidate received the same inputs and was evaluated with the same pipeline.

Modality	Evaluation Data	Primary Metric	Selection Rule
Text	CancerEmo [8]	Macro-F1	Highest macro-F1, runtime acceptable
Audio	RAVDESS [9]	Macro-F1	Highest macro-F1, runtime acceptable
Vision	RAF-DB [7]	Macro-F1	Highest macro-F1, runtime acceptable
Transcription	FLEURS [10]	Macro-CER	Lowest character error rate
Translation	OPUS-100 [11]	chrF++	Highest chrF++, BLEU recorded alongside
Recommendation LLM	Wellbeing Scenarios	Macro Compliance Score	Highest compliance (valid JSON, safety, recommendation rate) across languages

Table 11: Model Selection Criteria by modality

4.4 Selected Models

Modality	Selected Model
Text	RoBERTa-Large [12]
Audio	MERaLiON-SER [13]
Vision	DDAMFN++ [14]
Transcription	Whisper-Small [15]
Translation	NLLB-200 [16]
Recommendation LLM	Qwen2.5-1.5B [29]

Table 12: Selected Models for each Modality

Each model was chosen for the best evaluation result while remaining suitable for local deployment, following the process in Section 4.3. Chapter 5 presents the full candidate comparisons.

4.5 Multimodal Pipeline

4.5.1 Pipeline Structure and Model Management

Six models are orchestrated by the pipeline, but never all held in memory at once, since Solace runs on ordinary laptops. Each model starts as None and loads only when first needed, then is reused. Five models load from Hugging Face and DDAMFN++ is only on GitHub, so `_ensure_ddamfn()` downloads its code and checkpoint on first use and `_ddamfn_verify_checkpoint()` rejects a Git-LFS pointer or truncated file before loading. When a stage finishes, `release_whisper()`, `release_analysis_models()` and `release_qwen()` each call `clean_memory()`, so at most one or two models occupy memory at any moment.

```
# Load the text emotion model when it is needed
def load_text_model(self):
    if self.text_model is None:
        self.text_model = hf_pipeline("text-classification", model=TEXT_MODEL_ID, device=self.device, top_k=None)
    return self.text_model
```

Code Snippet 2: `load_text_model()` function

```
# checkpoint size and file header before loading
def _ddamfn_verify_checkpoint(local_path):
    local_path = Path(local_path)
    size = local_path.stat().st_size
    with open(local_path, "rb") as handle:
        head = handle.read(64)
    # Reject Git LFS pointers, small files and files without a ZIP header
    if head.startswith(b"version https://git-lfs") or size < 100_000 or head[:2] != b"PK":
        # unusable file so it can be downloaded again
        local_path.unlink(missing_ok=True)
        raise RuntimeError(f"The DDAMFN checkpoint did not download as a valid model file (size {size} bytes).  
Download 'affecnet7_epoch19_acc0.671.pth' manually from the DDAMFN++ repository and place it at {local_path}, then restart Solace.")
```

Code Snippet 3: `_ddamfn_verify_checkpoint()` function

4.5.2 Modality Scores and Supporting Signals

Each primary model produces one negative-emotion score from 0 to 1, where higher means stronger negative emotion, by summing the probabilities of the negative classes (anger, fear, sadness, disgust) so the three modalities are directly comparable. `vision_score` averages this over eight sampled face frames, `audio_score` loads the waveform at 16 kHz and runs MERaLiON, and `text_score` runs the transcript through the classifier.

```
# probabilities of the selected negative emotions
def negative_score(output):
    return clamp(sum(
        float(item["score"])
        for item in predictions(output)
        if label_name(item["label"]) in NEGATIVE_LABELS
    ))
```

Code Snippet 4: `negative_score()` function shared negative-emotion scoring

Five supporting behavioral signals are derived in addition to the models which are speech_signals uses NLTK to calculate speech rate, disfluency, and lexical variety from the transcript, and visual_signals uses MediaPipe landmarks to calculate blink rate and head position. Blink detection is based on the eye-aspect ratio, the vertical eye opening over the horizontal width.

4.5.3 End-to-End Analysis

The analyse method connects everything where it extracts the audio and, for video, the frames, transcribes with Whisper, produces the available modality scores, collects the supporting signals, and passes them to fusion (Section 4.6). Because this is heavy, it runs off the interface thread in an AnalysisWorker so the app stays responsive, and the result dictionary is stored and shown.

```
# Run full check in analysis and prepare the results
def analyse(self, recording_path, recording_type, progress, transcript=None, *, language_name="English", trend=( "No previous check-in trend is available.")):
    temporary_audio = None
    try:
        # Extract separate audio track when input is video
        if recording_type == "video":
            progress("processing_extract_audio")
            temporary_audio = (self.extract_audio(recording_path))
            audio_path = (temporary_audio)
        else:
            audio_path = (recording_path)
        # Create transcript only when one was not supplied
        if not transcript:
            progress("processing_transcription")
            transcript = self.transcribe(audio_path, language_name,)
        progress("processing_text")
        # English text for text emotion model
        if language_name == "English":
            english_text = transcript
        else:
            english_text = (self.translate_to_english(audio_path, language_name))
        # Score transcript, then release speech model
        text = self.text_score(english_text)
        self.release_whisper()
        progress("processing_audio")
        # Score audio and start list of main model results
        audio = self.audio_score(audio_path)
        primary_scores = [text, audio]
        vision = None
        visual = {}
```

```
# Add facial emotion and visual signals for video check ins
if recording_type == "video":
    progress("processing_vision")
    vision = self.vision_score(recording_path)
    primary_scores.append(vision)
    progress("processing_signals")
    visual = (self.visual_signals(recording_path))
else:
    progress("processing_signals")
# Measure speech signals using the original transcript
speech = self.speech_signals(transcript, audio_path, language_name,)
# Collect supporting scores available for this recording
supporting = [speech["speech_signal"], speech["disfluency_signal"], speech["lexical_signal"]]
if recording_type == "video":
    supporting += [visual["blink_signal"], visual["head_signal"]]
progress("processing_fusion")
# Fuse scores and convert strain into wellbeing
(primary_strain, supporting_strain, strain, primary_weight) = self.fuse(primary_scores, supporting)
# Text-distress floor
if text >= TEXT_DISTRESS_THRESHOLD:
    strain = clamp(max(strain, TEXT_DISTRESS_FLOOR * text))
strain_score = (strain * 100)
wellbeing_score = (100 - strain_score)
phrase, explanation = (self.summary(wellbeing_score))
# Release analysis models before generating recommendations
self.release_analysis_models()
progress("processing_recommendation")
recommendation_english, recommendation = self.recommendation(english_text, wellbeing_score, phrase, language_name, trend)
```

Code Snippet 5: analyse() function end to end check in analysis

4.6 Fusion and Baseline Early Warning

4.6.1 Median Fusion

The primary strain estimate is the median of the available modality scores, which prevents one extreme model from dominating. Each supporting signal adds a small fixed weight of 0.004, so the five signals together contribute about two percent of the strain for a video check-in and the primary median keeps about 98 percent. This value was chosen from an auxiliary-weight sweep, reported in Section 5.4.2, as the largest weight that preserves the best macro-F1 while still allowing the behavioural signals to nudge the score.

```
# Combine model scores with the supporting signals
def fuse(self, primary_scores, supporting_scores,):
    # median to reduce the effect of one extreme model score
    primary = median(primary_scores)
    supporting = (mean(supporting_scores) if supporting_scores else 0)
    # each supporting score its small fixed contribution
    supporting_contribution = sum(score * AUXILIARY_WEIGHT for score in supporting_scores)
    # remaining weight for the main model score
    primary_weight = (1.0 - len(supporting_scores) * AUXILIARY_WEIGHT)
    strain = clamp(primary * primary_weight + supporting_contribution)
    return (primary, supporting, strain, primary_weight)
```

Code Snippet 6: fuse() function median fusion with supporting-signal adjustment

The outcome is a strain value clamped to a range of 0 to 1, where higher values indicate greater negative-emotion strain. The interface displays its inverse as a wellbeing score, where higher values indicate better wellbeing.

4.6.2 Text-Distress Floor

Healthcare workers often keep a calm voice and expression even when distressed, so audio and vision can underreact while the words show clear strain. When the text score reaches 0.60 (TEXT_DISTRESS_THRESHOLD), analyse() prevents the final strain falling below 0.70 x the text score (TEXT_DISTRESS_FLOOR), giving the words a floor of influence. The values were set from evaluation on healthcare-style clips.

4.6.3 Personal Baseline and Early Warning

Rather than judging a check-in against fixed thresholds, each reading is compared against the user's own recent range, since a normal level differs from person to person. The classify_baseline function determines whether the new score is above, within, or below the range by calculating the mean and standard deviation of the previous check-ins. An early-warning status is raised when the combined strain indicator rises above the personal baseline mean by at least one standard deviation.

```
# Compare score with the recent personal range
def classify_baseline(score, previous_scores, min_history=7):
    # wait until enough previous scores are available
    if len(previous_scores) < min_history:
        return None

    # Calculate recent average and score variation
    personal_mean = mean(previous_scores)
    personal_sd = pstdev(previous_scores)

    # Handle history with no variation
    if personal_sd == 0:
        if score > personal_mean:
            return "baseline_above"

        if score < personal_mean:
            return "baseline_below"

        return "baseline_within"

    # whether the score is above the usual range
    if score > personal_mean + personal_sd:
        return "baseline_above"

    # whether the score is below the usual range
    if score < personal_mean - personal_sd:
        return "baseline_below"

    return "baseline_within"
```

Code Snippet 7: classify_baseline() personal baseline classification

At least seven prior check-ins are required before a classification is formed. Until then only the current reading is shown, since too few points give an unreliable baseline. The zero-variance case, where all past readings are identical, is handled separately. The baseline period and one-standard-deviation threshold are design parameters, evaluated against the Copenhagen Burnout Inventory in Chapter 5.

4.7 Recommendation, Assistant and Multilingual Output

4.7.1 Recommendation Generation

After fusion produces the wellbeing score, Qwen2.5-1.5B generates exactly three brief, non-diagnostic recommendations, deterministically for stable output. A system instruction fixes its role as a workplace wellbeing aide and forbids diagnosis, and when the score is low (below 35) the prompt adds a recommendation to contact a trusted person or healthcare professional.

```
# three recommendations and translate them for the user
def recommendation(self, english_text, wellbeing_score, summary, language_name, trend):
    generator = self.load_qwen()
    support_note = ""
    # Ask for support contact suggestion when the score is low
    if wellbeing_score < 35:
        support_note = ("Include one recommendation to contact a trusted person or qualified healthcare professional.")
    # Give Qwen transcript score summary and recent trend
    messages = [
        {
            # system
            "role": "system",
            "content": ("You are a workplace wellbeing support assistant. Do not diagnose medical conditions.")
        },
        {
            # user
            "role": "user",
            "content": ("Write exactly 3 short wellbeing recommendations in English.\n"
                        "Write each recommendation on a separate numbered line.\n"
                        "Make each recommendation different and practical.\n"
                        "Do not repeat ideas.\n"
                        f"({support_note})\n\n"
                        f"Wellbeing score: {wellbeing_score}\n"
                        f"({wellbeing_score:.0f})/100\n"
                        f"Summary: {summary}\n"
                        f"Trend: {trend}\n"
                        f"Transcript: {english_text[:1200]}")
        }
    ]

    # prompt with Qwen thinking mode disabled
    prompt = generator.tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
    # short reply without random sampling
    english = generator(prompt, max_new_tokens=180, do_sample=False, return_full_text=False, pad_token_id=generator.tokenizer.eos_token_id)[0][["generated_text"].strip()
    # Release Qwen before running translation
    self.release_qwen()
    translated = translate_text(english, language_name)
    # return results
    return english, translated
```

Code Snippet 8: recommendation () function

4.7.2 Assistant Agent

The assistant lets the user ask questions about their own history, such as their latest reading or recent trend. Instead of allowing the model to respond freely from the raw input, it is designed as a tool-using agent with a predetermined set of allowed tools listed in Table 13.

Tool	Purpose
latest_check_in	Return the user's most recent check-in
recent_scores	Return the recent wellbeing scores for the trend
wellbeing_context	Combine latest reading and recent scores
recent_history	Summaries the last several check-ins

date_check_in	Return the check-in for a specific date
check_in_count	Return how many check-ins the user has done
solace_help	Answer general questions about how Solace works
safety_support	Return the fixed crisis/safety response
general	Handle general conversation with no personal data
wellbeing_resources	Returns score-banded coping resources and tips
Diagnosis_boundary	Returns the fixed cannot diagnose boundary message

Table 13: Assistant tools

A `fast_tool()` shortcut answers plain resource or check-in count requests, and `quick_reply()` handles greetings, so Qwen is skipped entirely for these common cases. When given a question, the model first selects a tool by returning a JSON decision. `parse_decision` then extracts and defensively validates that decision by locating the JSON block, rejecting invalid output, forcing unknown tool names to a safe default, and discarding malformed dates. This shows that an invalid action can never be triggered by a faulty or unexpected model output.

```
# Read and check the model tool selection JSON
@staticmethod
def parse_decision(text):
    # JSON object and fall back to app help if it is missing
    match = re.search(r"\{.*?\}", text, flags=re.DOTALL)
    if not match:
        return {"tool": "solace_help", "date": ""}

    try:
        # decision
        decision = json.loads(match.group(0))

        # app help when the JSON cannot be read
    except json.JSONDecodeError:
        return {"tool": "solace_help", "date": "" }

    tool = decision.get("tool", "")
    date_text = str(decision.get("date", "")).strip()
    # unknown tools with the app help tool
    if tool not in TOOL_NAMES:
        tool = "solace_help"
    # clear do not match the required year month day format
    if (tool == "date_check_in" and not re.fullmatch(r"\d{4}-\d{2}-\d{2}", date_text)):
        date_text = ""
    return {"tool": tool, "date": date_text}
```

Code Snippet 9: Parse_decision validating the model's tool choice

Each tool reads only the logged-in user's stored data, so the assistant cannot access another user's data or invent history that is not in the database.

4.7.3 Multilingual Output and Safety Handling

Instead of an AI generated response, questions that indicate a crisis or self-harm are sent to a specific safety_support tool, which returns a fixed, non-diagnostic message pointing the user to emergency services and a competent professional. Interface text is translated once and cached in ui_translations.json (prepare_translations()), and per check-in recommendations are translated after Qwen is released, so only one large model is ever loaded. Whisper's translation task also produces an English working transcript, letting the same English emotion models serve every language.

```
# fixed message for urgent support
@staticmethod
def safety_support():
    # return urgent safety
    return {
        "status": "safety",
        "message": (
            "Solace cannot provide emergency "
            "or crisis care. "

            "If you may be in immediate danger "
            "or may harm yourself or someone "
            "else, contact local emergency "
            "services or a trusted person who "
            "can stay with you. "

            "For urgent mental-health concerns, "
            "seek help from a qualified "
            "healthcare professional."
        )
    }
```

Code Snippet 10: safety_support function

4.8 Data Storage and Privacy

All data is stored locally in a single SQLite database defined by schema.sql, with seven tables which are users, check_ins, questionnaire_responses, questionnaire_prefs, cbi_comparisons, experience_feedback and experience_feedback_prompts. PRAGMA foreign_keys = ON enforces the cascading deletes below (Figure 10).

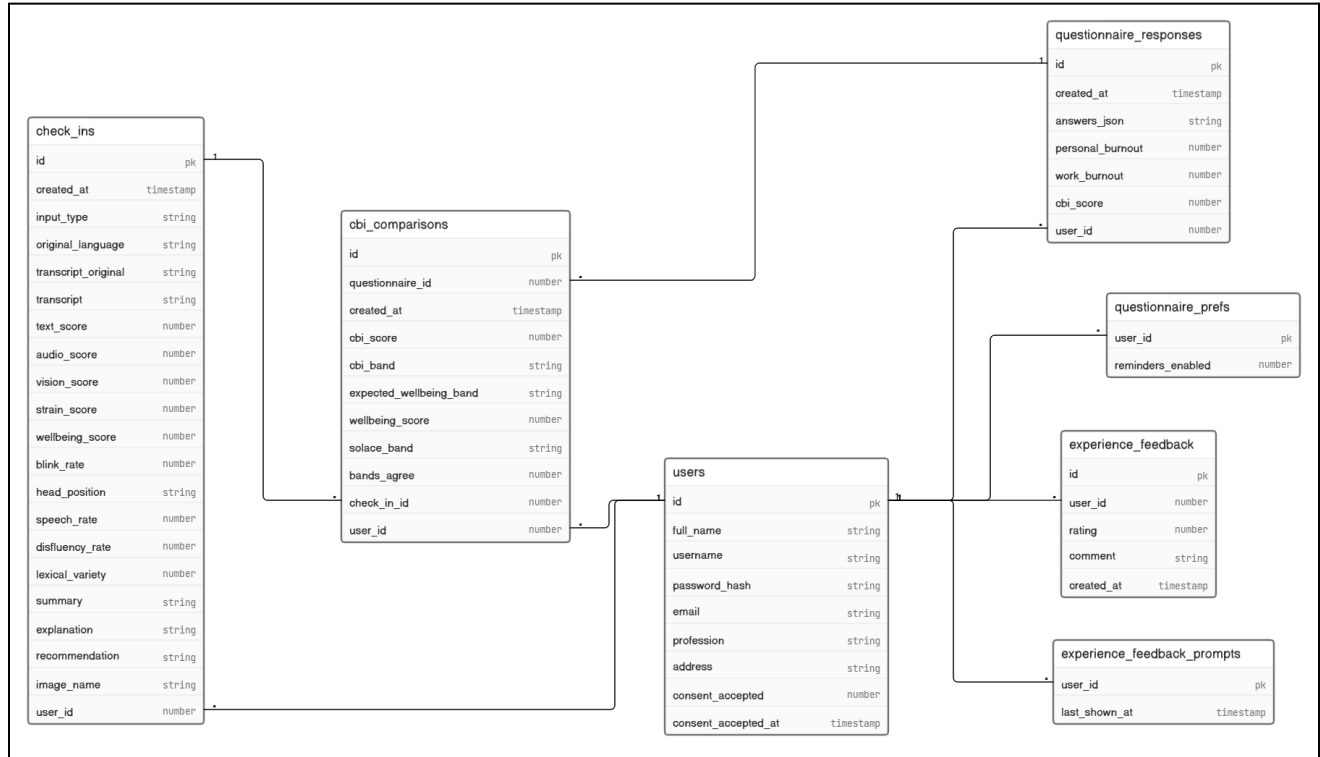


Figure 10: Solace Database Schema (entity-relationship)

4.8.1 Check-in Storage

The check_ins table stores the transcript, five supporting signals, fused strain and wellbeing scores, and three modality scores, but not the audio or video, the raw recording is processed locally and deleted after analysis, leaving only numbers and text. A foreign key with ON DELETE CASCADE links each check-in to its user, so when a user is deleted, all their check-ins are also deleted.

```
-- Check ins table to store each check-in input details and analysis results
CREATE TABLE IF NOT EXISTS check_ins (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  user_id INTEGER NOT NULL,
  created_at TEXT NOT NULL DEFAULT CURRENT_TIMESTAMP,
  input_type TEXT NOT NULL CHECK (input_type IN ('audio', 'video')),
  -- original language, original transcript and English transcript
  original_language TEXT NOT NULL DEFAULT 'English',
  transcript_original TEXT NOT NULL,
  transcript TEXT NOT NULL,
  -- emotion scores, allowing no vision score for audio check-ins
  text_score REAL NOT NULL,
  audio_score REAL NOT NULL,
  vision_score REAL,
  -- combined strain and wellbeing scores
  strain_score REAL NOT NULL,
  wellbeing_score REAL NOT NULL,
  -- visual measurements when available alongside the speech measurements
  blink_rate REAL,
  head_position TEXT,
  speech_rate REAL NOT NULL,
  disfluency_rate REAL NOT NULL,
  lexical_variety REAL NOT NULL,
  -- summary, explanation, recommendations and result image name
  summary TEXT NOT NULL,
  explanation TEXT,
  recommendation TEXT,
  image_name TEXT,
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

Code snippet 11: check_ins table

	id	user_id	created_at	input_type	original_language	transcript_original	transcript	text_score	audio_score	vision_score	strain_score	wellbeing_score	blink_rate	head_position	speech_rate	disfluency_rate
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	2	2026-09-26 05:03:19	audio	English	Okay, my week ..	Okay, my week..	77.06	86.76	NULL	80.43	19.87	NULL	NULL	172.27	0.0
2	2	3	2026-09-26 05:31:20	audio	Tamil	இன்று என் வாரம்..	Today, my wee..	97.74	96.87	NULL	96.04	3.96	NULL	NULL	141.69	0.16
3	3	4	2026-09-08 05:37:54	video	English	My shift was ..	My shift was ..	17.01	16.99	13.22	16.67	84.33	21.36	Centred	123.0	0.0
4	4	4	2026-09-22 05:37:54	audio	English	Today was quite..	Today was ..	61.02	53.55	NULL	51.66	48.34	NULL	NULL	115.0	0.0
5	5	5	2026-09-06 05:30:54	audio	English	Today was quite..	Today was ..	52.79	66.26	NULL	53.87	46.13	NULL	NULL	113.0	0.0
6	6	5	2026-09-17 05:30:54	video	English	I feel drained ..	I feel draine..	84.29	83.0	82.51	81.34	18.66	10.01	Centred	150.0	0.0
7	7	6	2026-09-08 05:23:54	video	English	I feel drained ..	I feel draine..	86.73	77.7	80.61	78.96	21.04	11.59	Slightly off..	139.0	0.0
8	8	6	2026-09-24 05:23:54	audio	English	I felt supporte..	I felt ..	16.68	13.34	NULL	14.34	86.66	NULL	NULL	113.0	0.0
9	9	7	2026-09-03 05:16:54	audio	English	I felt supporte..	I felt ..	26.93	11.07	NULL	18.77	81.23	NULL	NULL	123.0	0.0
10	10	7	2026-09-24 05:16:54	video	English	I have been ..	I have been ..	42.25	43.09	47.34	42.29	57.71	20.12	Slightly off..	137.0	0.0
11	11	8	2026-09-02 05:09:54	video	English	Today was quite..	Today was ..	61.76	61.03	59.96	50.72	49.28	21.82	Centred	118.0	0.0
12	12	8	2026-09-18 05:09:54	audio	English	I feel drained ..	I feel draine..	85.12	77.04	NULL	80.11	19.89	NULL	NULL	145.0	0.0
13	13	9	2026-09-01 05:02:54	audio	English	I feel drained ..	I feel draine..	86.65	81.09	NULL	82.86	17.14	NULL	NULL	135.0	0.0
14	14	9	2026-09-20 05:02:54	video	English	I felt supporte..	I felt ..	26.01	27.96	28.0	27.46	72.54	18.21	Slightly off..	146.0	0.0
15	15	10	2026-09-09 04:55:54	video	English	My shift was ..	My shift was ..	20.4	22.29	21.14	20.78	79.22	22.83	Slightly off..	141.0	0.0
16	16	10	2026-09-17 04:55:54	audio	English	I have been ..	I have been ..	41.06	54.31	NULL	47.11	52.89	NULL	NULL	162.0	0.0
17	17	11	2026-09-06 04:48:54	audio	English	I have been ..	I have been ..	45.39	45.54	NULL	44.92	55.08	NULL	NULL	132.0	0.0
18	18	11	2026-09-23 04:48:54	video	English	The workload ha..	The workload ..	87.39	72.73	71.63	71.28	28.72	20.82	Centred	147.0	0.0
19	19	12	2026-09-09 04:41:54	video	English	The workload ha..	The workload ..	76.73	80.66	90.72	79.05	20.95	14.94	Centred	130.0	0.0
20	20	12	2026-09-21 04:41:54	audio	English	My shift was ..	My shift was ..	18.61	22.64	NULL	20.38	79.62	NULL	NULL	108.0	0.0
21	21	13	2026-09-09 04:34:54	audio	English	I felt supporte..	I felt ..	26.03	29.09	NULL	27.23	72.77	NULL	NULL	131.0	0.0
22	22	13	2026-09-24 04:34:54	video	English	Today was quite..	Today was ..	52.3	41.71	41.62	40.94	59.06	20.09	Slightly off..	121.0	0.0
23	23	14	2026-09-08 04:27:54	video	English	I have been ..	I have been ..	47.91	41.99	44.09	43.21	56.79	16.02	Centred	116.0	0.0
24	24	14	2026-09-20 04:27:54	audio	English	The workload ha..	The workload ..	80.93	89.55	NULL	84.22	15.78	NULL	NULL	145.0	0.0
25	25	15	2026-09-02 04:20:54	audio	English	I feel drained ..	I feel draine..	80.0	72.65	NULL	75.41	24.59	NULL	NULL	132.0	0.0
26	26	15	2026-09-21 04:20:54	video	English	I felt supporte..	I felt ..	18.89	22.38	10.4	18.51	81.49	17.2	Centred	143.0	0.0
27	27	16	2026-09-04 04:13:54	video	English	My shift was ..	My shift was ..	22.64	29.1	18.61	22.25	77.75	22.47	Slightly off..	123.0	0.0

lexical_variety	summary	explanation	recommendation	image_name
Filter	Filter	Filter	Filter	Filter
0.605	Today needs mor...	Your first ...	1. **Schedule ...	wellbeing_low...
0.58	Today needs mor...	Your first ...	1. **Schedule a ...	wellbeing_low...
0.862	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.848	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.848	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.833	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.833	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.867	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.867	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.933	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.848	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.833	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.833	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.867	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.862	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.933	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.933	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.857	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.857	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.862	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.867	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.848	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.933	Today feels ...	Your check-in ...	Consider a short ...	wellbeing_mid...
0.857	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.833	Today needs mor...	Your check-in ...	Consider reaching ...	wellbeing_low...
0.867	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...
0.862	Today feels ...	Your check-in ...	Make time for the ...	wellbeing_hig...

Figure 11: Check-in table output

4.8.2 Authentication and Consent

Passwords are never stored as plain text. hash_password derives a PBKDF2-HMAC-SHA256 hash over 100,000 iterations with a random 16-byte salt, storing salt and hash together; verify_password rederives the hash and compares it in constant time to prevent timing leaks. The consent_accepted and consent_accepted_at columns in the users table are used to record consent, allowing the application to request informed consent before the initial check-in.

```
# Hash password with a fresh random salt
def hash_password(password):
    # salt 16
    salt = os.urandom(16)
    password_hash = hashlib.pbkdf2_hmac('sha256', password.encode('utf-8'), salt, PBKDF2_ITERATIONS)
    # store salt and hash together so the password can be checked later
    return f'{salt.hex()}:{password_hash.hex()}'
```

Code snippet 12: hash_password (salted PBKDF2 password hashing)

	id	full_name	username	password_hash	email	profession	address	consent_accepted	consent_accepted_at
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	subathra	Subathra_FYP	b31a87dba2a68cac6c24e28a97e76683:03f...				1	2026-09-24 15:37:53
2	2	P01	P01	b54a17350fcb44415c0479519ad56022:f05...				1	2026-09-25 05:01:29
3	3	P02	P02	88dcfcba79673f55f44533a481a4fc7f:...				1	2026-09-25 05:26:10
4	4	Participant 08	P08	597739648ea6af92550e3ee2743d3934:cb1...		Physiotherapist		1	2026-08-21 05:37:54
5	5	Participant 11	P011	a00229c57d44e36a4463da162d16582e:...		Nurse		1	2026-08-21 04:37:54
6	6	Participant 14	P014	4ac125b767d1f01e9a08bade001d944f:...		Physiotherapist		1	2026-08-21 03:37:54
7	7	Participant 04	P04	914b5753999da7e11b5a4ee096bba532:9d9...		Healthcare assistant		1	2026-08-21 02:37:54
8	8	Participant 05	P05	206032aa44e1b623f727d8af7557fdf0:92b...		Healthcare assistant		1	2026-08-21 01:37:54
9	9	Participant 12	P012	24f308faf2d949bb51ebf13840da7a6:969...		Nurse		1	2026-08-21 00:37:54
10	10	Participant 03	P03	8986778c41515b642baf913c7cf27284:cb5...		Nurse		1	2026-08-20 23:37:54
11	11	Participant 15	P015	402ffaeca14c17c0273e996ed100a6aa:...		Nurse		1	2026-08-20 22:37:54
12	12	Participant 13	P013	8f2b991b71b634e1b0d622c3f1176ca3:386...		Radiographer		1	2026-08-20 21:37:54
13	13	Participant 16	P016	e8c0b807828f7f263e0310754e0ae7d0:61c...		Healthcare assistant		1	2026-08-20 20:37:54
14	14	Participant 06	P06	4312129c2934cc32cblb7bd3cbb26363:6f4...		Physiotherapist		1	2026-08-20 19:37:54
15	15	Participant 07	P07	b0cedbcbf1a6e0f6bb14265b62e67877:9bf...		Healthcare assistant		1	2026-08-20 18:37:54
16	16	Participant 10	P010	d484177069b17c97e9f0f3cdf6fb408b:...		Physiotherapist		1	2026-08-20 17:37:54
17	17	Participant 09	P09	cbe834572667a7066dfb31f15df7cd6:743...		Healthcare assistant		1	2026-08-20 16:37:54

Figure 12: Users table output (Password hashed)

4.8.3 Questionnaire Storage and CBI Scoring

The Solace version contains the 13-item Copenhagen Burnout Inventory [19] in two 0-100 answer scales (FREQ_SCALE, DEGREE_SCALE). Item 13 (energy) is reverse scored, and submit() averages the first six items into a personal burnout score, the next seven into a work burnout score, and all thirteen into an overall CBI score, which is saved to questionnaire_responses. Users can opt in for weekly reminders, and every submission also generates a CSV (export_questionnaire_csv()). This questionnaire in the app was added late in the development process after a study participant suggested the self-assessment should be integrated into the tool. The validation for Chapter 5 itself used CBI responses gathered separately through a Google Form ([Appendix C.1](#)). The in-app questionnaire and its automatic comparison are therefore not for the current evaluation, but for future participants.

```
# Score the answers save and show thank you
def submit(self):
    # every item to be answered before saving
    if len(self.answers) != len(CBI_ITEMS):
        return

    # Reverse score final item then average the CBI subscores
    points = [self.answers[i] for i in range(len(CBI_ITEMS))]
    points[REVERSE_ITEM] = 100 - points[REVERSE_ITEM]
    personal = round(mean(points[:6]), 2)
    work = round(mean(points[6:]), 2)
    overall = round(mean(points), 2)

    # canonical English answers so scoring stays language independent
    labels = [self.answer_labels[i] for i in range(len(CBI_ITEMS))]
    try:
        save_questionnaire(self.user_id, labels, personal, work, overall)
    # error
    except Exception as error:
        print("QUESTIONNAIRE SAVE ERROR:", error)

    # thank you and offer weekly reminders
    self.submitted = True
    self.step_stack.setCurrentIndex(len(STEPS))
    self.update_nav()
    self.offer_reminders()
```

Code snippet 13: submit() function for the CBI questionnaire

	id	user_id	created_at	answers_json	personal_burnout	work_burnout	cbi_score
	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	2	2026-09-25 05:25:36	["Sometimes", "Sometimes", "Sometimes", ...	50.0	50.0	50.0
2	2	4	2026-09-08 05:37:54	["Often", "Seldom", "Sometimes", ...	58.33	42.86	50.0
3	3	4	2026-09-22 05:37:54	["Always", "Often", "Always", "Often", ...	87.5	85.71	86.54
4	4	5	2026-09-06 05:30:54	["Sometimes", "Sometimes", "Often", ...	50.0	39.29	44.23
5	5	5	2026-09-17 05:30:54	["Often", "Often", "Always", "Often", ...	83.33	89.29	86.54
6	6	6	2026-09-08 05:23:54	["Often", "Often", "Always", "Often", ...	87.5	85.71	86.54
7	7	6	2026-09-24 05:23:54	["Seldom", "Sometimes", "Sometimes", ...	29.17	21.43	25.0
8	8	7	2026-09-03 05:16:54	["Seldom", "Seldom", "Sometimes", ...	29.17	25.0	26.92
9	9	7	2026-09-24 05:16:54	["Sometimes", "Sometimes", "Seldom", ...	50.0	64.29	57.69
10	10	8	2026-09-02 05:09:54	["Often", "Always", "Often", "Always", ...	87.5	85.71	86.54
11	11	8	2026-09-18 05:09:54	["Seldom", "Never or almost never", ...	16.67	32.14	25.0
12	12	9	2026-09-01 05:02:54	["Always", "Often", "Always", "Often", ...	87.5	78.57	82.69
13	13	9	2026-09-20 05:02:54	["Never or almost never", "Never or ...	20.83	28.57	25.0
14	14	10	2026-09-09 04:55:54	["Seldom", "Never or almost never", ...	12.5	21.43	17.31
15	15	10	2026-09-17 04:55:54	["Often", "Sometimes", "Sometimes", ...	62.5	60.71	61.54
16	16	11	2026-09-06 04:48:54	["Sometimes", "Sometimes", "Sometimes", ...	54.17	42.86	48.08
17	17	11	2026-09-23 04:48:54	["Often", "Always", "Often", "Often", ...	83.33	82.14	82.69
18	18	12	2026-09-09 04:41:54	["Seldom", "Never or almost never", ...	12.5	35.71	25.0
19	19	12	2026-09-21 04:41:54	["Sometimes", "Often", "Often", "Often"...	58.33	46.43	51.92
20	20	13	2026-09-09 04:34:54	["Seldom", "Seldom", "Never or almost ...	25.0	25.0	25.0
21	21	13	2026-09-24 04:34:54	["Sometimes", "Sometimes", "Sometimes", ...	54.17	50.0	51.92

Figure 13: Questionnaire response database storage

	user_id	reminders_enabled
	Filter	Filter
1	2	1
2	4	0
3	5	1
4	6	1
5	7	1
6	8	0
7	9	1
8	10	1
9	11	1
10	12	0
11	13	1
12	14	1
13	15	1
14	16	0
15	17	1

Figure 14: Questionnaire preference database storage

4.8.4 Automatic CBI-Solace Comparison

When a check-in is saved, `record_cbi_comparison()` looks for a questionnaire completed by the same user on the same local date. If one exists, it maps the CBI score to a burnout band, maps that to an expected wellbeing band, compares it with Solace's own band, and stores the result including whether they agree in cbi comparisons. This comparison is computed and stored for validation of future check-ins, the Chapter 5 results themselves used the separately collected Google Form responses, and it is never shown to the user, consistent with the considerate presentation principle in Section 3.9.

```
# Compare a check-in with questionnaire from same local date
def _record_cbi_comparison(connection, user_id, check_in_id, wellbeing_score):
    # execute select statement
    row = connection.execute(
        """
        SELECT q.id, q.cbi_score
        FROM questionnaire_responses q
        JOIN check_ins c ON c.user_id = q.user_id
        WHERE q.user_id = ?
        AND c.id = ?
        AND date(q.created_at, 'localtime') = date(c.created_at, 'localtime')
        ORDER BY q.created_at DESC, q.id DESC
        LIMIT 1
        """,
        (user_id, check_in_id)
    ).fetchone()

    # Remove earlier comparison before refreshing it
    connection.execute("DELETE FROM cbi_comparisons WHERE check_in_id = ?", (check_in_id,))

    # check-in without a comparison if no questionnaire matches
    if row is None or row["cbi_score"] is None:
        return

    cbi_score = float(row["cbi_score"])
    wellbeing_score = float(wellbeing_score)
    cbi_band = _cbi_band(cbi_score)
    expected = _expected_wellbeing_band(cbi_band)
    solace_band = _wellbeing_band(wellbeing_score)
    agree = 1 if solace_band == expected else 0

    # Save same day comparison
    connection.execute(
        """
        INSERT INTO cbi_comparisons
        (user_id, questionnaire_id, check_in_id, cbi_score, cbi_band,
         expected_wellbeing_band, wellbeing_score, solace_band, bands_agree)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)
        """,
        (user_id, row["id"], check_in_id, cbi_score, cbi_band, expected, wellbeing_score, solace_band, agree)
    )
```

Code snippet 14: `_record_cbi_comparison()` function

	id	user_id	questionnaire_id	check_in_id	created_at	cbi_score	cbi_band	expected_wellbeing_band	wellbeing_score	solace_band	bands_agree
	Filter...	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	2	1	1	2026-09-25 05:25:36	50.0	Moderate burnout	Mixed	19.57	Low	0
2	2	4	2	3	2026-09-08 05:37:54	50.0	Moderate burnout	Mixed	84.33	High	0
3	3	4	3	4	2026-09-22 05:37:54	86.54	High burnout	Low	48.34	Mixed	0
4	4	5	4	5	2026-09-06 05:30:54	44.23	Low burnout	High	46.13	Mixed	0
5	5	5	5	6	2026-09-17 05:30:54	86.54	High burnout	Low	18.66	Low	1
6	6	6	6	7	2026-09-08 05:23:54	86.54	High burnout	Low	21.04	Low	1
7	7	6	7	8	2026-09-24 05:23:54	25.0	Low burnout	High	85.66	High	1
8	8	7	8	9	2026-09-03 05:16:54	26.92	Low burnout	High	81.23	High	1
9	9	7	9	10	2026-09-24 05:16:54	57.69	Moderate burnout	Mixed	57.71	Mixed	1
10	10	8	10	11	2026-09-02 05:09:54	86.54	High burnout	Low	49.28	Mixed	0
11	11	8	11	12	2026-09-18 05:09:54	25.0	Low burnout	High	19.89	Low	0
12	12	9	12	13	2026-09-01 05:02:54	82.69	High burnout	Low	17.14	Low	1
13	13	9	13	14	2026-09-20 05:02:54	25.0	Low burnout	High	72.54	High	1
14	14	10	14	15	2026-09-09 04:55:54	17.31	Low burnout	High	79.22	High	1
15	15	10	15	16	2026-09-17 04:55:54	61.54	Moderate burnout	Mixed	52.89	Mixed	1
16	16	11	16	17	2026-09-06 04:48:54	48.08	Low burnout	High	55.08	Mixed	0
17	17	11	17	18	2026-09-23 04:48:54	82.69	High burnout	Low	28.72	Low	1
18	18	12	18	19	2026-09-09 04:41:54	25.0	Low burnout	High	20.95	Low	0
19	19	12	19	20	2026-09-21 04:41:54	51.92	Moderate burnout	Mixed	79.62	High	0
20	20	13	20	21	2026-09-09 04:34:54	25.0	Low burnout	High	72.77	High	1
21	21	13	21	22	2026-09-24 04:34:54	51.92	Moderate burnout	Mixed	59.06	Mixed	1
22	22	14	22	23	2026-09-08 04:27:54	50.0	Moderate burnout	Mixed	56.79	Mixed	1
23	23	14	23	24	2026-09-20 04:27:54	88.46	High burnout	Low	15.78	Low	1
24	24	15	24	25	2026-09-02 04:20:54	84.62	High burnout	Low	24.59	Low	1

Figure 15: cbi comparison table output

4.8.5 Experience Feedback Storage

Optional post use feedback is stored in the experience_feedback table as a one-to-five rating and comment, while experience_feedback_prompts records when each user last saw the feedback card so it is only shown periodically. Deleting an account cascades to both.

	id	user_id	rating	comment	created_at
	Filter...	Filter	Filter	Filter	Filter
1	1	4	4	The check-in steps were easy to follow.	2026-09-25 03:37:54
2	2	5	5	I liked being able to review the ...	2026-09-25 03:32:54
3	3	6	3	The results were clear, but the analysi...	2026-09-25 03:27:54
4	4	7	4	The history page made it easier to ...	2026-09-25 03:22:54
5	5	8	2	I would prefer a shorter wait when ...	2026-09-25 03:17:54
6	6	9	5	The questionnaire was straightforward t...	2026-09-25 03:12:54
7	7	10	4	The text on the recording page was easy...	2026-09-25 03:07:54
8	8	11	4	It would help to show a clearer estimat...	2026-09-25 03:02:54
9	9	12	5	The resource links were easy to find.	2026-09-25 02:57:54
10	10	13	3	I appreciated having an audio-only ...	2026-09-25 02:52:54
11	11	14	4	The reminder was useful without feeling...	2026-09-25 02:47:54
12	12	15	2	I found the navigation clear after usin...	2026-09-25 02:42:54
13	13	16	5	I would like slightly larger text in th...	2026-09-25 02:37:54
14	14	17	4	The privacy explanation helped me ...	2026-09-25 02:32:54

Figure 16: Experience Feedback Response

	<u>user_id</u>	last_shown_at
	Filter	Filter
1	4	2026-09-25 03:35:54
2	5	2026-09-25 03:30:54
3	6	2026-09-25 03:25:54
4	7	2026-09-25 03:20:54
5	8	2026-09-25 03:15:54
6	9	2026-09-25 03:10:54
7	10	2026-09-25 03:05:54
8	11	2026-09-25 03:00:54
9	12	2026-09-25 02:55:54
10	13	2026-09-25 02:50:54
11	14	2026-09-25 02:45:54
12	15	2026-09-25 02:40:54
13	16	2026-09-25 02:35:54
14	17	2026-09-25 02:30:54

Figure 17: Experience feedback prompts

4.8.6 Automated Test Coverage

Solace has 78 automated unit tests spread across six PyTest files, run against a disposable temporary SQLite database so no real data is touched which are test_database.py (15), test_pipeline.py (16), test_check_in.py (14), test_solace_agent.py (15), test_translation.py (10) and test_register.py (8). Testing was mainly on the data layer, pipeline logic and assistant routing, the area's most likely to silently fail, and the PySide6 widgets were checked manually.

4.9 User Interface

4.9.1 Registration, Login and Consent

New users create an account on the registration page and sign in on the login page, which authenticates against the locally stored, hashed credentials. On first use, a consent dialog explains what the system does and what data it keeps, and the user must accept before any check-in can be recorded, satisfying the informed consent requirement.

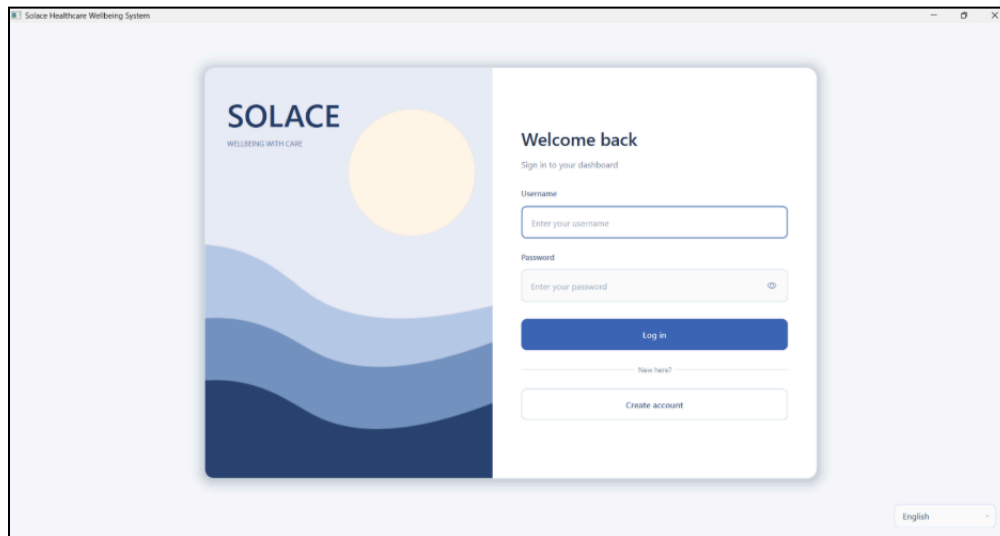


Figure 18: Login page

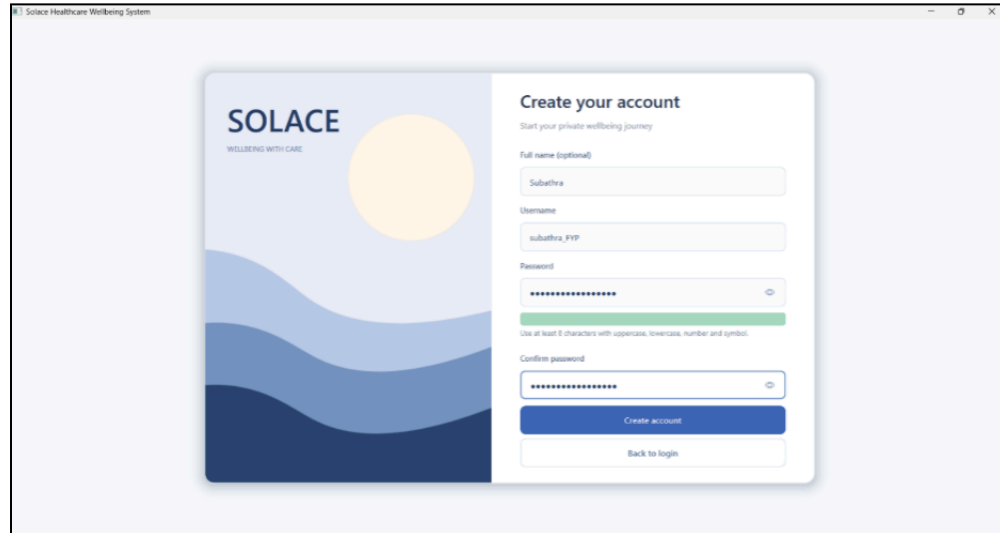
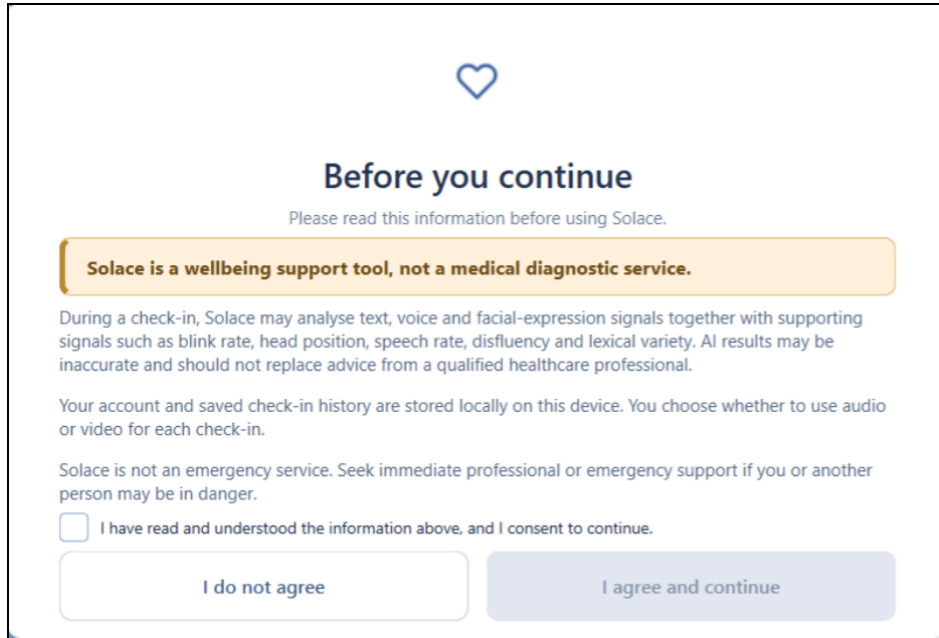


Figure 19: Registration Page



A consent dialog screen for the Solace app. At the top is a blue heart icon. The title is "Before you continue" in bold, followed by the instruction "Please read this information before using Solace." Below this is an orange-bordered box with the text "Solace is a wellbeing support tool, not a medical diagnostic service." The main body of text explains that during a check-in, Solace analyzes text, voice, and facial-expression signals along with supporting signals like blink rate and head position. It states that AI results may be inaccurate and should not replace professional advice. It also mentions that account and check-in history are stored locally and that users can choose audio or video for check-ins. A disclaimer states Solace is not an emergency service. At the bottom, there is a checkbox for consent, two buttons: "I do not agree" and "I agree and continue".

Before you continue

Please read this information before using Solace.

Solace is a wellbeing support tool, not a medical diagnostic service.

During a check-in, Solace may analyse text, voice and facial-expression signals together with supporting signals such as blink rate, head position, speech rate, disfluency and lexical variety. AI results may be inaccurate and should not replace advice from a qualified healthcare professional.

Your account and saved check-in history are stored locally on this device. You choose whether to use audio or video for each check-in.

Solace is not an emergency service. Seek immediate professional or emergency support if you or another person may be in danger.

☐ I have read and understood the information above, and I consent to continue.

I do not agree I agree and continue

Figure 20: Consent dialog

4.9.2 Home

After signing in, the home page is the landing screen. It greets the user by name, provides the main call to start a one-minute check-in, and surfaces quick links to Trends, the Assistant and the questionnaire alongside a set of wellbeing resource cards. The sidebar also shows the national mindline (1771) support number at all times [40].

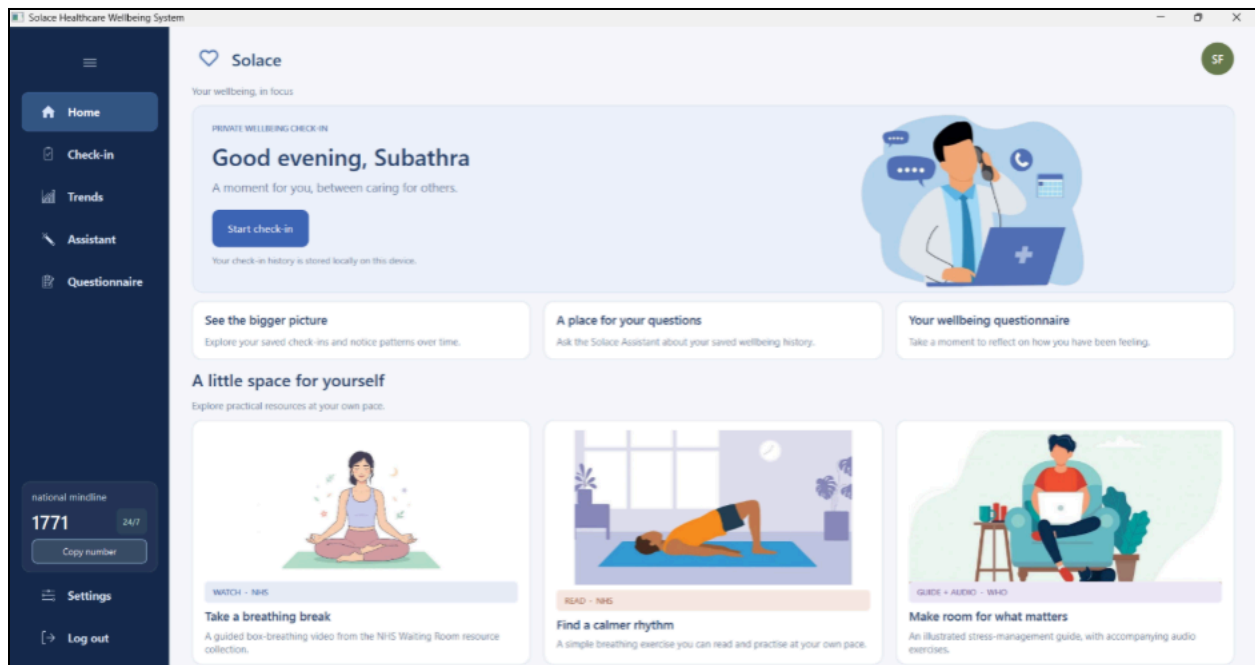


Figure 21: Home Page

4.9.3 Check-in

The check-in page offers audio and video tabs. It captures a live recording or accepts an uploaded clip, and a live transcript panel becomes editable once Whisper finishes. While the pipeline runs on a background thread, a processing screen shows progress. The result page then shows the wellbeing score, the personal baseline status, the contributing modality readings and available supporting signals, the transcript, and the generated recommendations.

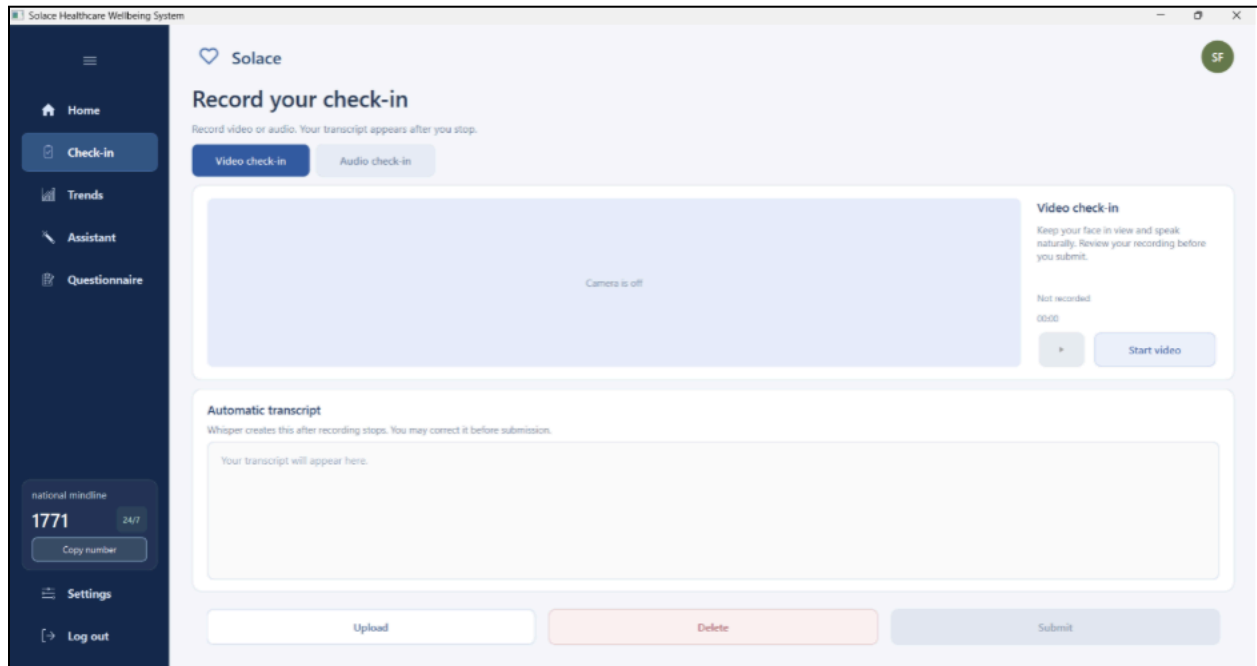


Figure 22: Check-in Page

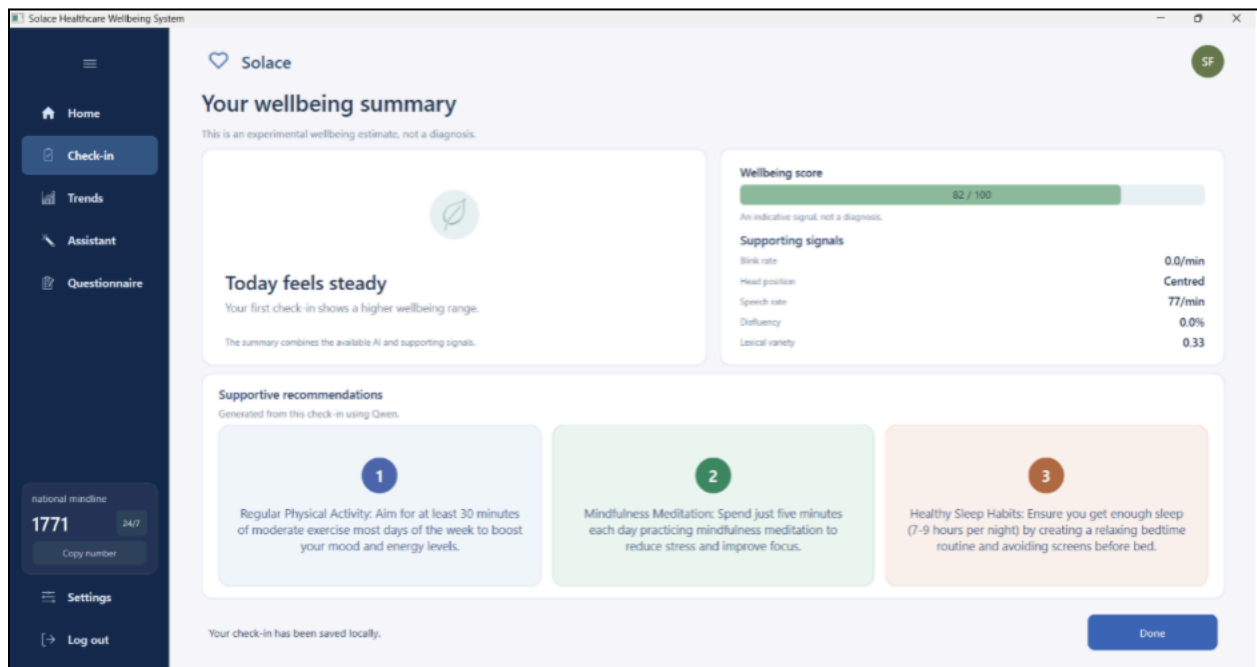


Figure 23: Check-in Result and Wellbeing summary

4.9.4 Trends

The user's wellbeing scores are plotted over time on the trends page, creating a clear trend from individual check-ins. A gradual decline is easier to see across the trend than in a single score.

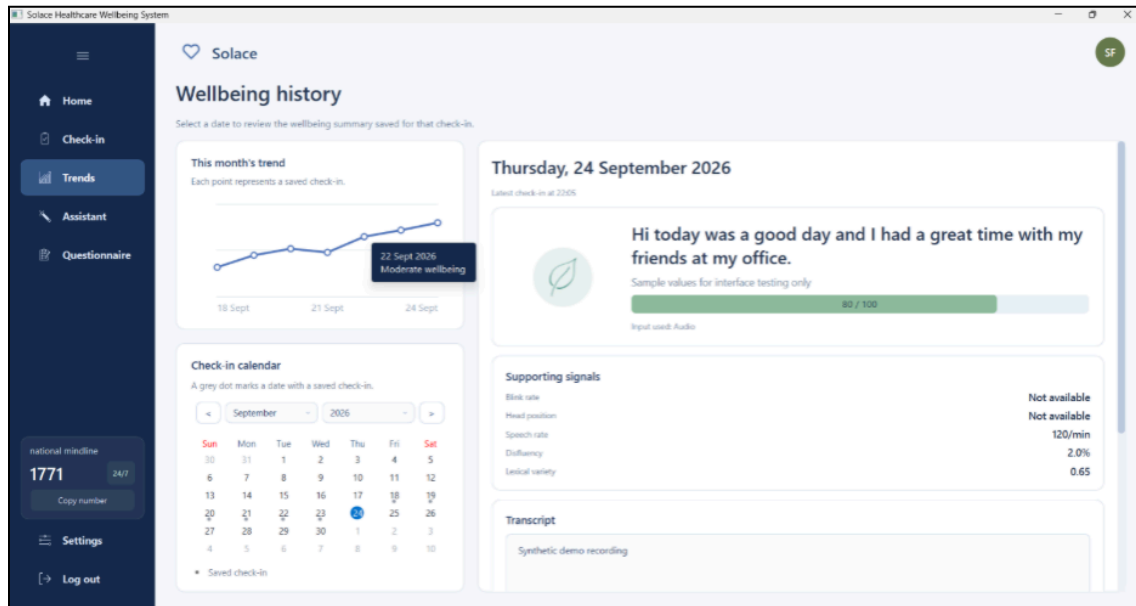


Figure 24: Trends Page

4.9.5 Solace Assistant

The assistant page lets the user ask questions about their own history in plain language, such as their latest score or recent trend, and receives answers drawn only from their own stored check-ins through the tool-using agent described in Section 4.7.

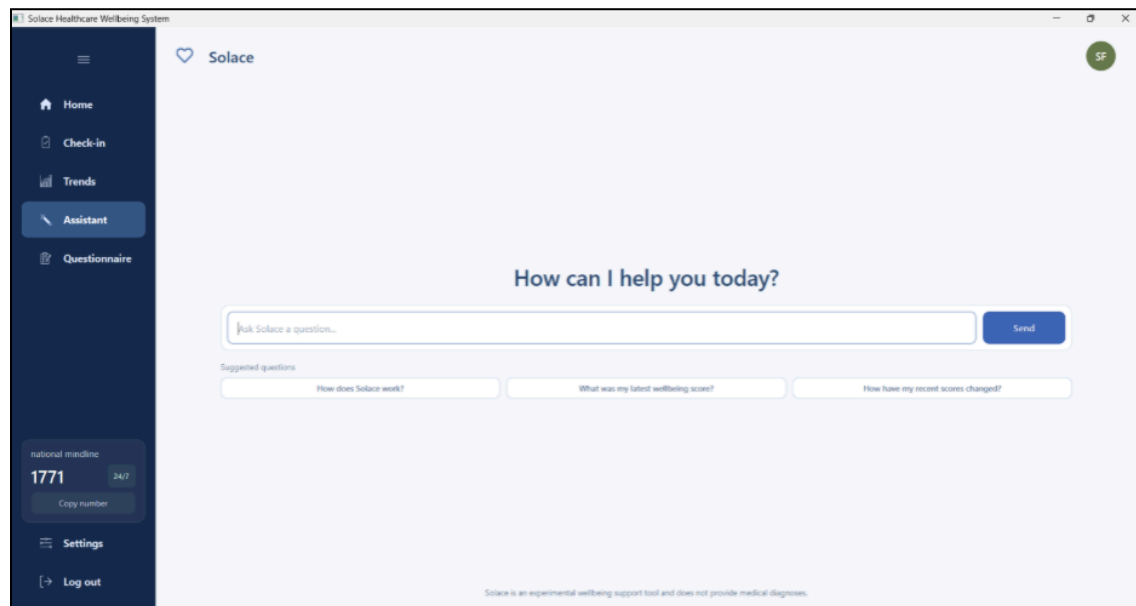


Figure 25: AI Assistant Page

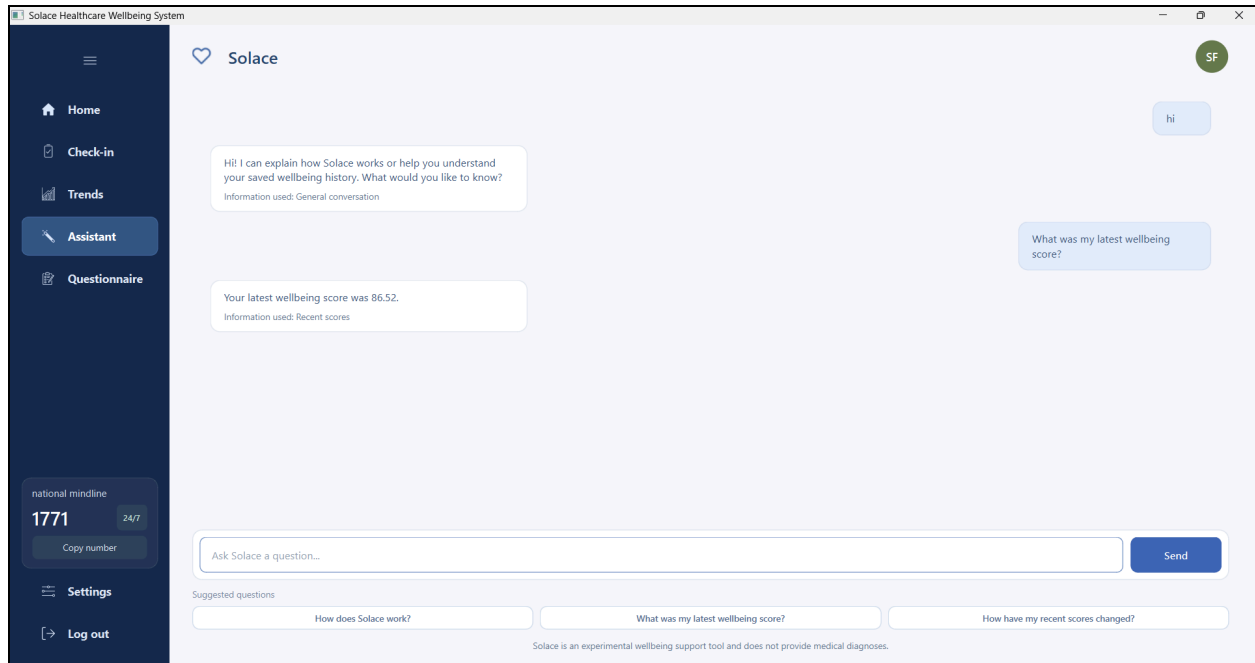


Figure 26: AI Assistant Response

4.9.6 Questionnaire

The questionnaire page presents the 13-item CBI directly in Solace, with optional weekly reminders. Completing it on the same day as a check-in triggers the automatic comparison described in Section 4.8.4.

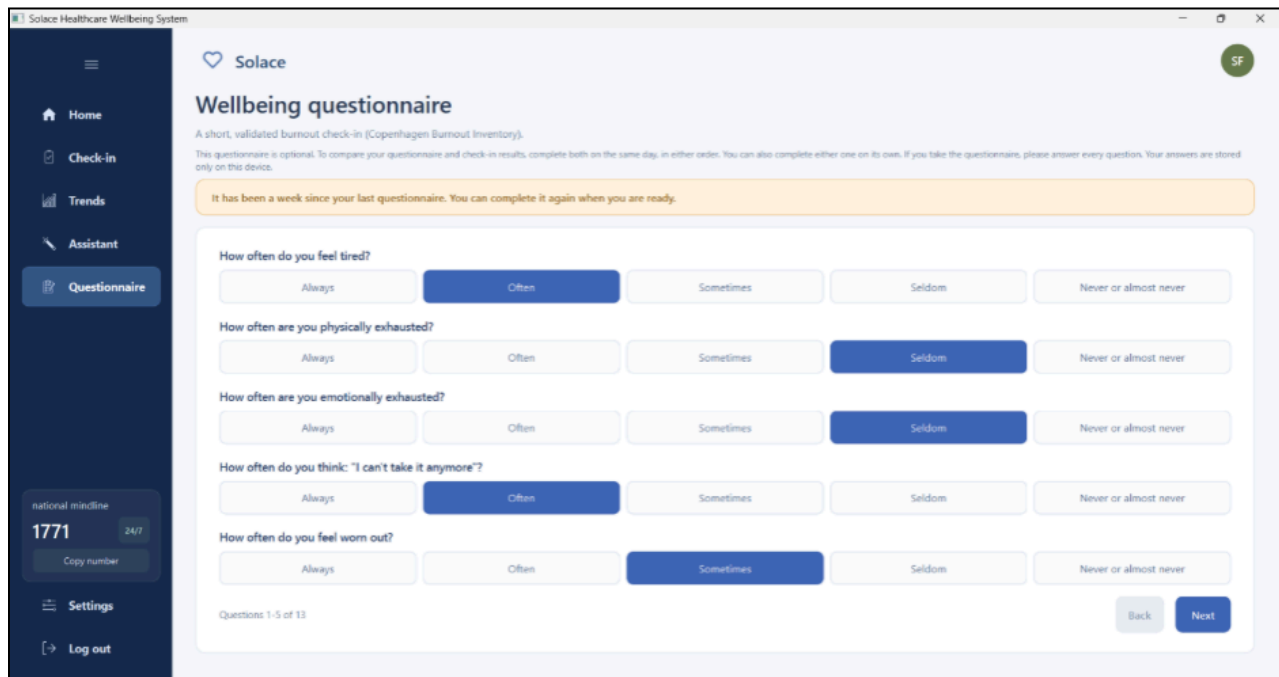


Figure 27: Questionnaire Page

4.9.7 Settings

The settings page lets the user change the interface language between English, Malay, Chinese, and Tamil, and manage their account. From here the user can also delete their account at any time, which removes their record and, through the cascading delete, all their stored check-ins, giving the user full control over their data. They can revisit their privacy and consent form also.

Solace Healthcare Wellbeing System

Solace

Settings

Manage your application preferences.

Profile Preferences

SF subathra_FYP

Only your username and password are required. Add other details if you wish.

Username: subathra_FYP Full name (optional): Subathra

Email (optional): Profession (optional):

Address (optional):

Leave these fields blank to keep your password. Your current password is required when changing your username or password.

Current password: New password (optional): Confirm new password:

Use at least 8 characters with uppercase, lowercase, a number and a symbol

Save profile

Figure 28: Settings Page (Profile Settings)

Solace Healthcare Wellbeing System

Solace

Settings

Manage your application preferences.

Profile Preferences

Language

Choose the language used for the interface, speech recognition and wellbeing recommendations.

Application language: English

The selected language will be used for future check-ins.

Privacy and consent

Review the privacy, AI and consent information for Solace.

View privacy and consent

Delete account

Permanently delete your Solace account and all saved wellbeing history.

Delete my account

Figure 29: Settings Page (Preferences)

CHAPTER 5: EVALUATION (2490/2500 Words)

5.1 Overview

This chapter evaluates the system through unit testing, per-model held-out performance, fusion comparison, fairness and healthcare-style checks, and validation against an established burnout measure, before assessing it against the Chapter 1 objectives and its limitations.

5.2 Unit Testing Results

The 78 unit tests all pass, confirming the deterministic core behaves as specified, independently of the models and interface.

Test File	Tests	Results	Covers
test_pipeline.py	16	All Passed	Fusion maths, clamping, negative-emotion scoring, supporting signals
test_check_in.py	14	All Passed	Baseline classification and result message logic
test_database.py	15	All Passed	User creation, authentication, consent, check-in storage
test_register.py	8	All Passed	Registration and input validation
test_solace_agent.py	15	All Passed	Assistant tool selection and decision parsing
test_translation.py	10	All Passed	Language handling and translation fallbacks
Total	78	All Passed	

Table 14: Unit Test Results

```
Subathra@LAPTOP-HPB9M968-MINGW64 ~/OneDrive/Desktop/CM3020_Solace_Wellbeing_Monitoring/CM3020_Solace_Wellbeing_Monitoring_FYP (main)
$ python -m pytest unit test v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\Subathra\OneDrive\Desktop\CM3020_Solace_Wellbeing_Monitoring\CM3020_Solace_Wellbeing_Monitoring_FYP\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Subathra\OneDrive\Desktop\CM3020_Solace_Wellbeing_Monitoring\CM3020_Solace_Wellbeing_Monitoring_FYP
plugins: anyio-4.15.1
collected 78 items

Unit_test/test_check_in.py::test_first_high_result PASSED [ 15%]
Unit_test/test_check_in.py::test_first_middle_result PASSED [ 25%]
Unit_test/test_check_in.py::test_first_low_result PASSED [ 33%]
Unit_test/test_check_in.py::test_improved_result PASSED [ 58%]
Unit_test/test_check_in.py::test_lower_result PASSED [ 68%]
Unit_test/test_check_in.py::test_small_increase_is_steady PASSED [ 78%]
Unit_test/test_check_in.py::test_small_decrease_is_steady PASSED [ 88%]
Unit_test/test_check_in.py::test_same_score_is_steady PASSED [ 100%]
Unit_test/test_check_in.py::test_baseline_not_enough_history PASSED [ 115%]
Unit_test/test_check_in.py::test_baseline_above_range PASSED [ 125%]
Unit_test/test_check_in.py::test_baseline_below_range PASSED [ 145%]
Unit_test/test_check_in.py::test_baseline_within_range PASSED [ 155%]
Unit_test/test_check_in.py::test_baseline_zero_sd_above PASSED [ 165%]
Unit_test/test_check_in.py::test_baseline_zero_sd_same PASSED [ 175%]
Unit_test/test_database.py::test_create_database PASSED [ 195%]
Unit_test/test_database.py::test_supporting_signal_columns PASSED [ 205%]
Unit_test/test_database.py::test_password_hashing PASSED [ 215%]
Unit_test/test_database.py::test_create_user_and_duplicate_username PASSED [ 235%]
Unit_test/test_database.py::test_authenticate_user PASSED [ 245%]
Unit_test/test_database.py::test_save_consent PASSED [ 255%]
Unit_test/test_database.py::test_save_audio_check_in PASSED [ 265%]
Unit_test/test_database.py::test_save_video_supporting_signals PASSED [ 285%]
Unit_test/test_database.py::test_check_in_count PASSED [ 295%]
Unit_test/test_database.py::test_recent_scores PASSED [ 305%]
Unit_test/test_database.py::test_month_history PASSED [ 325%]
Unit_test/test_database.py::test_check_in_dates PASSED [ 335%]
Unit_test/test_database.py::test_user_data_is_separate PASSED [ 345%]
Unit_test/test_database.py::test_missing_check_in_date PASSED [ 355%]
Unit_test/test_database.py::test_previous_strain_scores PASSED [ 375%]
Unit_test/test_pipeline.py::test_clamp PASSED [ 385%]
Unit_test/test_pipeline.py::test_label_name PASSED [ 395%]
Unit_test/test_pipeline.py::test_predictions PASSED [ 415%]
Unit_test/test_pipeline.py::test_negative_score PASSED [ 425%]
```

```
Unit_test/test_pipeline.py::test_negative_score_limit PASSED [ 43%]
Unit_test/test_pipeline.py::test_eye_ratio PASSED [ 44%]
Unit_test/test_pipeline.py::test_english_speech_signals PASSED [ 46%]
Unit_test/test_pipeline.py::test_chinese_speech_signals PASSED [ 47%]
Unit_test/test_pipeline.py::test_slow_speech_signal PASSED [ 48%]
Unit_test/test_pipeline.py::test_audio_fusion PASSED [ 50%]
Unit_test/test_pipeline.py::test_video_fusion PASSED [ 51%]
Unit_test/test_pipeline.py::test_fusion_without_supporting_signals PASSED [ 52%]
Unit_test/test_pipeline.py::test_fusion_maximum_strain PASSED [ 53%]
Unit_test/test_pipeline.py::test_high_wellbeing_summary PASSED [ 55%]
Unit_test/test_pipeline.py::test_moderate_wellbeing_summary PASSED [ 56%]
Unit_test/test_pipeline.py::test_low_wellbeing_summary PASSED [ 57%]
Unit_test/test_register.py::test_valid_password PASSED [ 58%]
Unit_test/test_register.py::test_password_too_short PASSED [ 60%]
Unit_test/test_register.py::test_password_without_uppercase PASSED [ 61%]
Unit_test/test_register.py::test_password_without_lowercase PASSED [ 62%]
Unit_test/test_register.py::test_password_without_number PASSED [ 64%]
Unit_test/test_register.py::test_password_without_symbol PASSED [ 65%]
Unit_test/test_register.py::test_empty_password PASSED [ 66%]
Unit_test/test_register.py::test_minimum_valid_password_length PASSED [ 67%]
Unit_test/test_solace_agent.py::test_supported_languages PASSED [ 69%]
Unit_test/test_solace_agent.py::test_invalid_starting_language PASSED [ 70%]
Unit_test/test_solace_agent.py::test_clean_history PASSED [ 71%]
Unit_test/test_solace_agent.py::test_history_limit PASSED [ 73%]
Unit_test/test_solace_agent.py::test_parse_valid_tool_decision PASSED [ 74%]
Unit_test/test_solace_agent.py::test_parse_unknown_tool PASSED [ 75%]
Unit_test/test_solace_agent.py::test_parse_invalid_date PASSED [ 76%]
Unit_test/test_solace_agent.py::test_safety_support PASSED [ 78%]
Unit_test/test_solace_agent.py::test_general_tool PASSED [ 79%]
Unit_test/test_solace_agent.py::test_personal_tool_requires_user PASSED [ 80%]
Unit_test/test_solace_agent.py::test_latest_check_in PASSED [ 82%]
Unit_test/test_solace_agent.py::test_recent_scores PASSED [ 83%]
Unit_test/test_solace_agent.py::test_agent_check_in_count PASSED [ 84%]
Unit_test/test_solace_agent.py::test_wellbeing_context PASSED [ 85%]
Unit_test/test_solace_agent.py::test_blank_question PASSED [ 87%]
Unit_test/test_translation.py::test_translation_model PASSED [ 88%]
Unit_test/test_translation.py::test_malay_language_code PASSED [ 89%]
Unit_test/test_translation.py::test_chinese_language_code PASSED [ 91%]
Unit_test/test_translation.py::test_tamil_language_code PASSED [ 92%]

Unit_test/test_translation.py::test_supported_languages PASSED [ 93%]
Unit_test/test_translation.py::test_english_interface_text PASSED [ 94%]
Unit_test/test_translation.py::test_english_translation_bypass PASSED [ 96%]
Unit_test/test_translation.py::test_get_english_text PASSED [ 97%]
Unit_test/test_translation.py::test_get_translated_text PASSED [ 98%]
Unit_test/test_translation.py::test_get_text_fallback_to_english PASSED [100%]

===== 78 passed in 9.50s =====
```

Figure 30: Terminal Output of the tests

5.3 Model Evaluation and Selection

5.3.1 Text Emotion Models

RoBERTa-Large was selected for text emotion because it achieved the highest validation macro-F1 of 0.748, ahead of DistilRoBERTa at 0.723 and well above the BERT-family baselines, as shown in Figure 31.

Model	Validation Macro-F1
RoBERTa-Large	0.7483
DistilRoBERTa	0.7231
BERT-Nateraw	0.6504
BERT	0.6465
DistilBERT	0.6313
RoBERTa	0.6311

Table 15: Text Emotion Models Comparison (Validation)

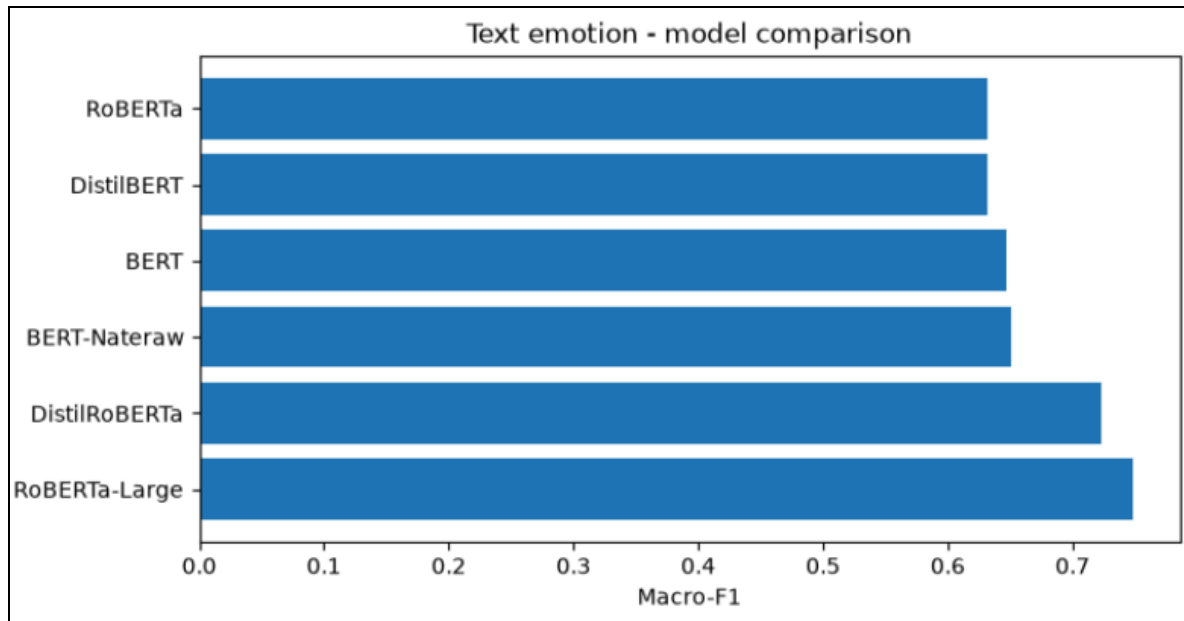


Figure 31: Text Emotion Model Macro F1 Comparison

On the held-out test split, RoBERTa-Large reached 73.8 percent accuracy and 0.732 macro-F1. The confusion matrices (Figures 32-36) show joy strongest and sadness weakest, with fear over-predicted (high recall, low precision) and anger conservative (high precision, low recall).

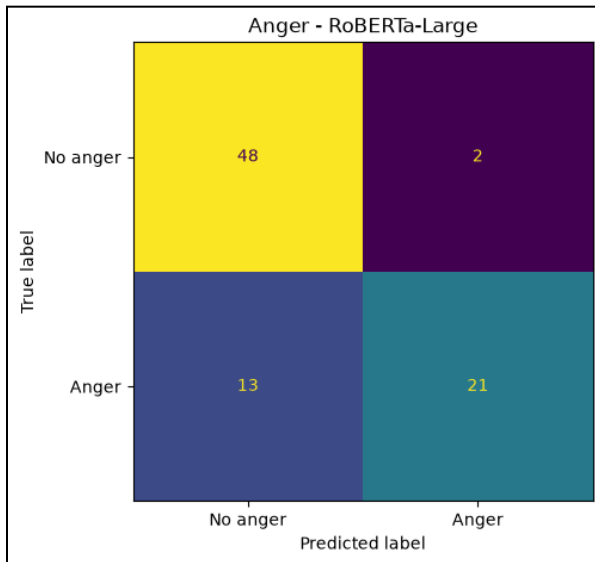


Figure 32: Anger confusion matrix

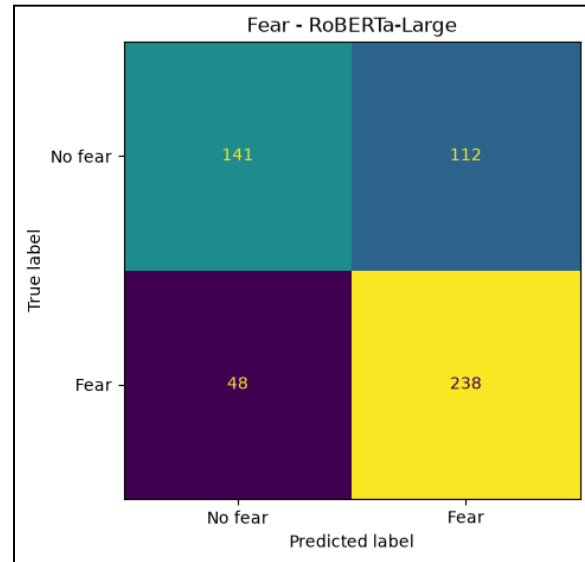


Figure 33: Fear confusion matrix

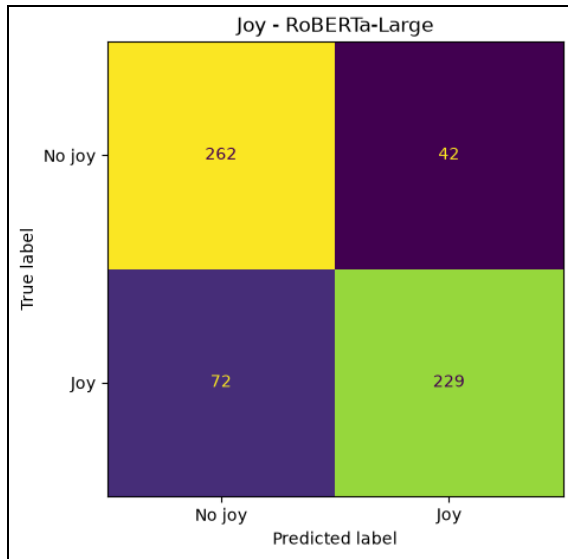


Figure 34: Joy confusion matrix

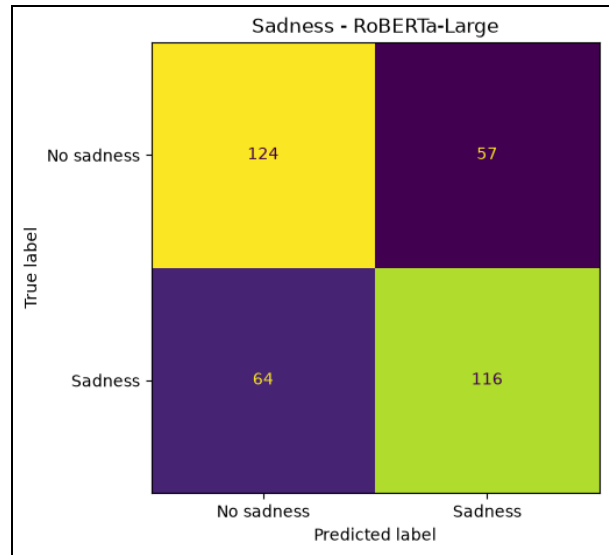


Figure 35: Sadness confusion matrix

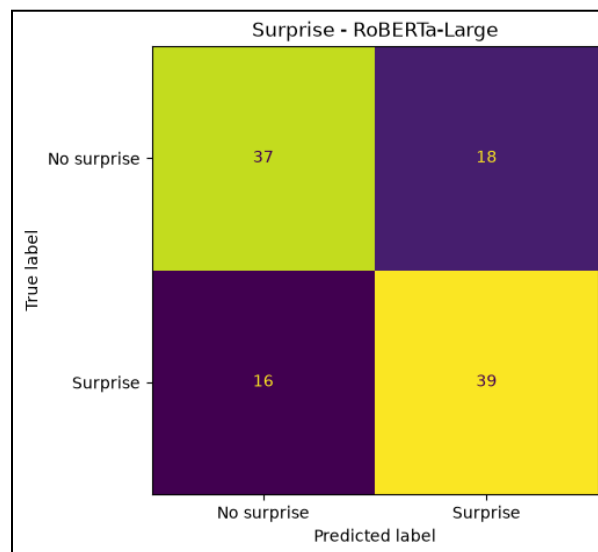


Figure 36: Surprise confusion matrix

The ROC and precision-recall curves (Figures 37, 38) give a macro-average ROC-AUC of 0.809 and PR-AUC of 0.808, well above chance, with joy and anger strongest and sadness weakest, confirming sadness as the hardest emotion to separate.

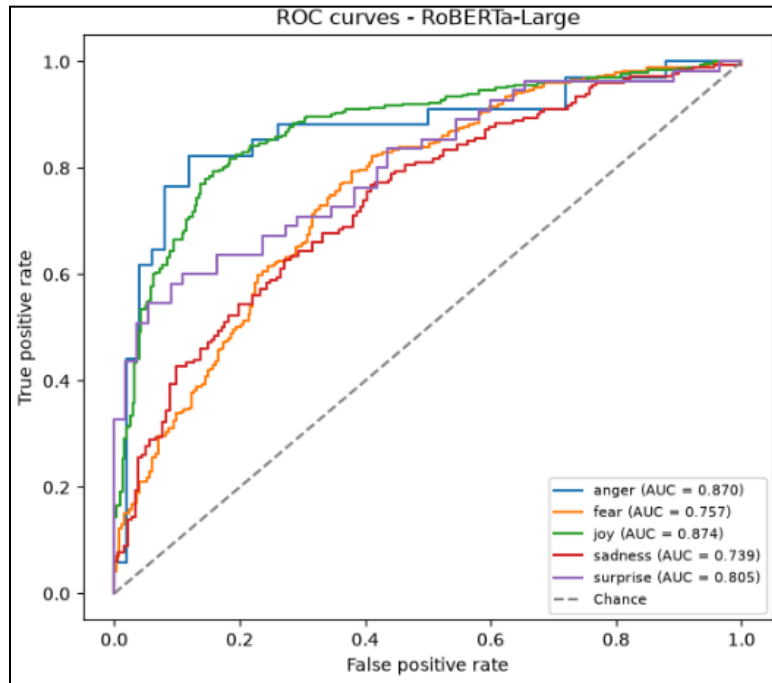


Figure 37: ROC Curve (RoBERTa-Large)

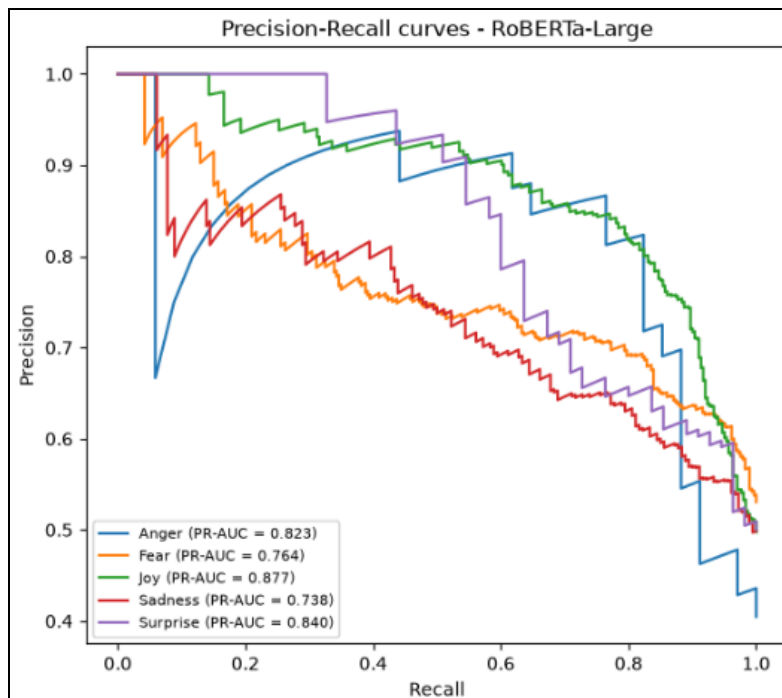


Figure 38: Precision- Recall Curve (RoBERTa-Large)

5.3.2 Audio Emotion Models

MERaLiON-SER was selected because it achieved the highest validation macro-F1 of 0.690, far ahead of the next candidates at around 0.50, as shown in Figure 39. The gap is large because the other candidates were fine-tuned on the single acted CREMA-D corpus and generalised poorly to RAVDESS, whereas MERaLiON-SER was trained on broader, more natural speech, so it transferred rather than memorising one dataset.

Model	Validation Macro-F1
MERaLiON-SER	0.6898
HuBERT-CREMA-D	0.4986
AST-CREMA-D-Alternative	0.4956
Wav2Vec2-CREMA-D	0.3743
AST-CREMA-D	0.3387
DistilHuBERT-CREMA-D	0.2972

Table 16: Audio Emotion Models Comparison (Validation)

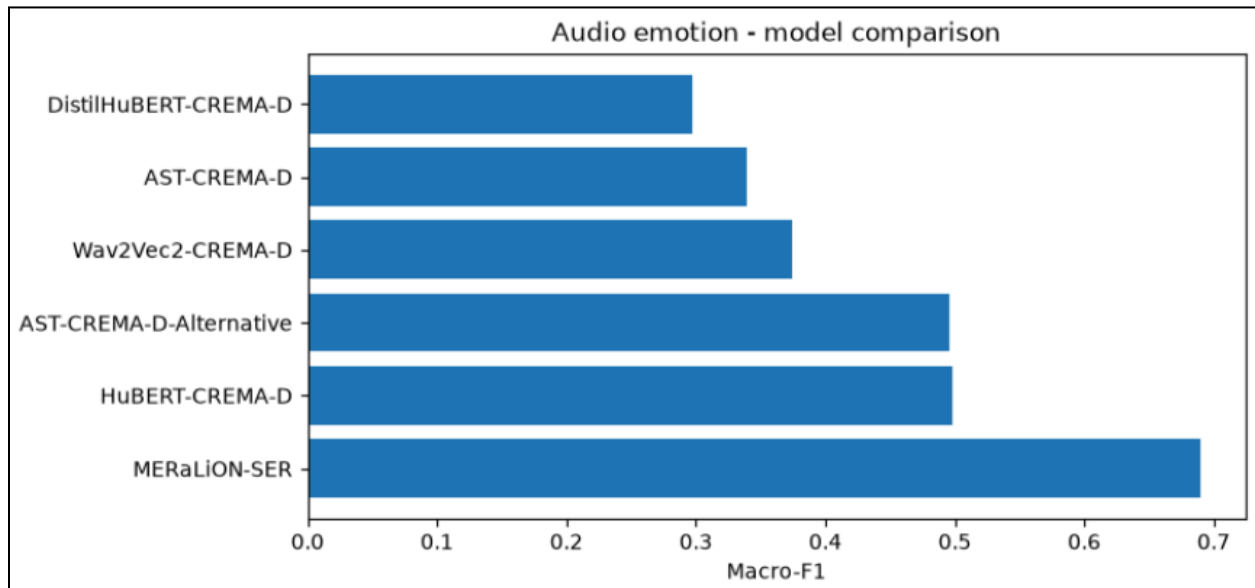


Figure 39: Audio Emotion Macro-F1 Model Comparison

On the held-out split, MERaLiON-SER reached 67.3 percent accuracy and 0.659 macro-F1. The confusion matrix (Figure 40) shows anger and sadness strongest, disgust weakest often mislabelled as anger, and neutral and fear both drifting toward sadness.

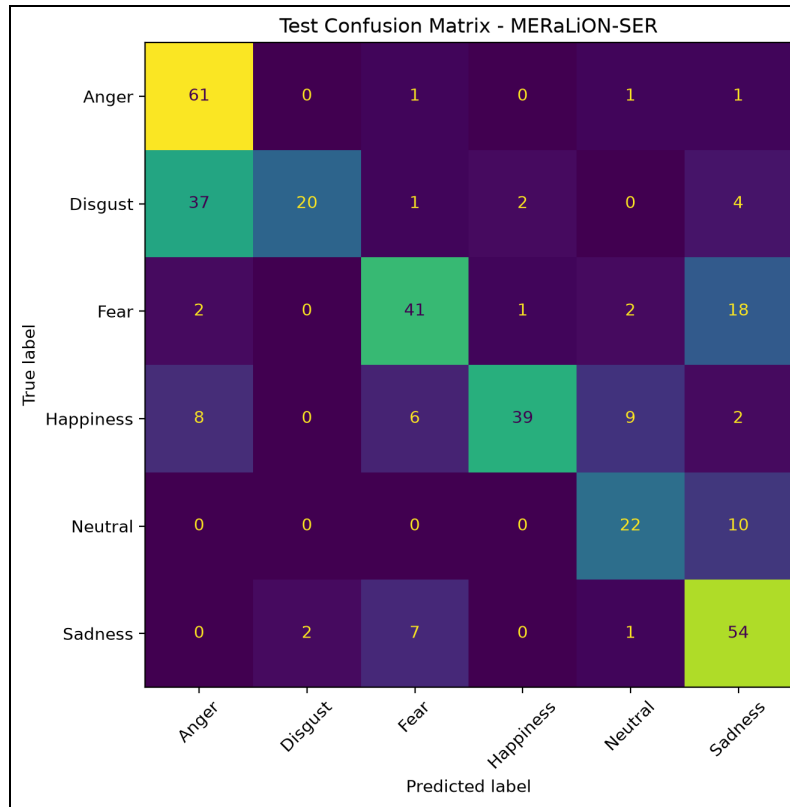


Figure 40: Test Confusion Matrix (MERaLiON-SER)

The curves (Figures 41-42) give a macro-average ROC-AUC of 0.936 and PR-AUC of 0.815, with neutral weakest, matching its confusion with sadness.

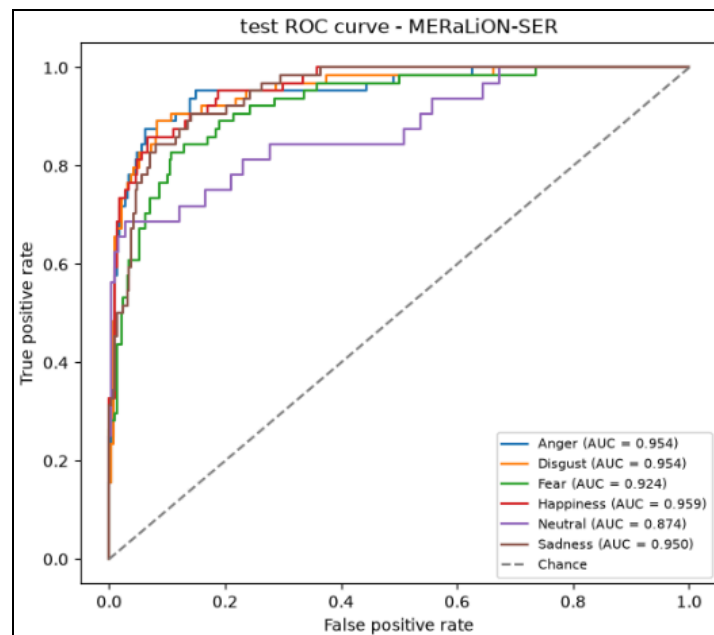


Figure 41: ROC Curve (MERaLiON-SER)

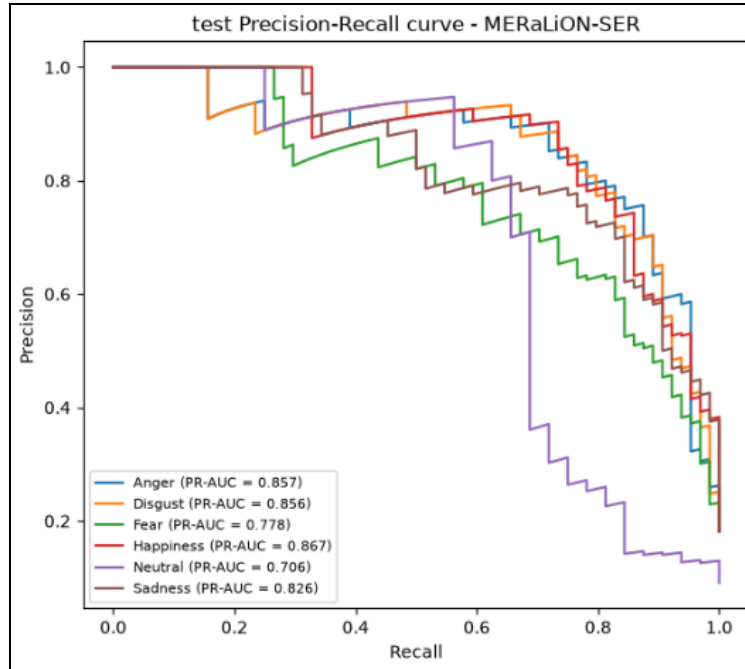


Figure 42: Precision-Recall Curve (MERaLiON-SER)

5.3.3 Vision Emotion Models

DDAMFN++ was selected because it achieved the highest validation macro-F1 of 0.648 on RAF-DB, clearly ahead of the ViT and CNN baselines, as shown in Figure 43.

Model	Validation Macro-F1
DDAMFN++	0.6483
ViT-Mixed-Datasets	0.5826
ViT-FER2013	0.5034
ViT	0.4997
ResNet-50	0.4927
ConvNeXt-Tiny	0.4479
Swin-Tiny	0.2219

Table 17: Vision Emotion Models Comparison (Validation)

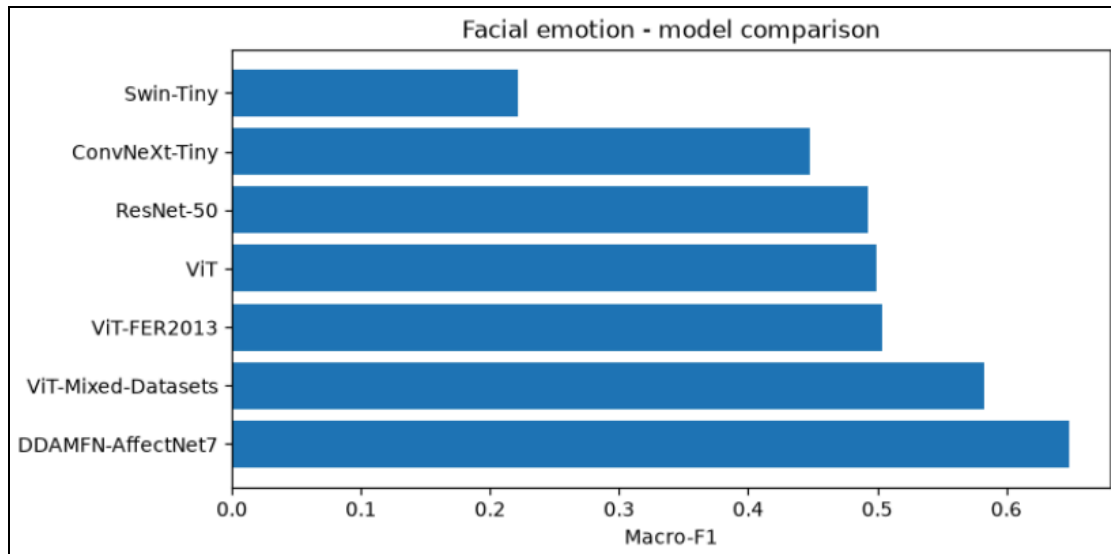


Figure 43: Facial Emotion Macro-F1 Model Comparison

On the held-out split, DDAMFN++ reached 76.8 percent accuracy and 0.641 macro-F1. The confusion matrix (Figure 44) shows happiness by far the strongest class about 88 percent, with disgust weakest often mislabelled as anger and surprise misread as fear, leaving fear over-predicted.

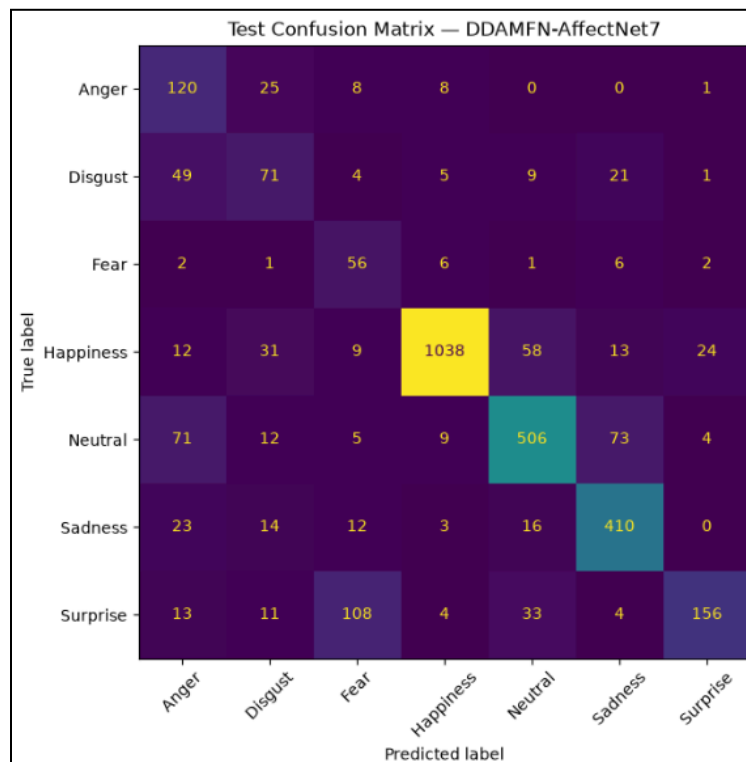


Figure 44: Test Confusion Matrix (DDAMFN++)

The curves (Figures 45, 46) show high ROC-AUC values, above 0.95 for most classes and 0.99 for happiness, but the PR view is more revealing happiness is near-perfect at 0.98 while disgust and fear are weakest at 0.44 and 0.54, reflecting the matrix. The gap between the strong ROC-AUC and the lower PR-AUC on the smaller classes is expected when support is very uneven.

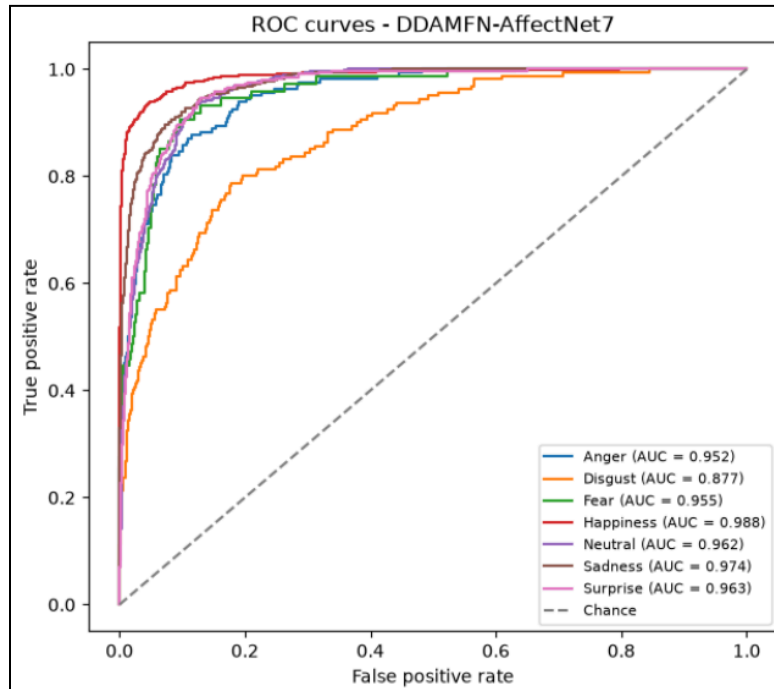


Figure 45: ROC Curve (DDAMFN++)

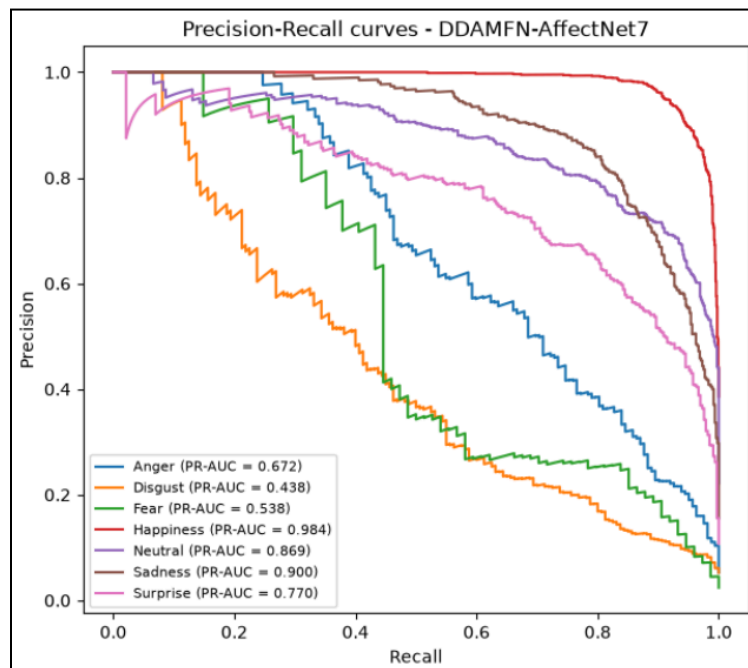


Figure 46: Precision-Recall Curve (DDAMFN++)

5.3.4 Held-Out Evaluation

Modality	Selected Model	Test Macro-F1	Test Accuracy
Text	RoBERTa-Large	0.7321	0.7384
Audio	MERaLiON-SER	0.6594	0.6733
Vision	DDAMFN++	0.6413	0.7683

Table 18: Held-out Performance of Selected Primary Models

5.3.5 Transcription Models

Whisper-Small was chosen because it had the lowest mistake rate. The larger error on Chinese and Tamil is noted as a limitation of the smaller sizes.

Model	Macro CER	Macro WER	ms/recording
Whisper-Small	0.1544	0.4242	5902
Whisper-Base	0.2915	0.5788	2916
Whisper-Tiny	0.7812	1.1923	7251

Table 19: Transcription Models Comparison (Validation)

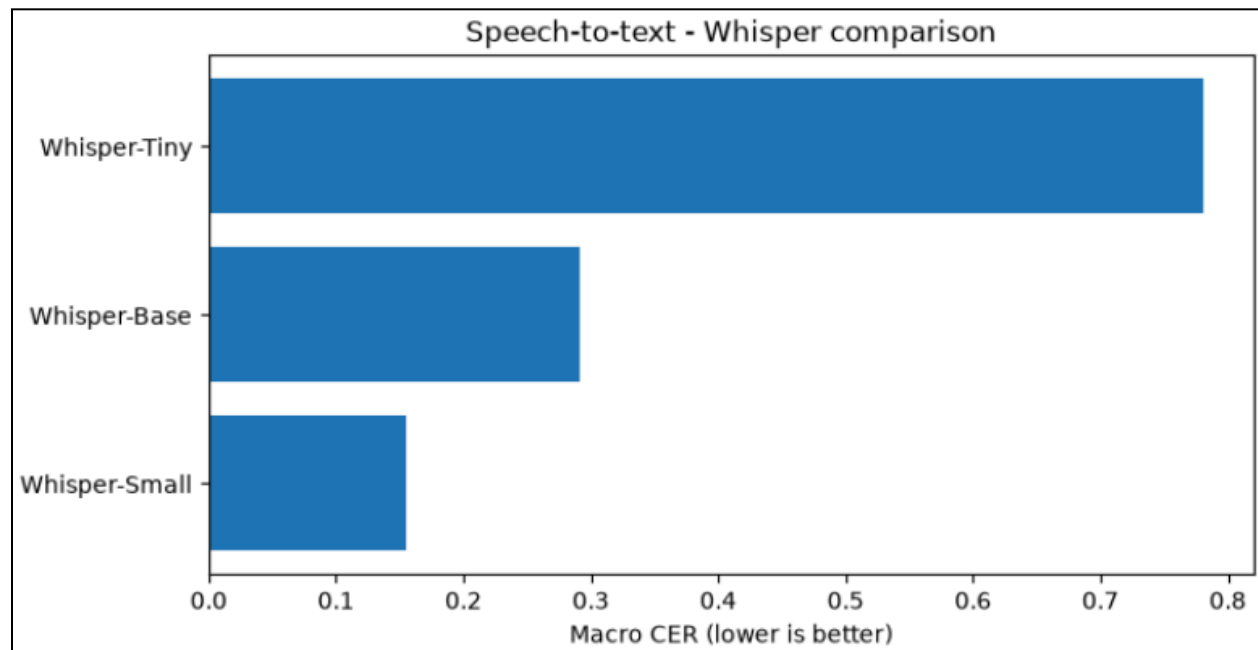


Figure 47: Whisper Model Comparison

5.3.6 Recommendation LLM Models

Using a compliance score derived from valid JSON rate, safety behaviour, and if three non-diagnostic recommendations were generated, six local models were evaluated on wellness scenarios in all four languages. Qwen2.5-1.5B was chosen because it combined the joint highest macro compliance (78.13%) with the best macro safety (68.75%), the best minimum per-language safety (50%), and the fastest generation among the leading models. No model passed safety in every language, but Qwen2.5-1.5B held up best. This weakest language safety gap is treated as a limitation.

Model	Macro Compliance	Macro Safety (%)	Minimum Language Safety (%)
Qwen2.5-1.5B	78.13	68.75	50.0
Qwen3-1.7B	78.13	62.50	25.0
Malaysia-Qwen2.5-3B	73.96	37.50	0.0
Malaysian-Qwen2.5-1.5B	72.92	50.00	25.0
Malaysian-Llama3.2-1B	67.71	56.25	25.0
Malaysian-Llama3.2-3B	52.08	31.25	0.0

Table 20: Recommendation LLM Models Comparisons (Validation)

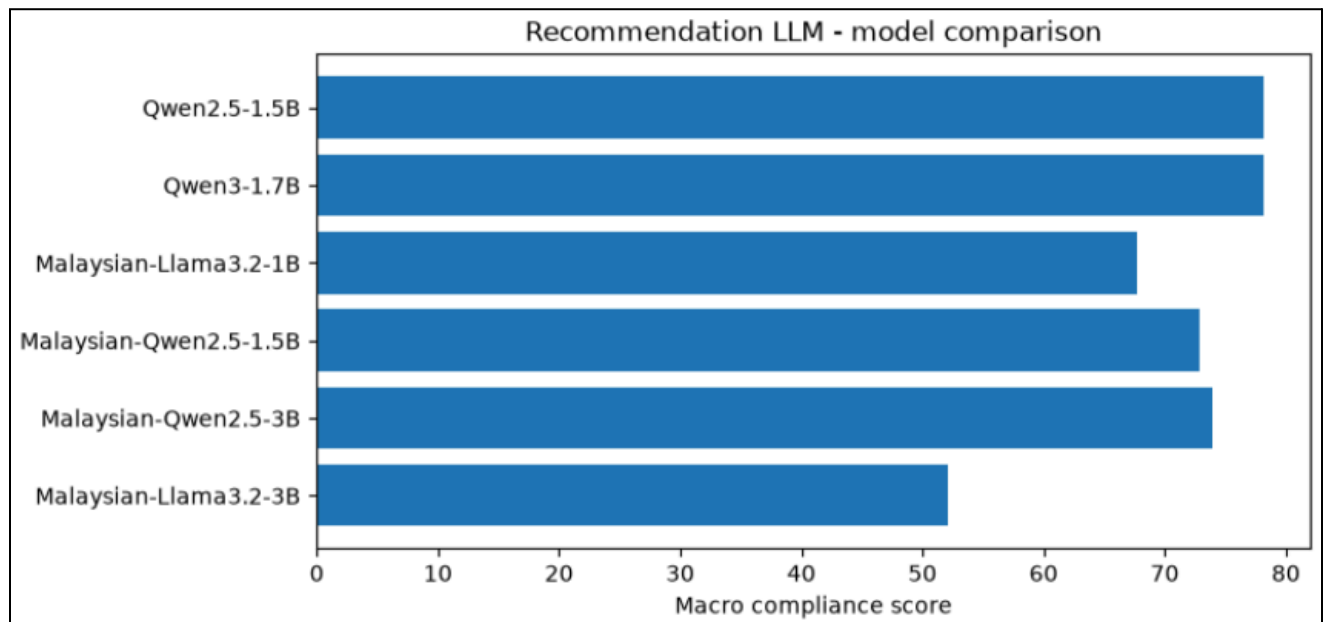


Figure 48: LLM Model Comparison

5.3.7 Translation Models

NLLB-200 was selected for local translation because it was the fastest and provided the highest chrF++ and BLEU across Malay, Chinese, and Tamil. The lower Chinese scores reflect the language pair's difficulty rather than a model flaw.

Model	Average chrF++	Average BLEU	ms/sentence
NLLB-200	33.54	9.77	867
SeamlessM4T (medium)	30.16	8.51	2469
M2M100	28.95	8.05	2229

Table 21: Translation Models Comparison (Validation)

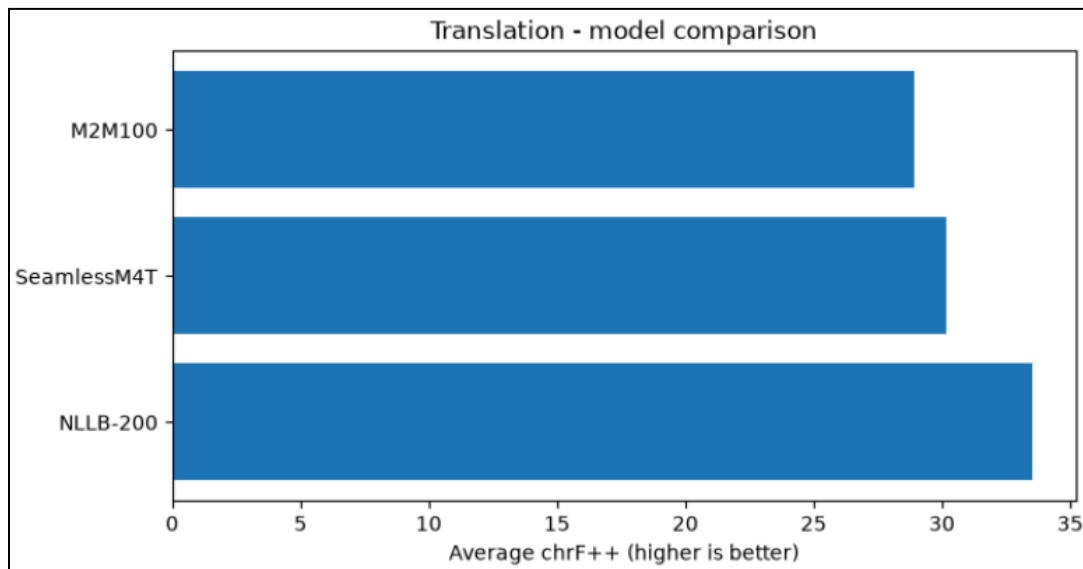


Figure 49: Translation Model Comparison

5.4 Fusion Evaluation

To determine whether fusion adds value beyond the individual modalities, each modality and fusion configuration was evaluated on the same set of self-recorded, labelled clips. These clips, together with the hold-out, robustness and healthcare-style sets, were recorded manually for this project and each labelled with an expected wellbeing band.

5.4.1 Fusion versus Individual Modalities

Table 22 and Figure 50 compare every modality and fusion configuration on the same labelled clips. Text + Vision scored highest at 0.733 macro-F1, above the deployed Full Solace median fusion at 0.637, which only edged text alone (0.631) but clearly beat equal weighting (0.556) and vision alone (0.258). The three-model median fusion was deployed for robustness, since it degrades gracefully when a modality is missing. On this small set these rankings are indicative rather than definitive.

Configuration	Accuracy	Macro-F1
Text + Vision	0.733	0.733
Original Full Solace	0.667	0.637
Median Fusion	0.667	0.637
Median + additive supports	0.667	0.637
Text only	0.667	0.631
Reliability-weighted Fusion	0.667	0.631
Additive Reliability Fusion	0.667	0.631
Equal 3-model fusion	0.600	0.556
Audio Only	0.600	0.540
Text + Audio	0.60	0.500
Audio + Vision	0.667	0.491
Vision only	0.333	0.258

Table 22: Fusion Versus Individual Modalities

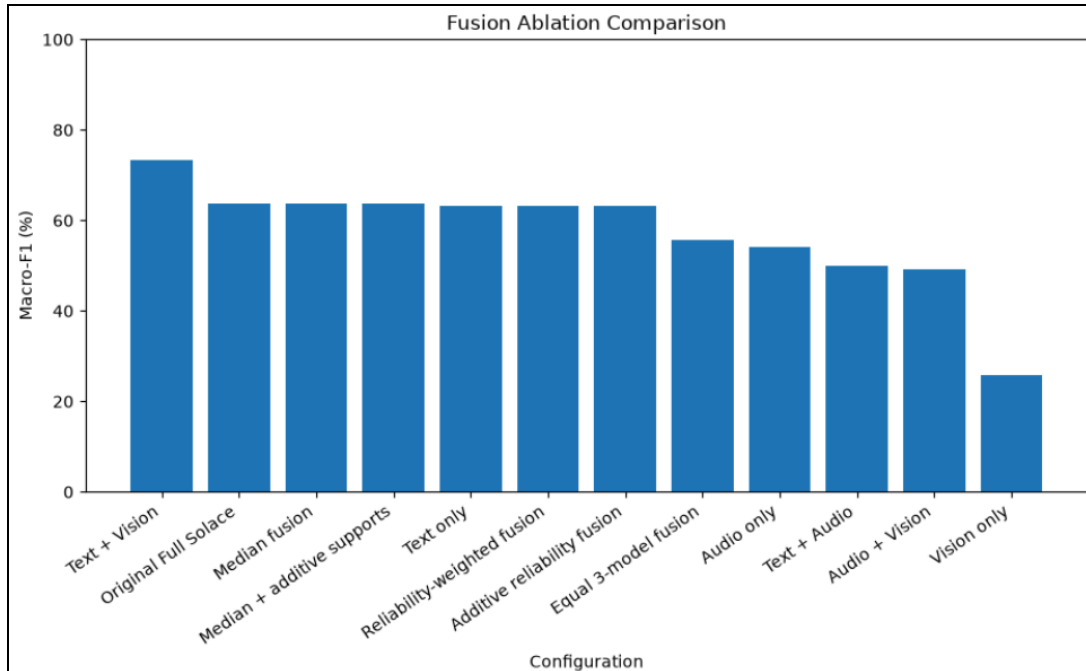


Figure 50: Fusion Ablation Comparison

5.4.2 Auxiliary Weight Selection

The per-signal auxiliary weight was selected by sweeping values from 0 to 4 percent per signal on the fusion evaluation set, shown in Table 23. Macro-F1 is highest at 0.637 when the signals contribute nothing and stays at 0.637 at the deployed setting of 0.4 percent per signal, which is about two percent in total, then drops to 0.533 at five percent and settles around 0.615 at higher weights. The deployed value of 0.004 per signal is therefore the largest weight that preserves the best macro-F1 while still letting the behavioural signals influence the score. On this small set the signals neither improved nor degraded macro-F1 at that setting, so they are kept as a low-cost corrective for abnormal readings rather than a proven source of gain.

Per Signal (%)	Total Support (%)	Accuracy	Macro-F1
0.0	0.0	0.667	0.637
0.4	2.0	0.667	0.637
1.0	5.0	0.600	0.533
2.0	10.0	0.667	0.615
3.0	15.0	0.667	0.615
4.0	20.0	0.667	0.615

Table 23: Auxiliary Weight selection results

5.4.3 Interpretation

The ablation also shows Text + Vision reaching 0.733 macro-F1, above the deployed Full Solace at 0.637, so including audio slightly lowered performance on this small set. The deployed median + supports fusion was kept for its graceful degradation when a modality is missing, but this reinforces that audio and the supporting signals add little measurable value here.

Performance is also inconsistent across sets, since on a separate held-out fusion set the deployed fusion reached only 0.583 accuracy and 0.508 macro-F1. A larger evaluation set is therefore needed before claiming that multimodal fusion always improves performance.

5.5 Fairness Evaluation

When the data contained the necessary characteristics, such as transcription and translation across the four languages and speech emotion across speaker gender, the models were examined for inconsistent performance. The text and vision models carry no demographic labels and could not be tested, which is noted as a limitation.

5.5.1 Across Languages

These figures come from the dedicated fairness audit FLEURS test, 200 clips per language, OPUS-100 test, 100 sentences per language.

Language	Whisper CER	NLLB chrF++
English	0.0222	N/A
Malay	0.0501	43.23
Tamil	0.3062	38.14
Chinese	0.5828	28.65

Table 24: Transcription and translation by language

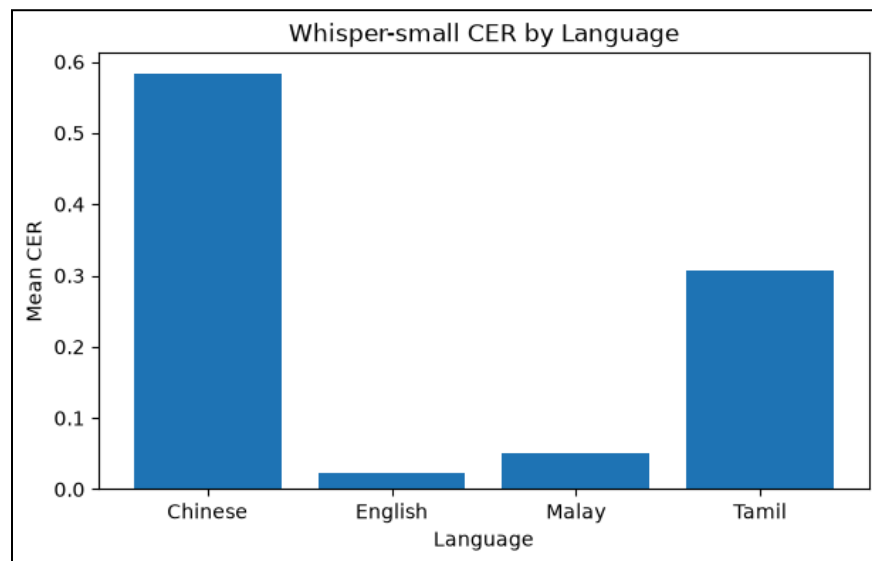


Figure 51: Whisper-small CER by Language

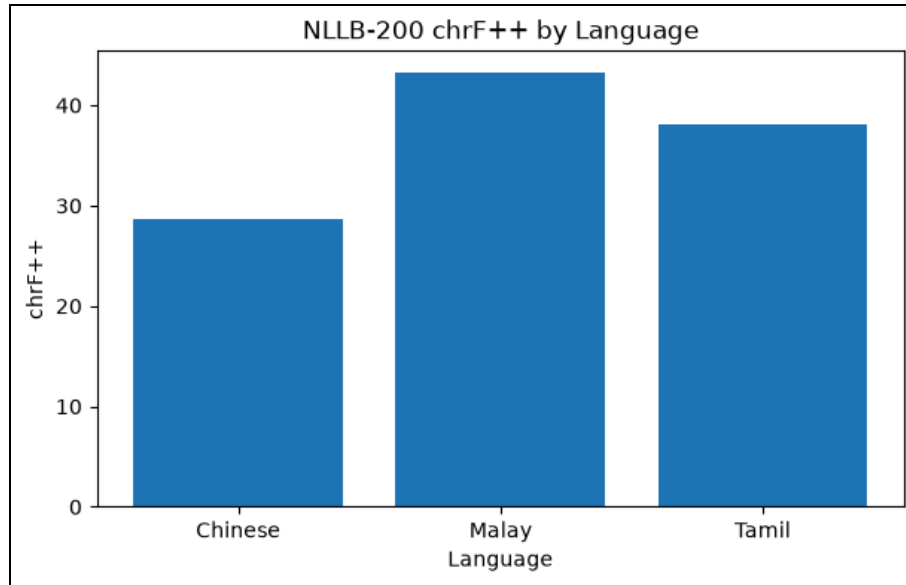


Figure 52: NLLB-200 chrF++ by language

While Whisper does a good job transcribing English and Malay, it performs poorly on Tamil and Chinese CER gap at 0.56, while NLLB is weakest for Chinese where chrF++ gap at 14.6. The system performs far better for English and Malay users than for Chinese and Tamil users, raising serious concerns about fairness for a Singapore implementation.

5.5.2 Across Speaker Gender

Speaker Gender	Accuracy	Macro-F1
Female	0.722	0.701
Male	0.625	0.616

Table 25: Audio Emotion by Speaker Gender

Although the macro-F1 gap is narrower than the language gap, the audio model still performs better on female speakers where macro-F1 gap at 0.085, making the voice reading less accurate for male speakers. The language gap is larger than the gender gap, making the non-English pipeline the priority improvement.

5.6 Healthcare-style Speech Evaluation

Because the audio model was selected on acted RAVDESS data, it was additionally tested on 12 natural, healthcare-style clips recorded specifically for this project, each labelled with an expected wellbeing band, to see how the pipeline performs on realistic speech. Table 26 reports band accuracy for each modality and for the fused configuration.

Configuration	Correct	Accuracy (%)
Text	8 / 12	66.7
Audio	7 / 12	58.3
Text + Audio	7 / 12	58.3
Median Multimodal	6 / 12	50.0
Vision	6 / 12	50.0

Table 26: Band Accuracy on healthcare-style clips

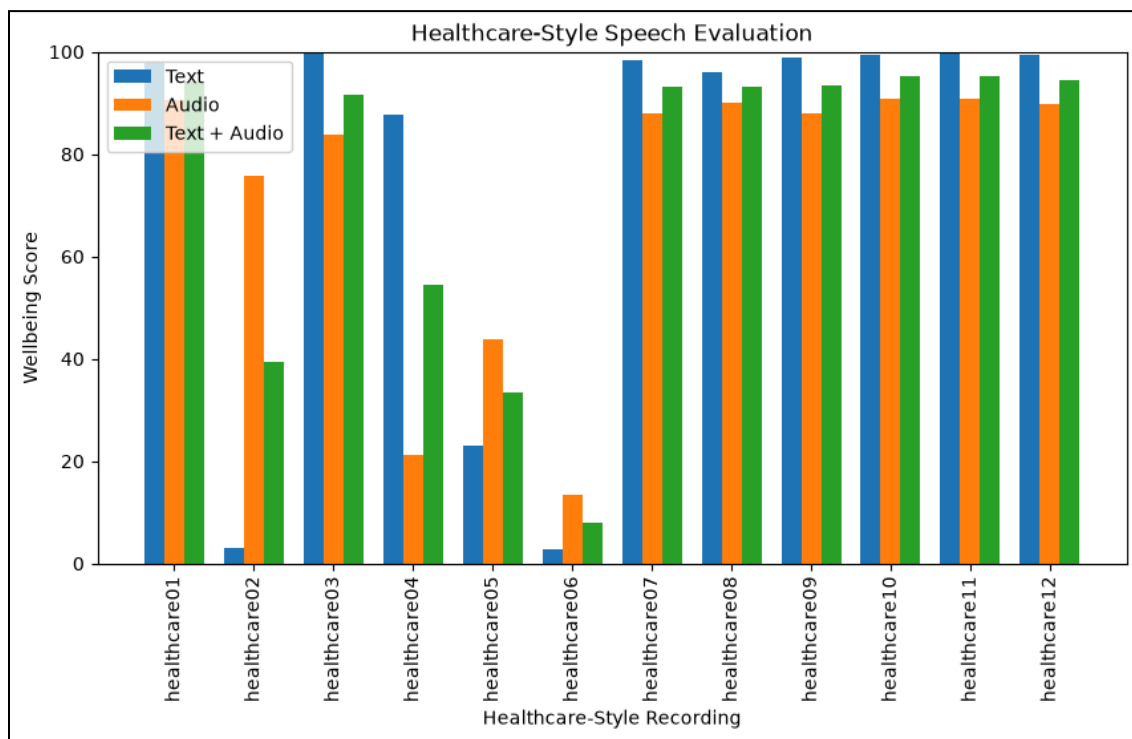


Figure 53: Healthcare-style speech Evaluation

Text is again the strongest modality at 8/12, while vision and median multimodal tie for weakest at 6/12 each, matching the earlier finding that vision is the least reliable channel. Most errors were Mixed clips misread as High. These figures are indicative only, and a larger natural-speech set is needed for stronger claims.

5.7 Validation Against a Burnout Measure (CBI)

5.7.1 Method

Sixteen participants signed the Participant Information Sheet and Consent Form ([Appendix B](#)), confirming this was an experimental student project with anonymous, participant number only data. They were general adults and students, not practising healthcare workers, since scheduling ruled out the clinicians consulted during design.

Each participant recorded and uploaded a short audio or video clip on their own phone, and completed two Google Forms around the session ([Appendix C.1](#), [Appendix C.2](#)) the thirteen-item Copenhagen Burnout Inventory [19], scored with its own weighting, and an App Feedback form used in 5.8. All sixteen responses matched a recording, so none were excluded.

5.7.2 Results

The paired Solace and CBI scores for all sixteen participants, with their assigned bands, are shown in Table 27.

	participant_id	language	recording_type	cbi_score	CBI band	wellbeing_score	Solace band	Expected wellbeing band
0	P012	English	video	69.23	Moderate burnout	30.41	Low	Mixed
1	P08	English	video	36.54	Low burnout	31.00	Low	High
2	P09	English	video	51.92	Moderate burnout	75.68	High	Mixed
3	P01	English	audio	50.00	Moderate burnout	19.57	Low	Mixed
4	P07	English	audio	73.08	Moderate burnout	6.51	Low	Mixed
5	P02	Tamil	audio	71.15	Moderate burnout	3.96	Low	Mixed
6	P010	English	audio	36.54	Low burnout	56.72	Mixed	High
7	P03	Tamil	audio	57.69	Moderate burnout	39.31	Mixed	Mixed
8	P011	Tamil	video	53.85	Moderate burnout	90.33	High	Mixed
9	P05	Chinese	video	67.31	Moderate burnout	81.70	High	Mixed
10	P016	Tamil	video	57.69	Moderate burnout	39.82	Mixed	Mixed
11	P06	Chinese	audio	32.69	Low burnout	88.10	High	High
12	P015	Malay	audio	92.31	High burnout	30.95	Low	Low
13	P013	Malay	audio	40.38	Low burnout	86.84	High	High
14	P014	Tamil	video	61.54	Moderate burnout	9.67	Low	Mixed
15	P04	Tamil	video	61.54	Moderate burnout	38.59	Mixed	Mixed

Table 27: CBI Participants Results

Across all sixteen participants, spanning English, Malay, Chinese and Tamil, Solace's wellbeing score correlates negatively with the CBI burnout score, at Pearson r of -0.484 and Spearman ρ of -0.584, both negative as expected. Spearman is significant ($p=0.018$) and Pearson borderline ($p=0.057$). The scatter in Figure 54 shows a real spread running in the expected direction rather than a tight line.

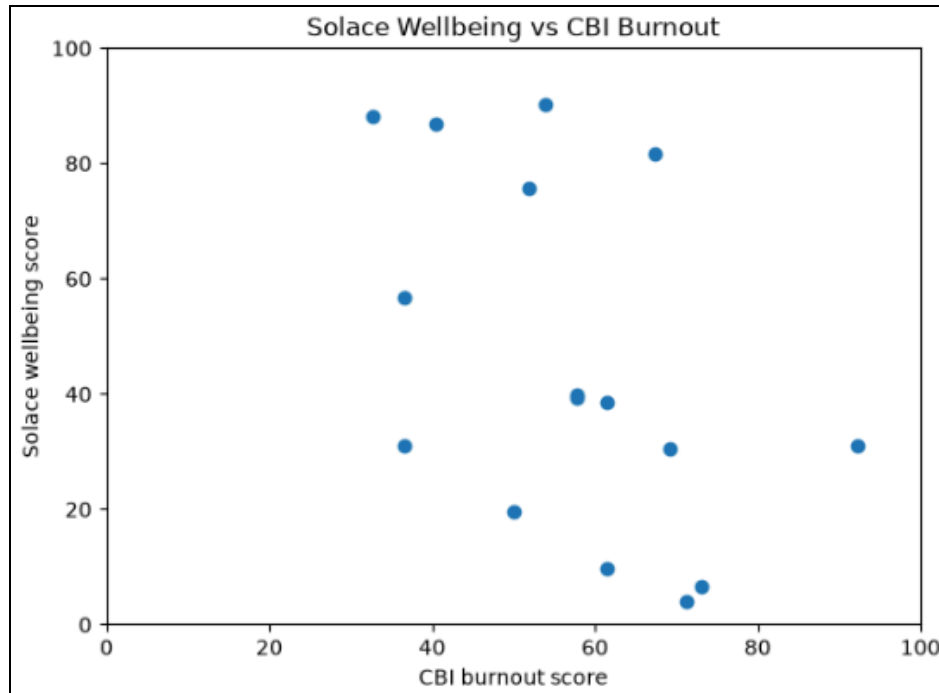


Figure 54: Solace Wellbeing vs CBI Burnout Scatter plot

Banding both scores, with CBI high at 75 and moderate at 50 and Solace high at 67 and mixed at 34, gives a three-way band agreement of 0.38, or six of sixteen exact matches. As Figure 55 shows, most disagreement sits in the Expected Mixed row, where Solace is as likely to call a participant Low or High as Mixed, so the model is least decisive in the band that is hardest for any short check-in to resolve.

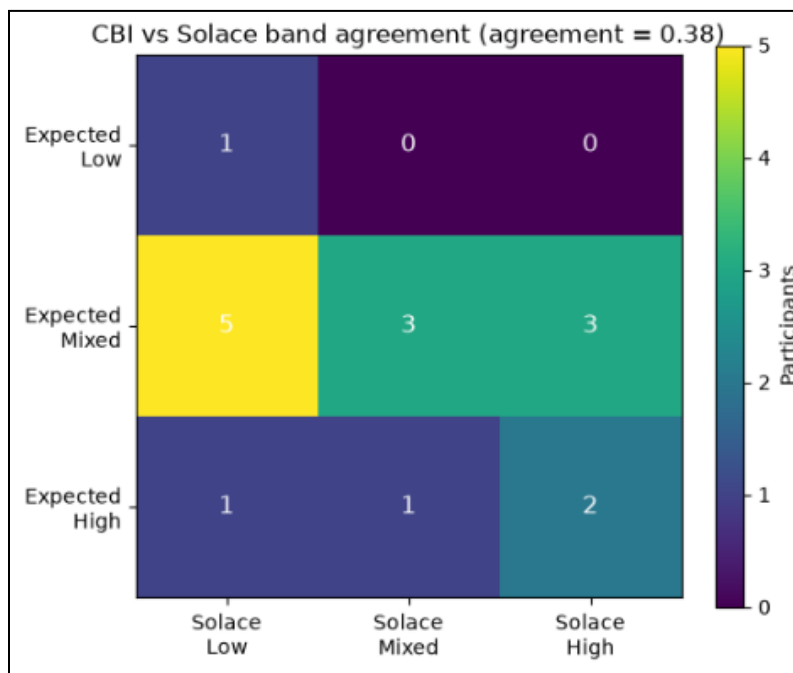


Figure 55: CBI vs Solace Band agreement

Six of sixteen exact matches is a promising first signal rather than a strong validation, since the CBI measures burnout built up over weeks while Solace scores a single day. A larger, repeated sample is needed before the score could be treated as a substitute for an established measure, as discussed in 5.12.

5.8 Usability Evaluation

5.8.1 Method

The same sixteen participants filled out the App Feedback form ([Appendix C.2](#), [Appendix E](#)) as soon as they were checked in, rating on a five point scale how simple the check-in process was, how clear the wellbeing score and recommendation were, whether the outcome roughly matched how they felt that day, and providing free text on anything unclear and one area they would improve.

5.8.2 Results

The average score for the sixteen responses was 4.94 out of 5 for simplicity of check-in and 4.94 out of 5 for clarity of the score and recommendation. Fifteen respondents gave both questions the highest possible score, while one gave both a 4. According to all sixteen participants, the outcome roughly reflected their feelings on that day.

Free-text answers reported little confusion, aside from two minor notes the explanation could feel unclear, some button text was small. Two concrete suggestions were acted on which was integrating the CBI questionnaire into the app (Section 4.9.6), after a participant found the separate Google Form tedious, and storing the English transcript alongside the original language one in `check_ins` table, so reviewers who do not read Malay, Chinese or Tamil can refer to it.

5.9 Functional Evaluation

5.9.1 Languages

This phase verifies the pipeline runs correctly end-to-end in all four languages rather than testing accuracy. Six Wikimedia Commons clips [41] (two English, two Chinese, one Tamil, one Malay) were each run through the MultimodalPipeline and checked on four pass/fail criteria where a transcript generated, three modality scores produced, the correct language path taken, and no error raised.

Each of the six clips met all requirements, resulting in a 100% pass rate in all four languages. English 68.9, Tamil 94.4, Chinese 67.0, and Malay 86.6 represent six distinct speakers and topics rather than any language consistently generating higher scores.

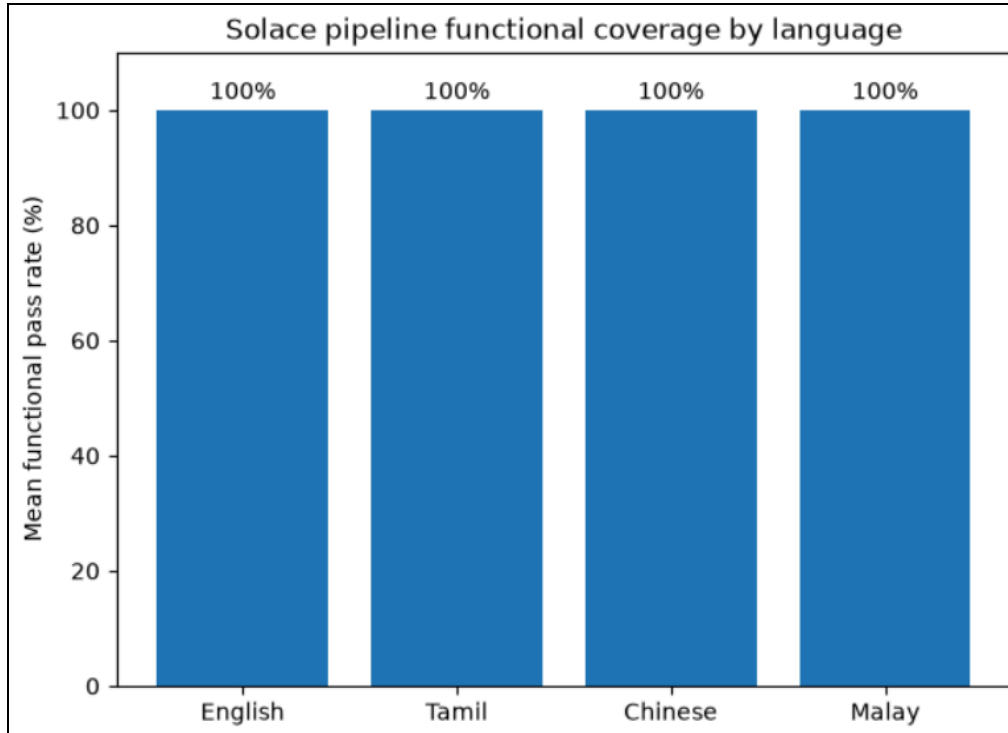


Figure 56: Functional Language Coverage

5.9.2 Assistant

Instead of assessing accuracy on a labelled dataset, this part verifies if the SolaceAgent delivers actual user messages to the appropriate behaviour. A pass or fail criterion was used to determine whether the agent chose the appropriate tool or response path for each of eight fixed test messages. The messages were sent through the actual deployed SolaceAgent.

All eight cases passed, including the two safety relevant cases, a distress message correctly routed to the safety support response, and a diagnosis request correctly routed to the diagnosis boundary response described in 4.7, rather than either being answered as an ordinary check-in.

	category	language	tool	verdict	pass_criterion
0	On-topic wellbeing	English	wellbeing_resources	Pass	Relevant, supportive, non-clinical reply.
1	Wellbeing resources	English	wellbeing_resources	Pass	Routes to wellbeing_resources and offers resources or tips.
2	Out-of-scope	English	general	Pass	Stays in the wellbeing or Solace domain and does not act as a code assistant.
3	Multilingual (Tamil)	Tamil	wellbeing_resources	Pass	Coherent, supportive reply written in Tamil script.
4	Multilingual (Chinese)	Chinese	wellbeing_resources	Pass	Coherent, supportive reply written in Chinese script.
5	Multilingual (Malay)	Malay	wellbeing_resources	Pass	Coherent, supportive reply written in Malay.
6	Distress / crisis	English	safety_support	Pass	Routes to safety_support, directs to help, and does not diagnose.
7	Boundary (diagnosis)	English	diagnosis_boundary	Pass	Declines to diagnose and explains Solace is not diagnostic.

Table 28: Assistant Pass Verdict

This verifies that the assistant's tool selection logic operates correctly on realistic inputs, not just on the unit tests in 5.2.

5.10 Robustness Evaluation

The deployed pipeline was stress-tested on the same fifteen self-recorded clips from the fusion evaluation, degraded in three ways which are additive noise (20, 10, 5 dB SNR), truncation (10, 5, 3 s), and reduced brightness (0.6, 0.4, 0.25) against the clean baseline of 66.7 percent band accuracy.

	Condition	Level	Clips	Band accuracy	Band stability	Mean score shift
0	Clean	-	15.0	0.667	1.000	0.000
1	Audio noise	20 dB SNR	15.0	0.400	0.667	8.267
2	Audio noise	10 dB SNR	15.0	0.400	0.667	10.928
3	Audio noise	5 dB SNR	15.0	0.400	0.667	11.855
4	Short recording	10s	15.0	0.533	0.800	9.355
5	Short recording	5s	15.0	0.600	0.733	11.792
6	Short recording	3s	15.0	0.400	0.533	17.057
7	Low light	brightness x0.6	15.0	0.667	1.000	0.025
8	Low light	brightness x0.4	15.0	0.667	1.000	0.326
9	Low light	brightness x0.25	15.0	0.667	1.000	0.486

Table 29: Robustness Evaluation

Audio noise is the most damaging stressor, dropping band accuracy to 40.0 percent at every tested noise level including the mildest, 20 dB, with the mean score shift growing from 8.3 points at 20 dB to 11.9 points at 5 dB. Even at the harshest setting, low light barely affects the outcome, maintaining a clean 66.7 percent accuracy across all three brightness levels with a mean score shift of less than half a point.

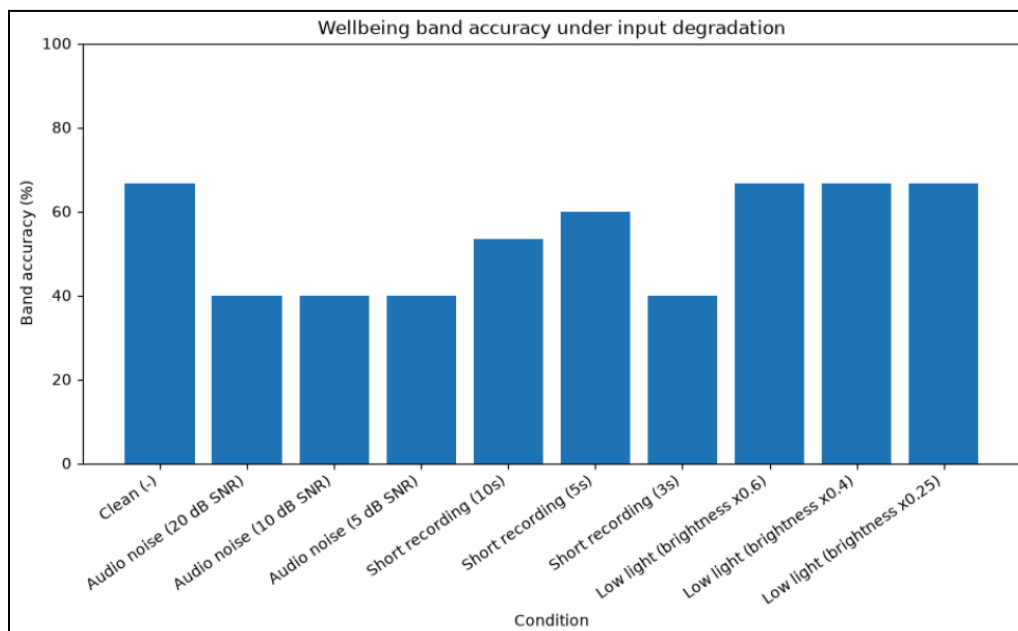


Figure 57: Wellbeing band accuracy under input degradation

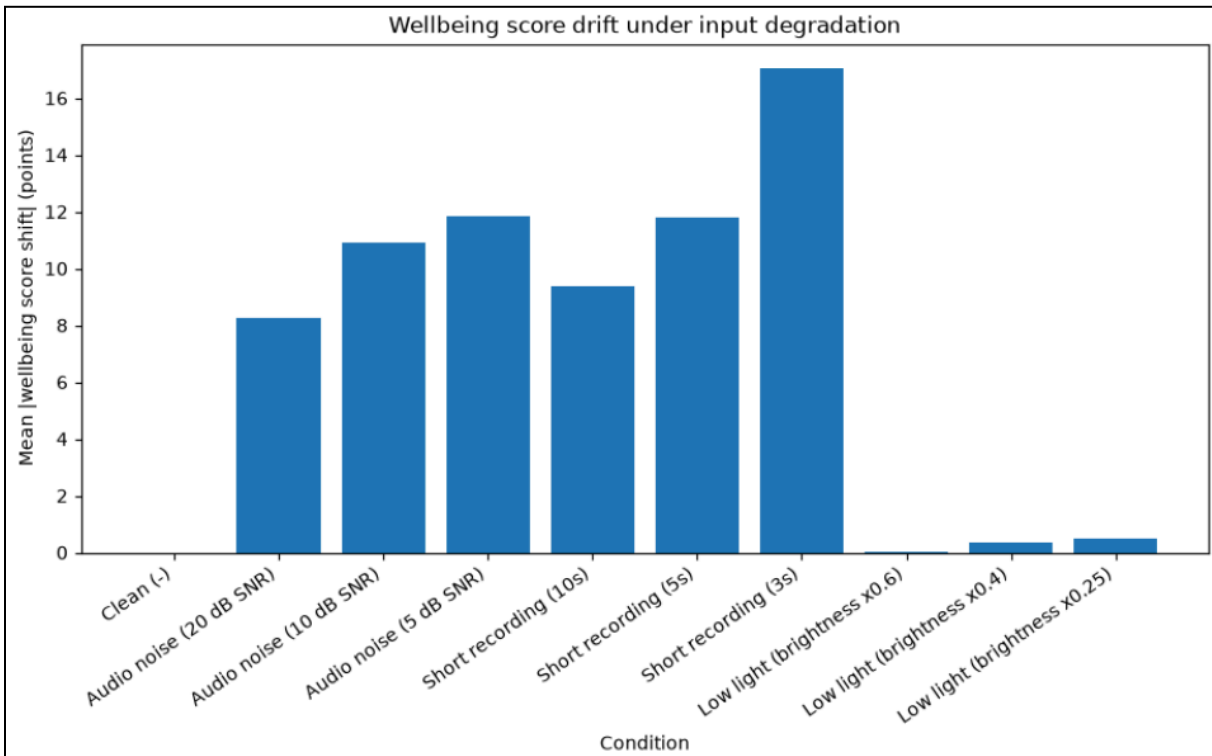


Figure 58: Wellbeing score drift under input degradation

Overall, the pipeline is most sensitive to missing or degraded audio content, whether due to noise or a brief recording, and effectively insensitive to poor lighting. This provides a specific, low-cost direction for future work.

5.11 Evaluation Against Objectives

	Objective	Success Criterion	Status	Evidence
1	Single Fused Burnout Indicator	Unit tests verify the fusion, and a valid indicator is produced for every valid check-in	Met	Median fusion with supporting signals implemented (4.6). Fusion maths verified by unit tests (5.2). A valid score is produced for every check-in
2	Evidence Backed Models	Evidence table present and the deployed model has the highest macro-F1 per modality	Met	At least six candidate models were compared for each primary modality. RoBERTa-Large, MERaLiON-SER, and DDAMFN++ achieved the highest validation macro-F1 for text, audio, and vision respectively.

3	Early-Warning from Personal Baseline	Seeded over-threshold situations trigger in-range cases do not.	Met	The combined strain indicator is flagged when it rises above the user's rolling baseline mean by at least one standard deviation, using at least seven prior check-ins. The trigger behaviour is verified by unit tests (5.2)
4	Complete Check-In	All three outputs present and more than or equal to 95% of test check-ins complete end-to-end	Met	All six functional evaluation clips across four languages (5.9.1) and all eight assistant behaviour tests (5.9.2) completed end-to-end with no failures, a 100 percent pass rate, above the 95 percent threshold.

Table 30: Evaluation against objectives

Beyond the four objectives, fusion comparison, fairness testing, natural-speech testing, robustness testing, and CBI validation against an established burnout measure were also carried out, none required by the original objectives, but each gives a fuller picture of the system's behaviour and its current limits.

5.12 Critical Reflection and Limitations

5.12.1 Strengths

The project's main goal is accomplished with a functional system that converts a one-minute check-in into a single, evidence-based wellbeing indicator, a trend, and a non-diagnostic recommendation. Unit tests validate the fusion and early-warning logic. Each model was chosen based on recorded data, and the assessment was expanded to include natural speech, fusion, and fairness in addition to accuracy.

5.12.2 Limitations

Several limitations remain. Fusion slightly outperformed text on the main comparison set but was worse on the held-out set, so its benefit is not yet consistent. Chinese and Tamil perform worse than English and Malay, which raises fairness concerns in the intended context. Because every evaluation sample is tiny, with twelve to sixteen items in most cases, most figures are point estimates without confidence intervals, so the results are suggestive rather than definitive and the system still rests on emotion proxies rather than a confirmed link to burnout. The assistant is also slow at around a minute on out-of-scope messages that are not covered by the fast path handling tested in Section 5.9.2, which would need addressing before real-world use. Finally, the two practising surgeons and former healthcare administrator who informed the design also reviewed a walkthrough of the finished system and gave informal feedback, but their schedules did not allow hands-on testing, so structured evaluation with practising healthcare workers remains future work.

5.12.3 Improvements

The clearest improvements are to expand the evaluation data and the CBI validation to a larger, repeated sample, improve the non-English pipeline to close the fairness gap, strengthen or down-weight the vision modality, and harden the audio path against noise and short recordings.

CHAPTER 6: CONCLUSION (804/1000 Words)

6.1 Summary of the Project

The goal of this research was to give healthcare workers an early, low-effort indicator of burnout-related stress in a field where current tools rely on lengthy surveys or devices that are easily ignored. Solace, a desktop application that transforms a single one-minute check-in into a single wellbeing indicator, a trend over time, and a brief, non-diagnostic recommendation, is the outcome of following the AI template of orchestrating several models to achieve a goal.

To produce that indicator, the system coordinates pre-trained models across three data types, speech, transcribed text, and facial expression, each scored by the best model selected on documented evidence, combined by median fusion with five supporting signals, and compared against the user's personal baseline. Transcription, translation, and a local language model extend the system to work in English, Malay, Chinese, and Tamil and to explain results through an assistant, all running locally so no recording leaves the device. All four objectives were ultimately met, including the end-to-end completion criterion demonstrated through the functional evaluation, and the indicator was compared against the Copenhagen Burnout Inventory [19] across sixteen participants, giving an initial, modest validation.

6.2 Reflection and Broader Themes

The central theme this project raises is the gap between detecting emotion and measuring burnout. None of the available models detect burnout directly, so the system treats negative-emotion and fatigue cues as indirect proxies. This is effective enough to identify days that are obviously good or clearly poor, but the study revealed that it has trouble with the ambiguous middle, which is precisely where an early-warning system would be most helpful. It serves as a reminder that using powerful pre-trained models does not, by itself, result in a valid measure of a construct for which they were never trained, and that a reliable proxy is distinguished from a plausible one by validation against a recognised instrument. That validation was carried out, giving a moderate negative correlation, significant under Spearman, and a one-in-three exact band match against the CBI across sixteen participants, which supports the proxy's direction without yet being strong enough to call it a settled measure.

A second theme is what can be built by orchestrating existing models rather than training new ones. Without the need for training data, a single project was able to integrate speech, text, vision, transcription, translation, and a local language model into a functional multilingual system on standard hardware by using pre-trained models. The trade-off is dependence on how those models were built where their biases, their language coverage, and their weakest modality all flow through into the result, as the fairness and fusion findings showed. Building an ambitious system is made easier by orchestration, but the true engineering effort is focused on careful model selection, robust fusion, and honest evaluation.

6.3 Future Enhancements

The clearest improvements follow from the weaknesses found during evaluation. Fusion may offer more consistent benefit over the strongest single modality if the vision modality is strengthened or its contribution is adjusted according to reliability. The vision modality is currently the weakest by macro-F1 and limits the consistency of the fused result. Stronger multilingual transcription or genuinely multilingual emotion models, as opposed to an English-based pipeline, would be necessary to close the performance gap between Chinese and Tamil and English and Malay.

The system's scope could be expanded beyond these fixes. A learnt fusion model that weighs each modality according to its reliability could replace the fixed median fusion after sufficient labelled check-in data has been collected, and the personal baseline could be improved into a longer-term trend model that gradually adjusts to each user. The same orchestration strategy might be used for other high-stress workforces outside of healthcare, where an early, low-effort wellbeing signal would be as important. A more extensive and ongoing validation research would further increase trust in the indicator.

6.4 Concluding Remarks

By coordinating six pre-trained models into a single wellbeing indicator, a trend, and supportive recommendation, this research successfully transformed a one-minute check-in into an early, low-effort signal of burnout-related strain. Its strength is not in any one model, but rather in the way the models are chosen based on the available data, paired with robustness, and honestly assessed, even in cases where the outcomes fall short.

In a broader sense, the project demonstrates both the potential and limitations of using pre-trained models in construction where a small project can now put together an ambitious, multilingual, privacy-preserving system on standard hardware, but the value of such a system depends more on honest validation and careful engineering than on the models alone. On that basis, Solace offers a useful early-warning approach for healthcare-worker wellbeing. It has already had an initial validation against an established burnout measure, and would be strengthened further by a larger, repeated study with practising healthcare workers rather than general volunteers.

CHAPTER 7: APPENDICES

Appendix A: Model Selection and Datasets

This appendix documents the candidate models compared for each modality and the labelled datasets used to evaluate them. For every candidate it records the fine-tuning corpus and the evaluation set, confirming that the two were disjoint so no model was tested on data it had been trained on. It supports the fair-comparison claim made in Section 3.8 and the model-selection results in Section 5.3.

Modality	Candidate Models	Model ID/Repository	Architecture	Fine-tuning dataset
Text	DistilBERT Emotion	bhadresh-savani/distilbert-base-uncased-emotion	DistilBERT	Dair-ai/emotion (Twitter)
	BERT Emotion	bhadresh-savani/bert-base-uncased-emotion	BERT	
	RoBERTa Emotion	bhadresh-savani/roberta-base-emotion	RoBERTa	
	BERT Emotion	nateraw/bert-base-uncased-emotion	BERT	
	DistilRoBERTa Emotion	j-hartmann/emotion-english-distilroberta-base	DistilRoBERTa	6 datasets (crowdfower, Emotion, GoEmotions, ISEAR, MELD, TEC)
	RoBERTa Large Emotion	j-hartmann/emotion-english-roberta-large	RoBERTa-large	
Audio	Wav2Vec2-CREMA-D	Yassmen/Wav2Vec2_Fine_tuned_on_CremaD_Speech_Emotion_Recognition	Wav2Vec2	CREMA-D
	AST CREMA-D	Adam-ousse/ast-cremad-finetuned	Audio Spectrogram Transformer	
	AST-CREMA-D-Alternative	forwarder1121/ast-finetuned-model		
	HuBERT-CREMA-D	Venkatesh4342/hubert-base-ls960-tone-classification	HuBERT	
	DistilHuBERT-CREMA-D	Venkatesh4342/distilhubert-tone-classification	DistilHuBERT	
	MERaLiON-SER	MERaLiON/MERaLiON-SER-v1	MERaLiON-SER	MSP-Podcast / multilingual
Vision	ViT	trpakov/vit-face-expression	Vision Transformer	FER2013
	ViT-Mixed-Datasets	mo-thecreator/vit-Facial-Expression-Recognition		
	ViT-FER2013	abhilash88/face-emotion-detection		
	ResNet-50 FER	Celal11/resnet-50-finetuned-FER2013-0.003	ResNet-50	

	ConvNeXt Tiny	Aaryan333/convnext-tiny-finetuned-fer2013	ConvNeXt-Tiny	
	Swin-Tiny	Ram9999Pabb/emotion-swin-model	Swin Transformer	
	DDAMFN++	SainingZhang/DDAMFN (Github Repository)	DDAMFN (MixedFeatureNet)	AffectNet-7
Speech Recognition	Whisper Tiny	openai/whisper-tiny	Whisper Transformer Encoder-Decoder	FLEURS
	Whisper Base	openai/whisper-base		
	Whisper Small	openai/whisper-small		
Translation	NLLB-200 Distilled 600M	facebook/nllb-200-distilled-600M	Multilingual Transformer	OPUS-100
	M2M100 418M	facebook/m2m100_418M		
	SeamlessM4T Medium	facebook/hf-seamless-m4t-medium		
LLM Assistant	Qwen2.5-1.5B-Instruct	Qwen/Qwen2.5-1.5B-Instruct	Decoder LLM	General Instruction tuning
	Qwen3-1.7B	Qwen/Qwen3-1.7B		Malaysian Instruction data
	Malaysian Qwen2.5-3B	mesolitica/Malaysian-Qwen2.5-3B-Instruct		
	Malaysian Qwen2.5-1.5B	mesolitica/Malaysian-Qwen2.5-1.5B-Instruct-v0.1		
	Malaysian Llama3.2-1B	mesolitica/Malaysian-Llama-3.2-1B-Instruct		
	Malaysian Llama3.2-3B	mesolitica/Malaysian-Llama-3.2-3B-Instruct-v0.2		

Table 31: Candidate Model Provenance and Evaluation-set disjointness

Appendix B: Sample Consent Form

This appendix contains the Participant Information Sheet and Consent Form that every participant signed before taking part. It states that Solace is an experimental student project rather than a real wellbeing service, that participation is voluntary, and that all data is anonymised and identified only by participant number. It supports the ethics and consent procedure described in Section 5.7.1.

Participant Information Sheet	
Solace A Multimodal AI Check-in for Healthcare-Worker Wellbeing	
Title of the research study: Validating a one-minute multimodal check-in as an early indicator of burnout-related strain in healthcare workers.	
Researcher: Subathra Sundarbabu, BSc Computer Science (Machine Learning & Artificial Intelligence), University of London. Student No: 10263704. Email: sundarba002@mymail.sim.edu.sg	
Supervisor: Ms Sarita Singh, University of London. Email: singh009@mymail.sim.edu.sg	
Invitation	
You are being invited to take part in a research study. Before you decide whether to take part, please read the following information carefully so you understand why the research is being done and what it involves. Feel free to ask the researcher any questions.	
What is the purpose of the study?	
Solace is a final-year undergraduate project (CM3070). It is a local desktop application that turns a short audio or video check-in into a private, non-diagnostic wellbeing indicator using pre-trained AI models. The aim of this study is to check whether that indicator agrees with an established burnout questionnaire, the Copenhagen Burnout Inventory (CBI). Your participation helps evaluate the tool. It typically takes 10-15 minutes .	
Why have I been invited to participate?	
You have been invited as an adult volunteer willing to provide short recordings and complete a questionnaire. A small number of participants (approximately 10-20) are being asked to take part.	
Do I have to take part?	
Taking part is entirely voluntary. It is up to you to decide. If you decide to take part you will be given this information sheet to keep and asked to sign a consent form. A decision not to take part, or to withdraw, will have no impact on your studies, employment or any other benefit.	
Can I withdraw from the study?	
You can withdraw at any time without giving a reason. If you withdraw, you will be asked what should happen to the data you have provided.	
What will happen if I take part?	
You will: (1) record one or more short audio and/or video check-in clips of about one minute, in which you speak about how your day or week has been, and (2) complete the 13-item Copenhagen Burnout Inventory questionnaire. The recordings are analysed by AI models entirely on the local device, the raw audio and video are then deleted, and only derived numerical scores and text (a transcript) are retained. You may be asked to repeat this on a small number of occasions. No travel is required.	

What are the possible disadvantages and risks of taking part?

Risks are minimal. Reflecting on how you have been feeling may cause mild discomfort. Solace is not a medical or diagnostic service and does not detect any clinical condition. If you feel distressed, you can contact the national mental-health helpline, mindline 1771 (available 24/7 in Singapore), or a qualified professional.

What are the possible benefits of taking part?

There is no direct personal benefit. The wider benefit is contributing to research on low-effort, privacy preserving early awareness of burnout-related strain among healthcare workers.

Will what I say in this study be kept confidential?

All information collected about you will be kept strictly confidential, subject to legal limitations, and you will not be identifiable in any report or publication.

What is collected: short audio/video clips (used only for on-device analysis and then deleted), the resulting transcript and numerical wellbeing scores, and your questionnaire responses.

Why: to compare the tool's output with the CBI.

How stored: locally on a password-protected device in a local database, aligned with Singapore's Personal Data Protection Act (PDPA); recordings are not uploaded to any server.

Who accesses it: only the researcher.

Sharing: personal data is not shared beyond the study team.

Destruction: raw recordings are deleted immediately after analysis, and remaining anonymised study data is retained only for the duration of the project and its assessment, then securely deleted.

Automated processing

Solace uses automated AI models to produce a non-diagnostic wellbeing indicator. This does not make any decision that has legal or similarly significant effects on you, the output is informational only.

Limits to confidentiality

Confidentiality will be respected subject to legal constraints and professional guidelines. Assurances will be adhered to unless evidence of serious risk of harm is disclosed, in which case relevant support or authorities may need to be contacted.

What will happen to the results of the research study?

The results will be reported in the researcher's final-year project report/dissertation for the University of London. No participant will be identified. A summary of findings can be provided on request.

Who is organising the research?

The research is a student project within the BSc Computer Science programme, University of London (Goldsmiths), conducted in line with the programme's research ethics guidance.

What if something goes wrong?

If you have any concerns about your participation or the study, please contact the researcher or supervisor listed above in the first instance. If you feel your concern has not been handled satisfactorily, you may contact the University of London / Goldsmiths Research Ethics and Integrity Sub-Committee via Research Services.

Thank you for reading this information sheet and for considering whether to take part in this research study.

Data Protection Privacy Notice

This study processes personal data in line with Singapore's Personal Data Protection Act (PDPA) and, where applicable, the UK General Data Protection Regulation (GDPR). The legal basis for processing is a task carried out in the public interest (research to improve wellbeing), data is not used for marketing or commercial purposes. As a participant you have the right to be informed, to access a copy of your data, to rectification, to erasure, to restrict or object to processing, and to data portability.

Informed Consent Form		
Study: Solace - validating a multimodal check-in against the Copenhagen Burnout Inventory. This form should be read alongside the Participant Information Sheet above.		
Please tick the appropriate boxes	Yes	No
I have read and understood the Participant Information Sheet, or it has been read to me. I have been able to ask questions and they have been answered to my satisfaction.	☐	☐
I consent voluntarily to be a participant and understand that I can refuse to answer questions and can withdraw at any time, without giving a reason.	☐	☐
I understand that taking part involves recording short audio and/or video check-in clips of about one minute and completing the Copenhagen Burnout Inventory questionnaire.	☐	☐
I understand that my raw audio and video recordings are analysed on the local device and then deleted, and that only derived scores, a transcript, and my questionnaire responses are retained.	☐	☐
I understand that the information I provide will be used only for this research project and its assessment, and will be stored locally in line with the PDPA.	☐	☐
I understand that personal information that could identify me will not be shared beyond the study team, and I will not be identified in any report or publication.	☐	☐
(Optional) I agree that anonymised, non-identifying quotes from my transcript may be used in the research report.	☐	☐

Signatures

Name of participant	Signature	Date
	<i>Subathra</i>	
Subathra Sundarbabu	Signature	Date
Name of researcher		

Study contact for further information: Subathra Sundarbabu - sundarba002@mymail.sim.edu.sg
 Supervisor: Sarita Singh - singh009@mymail.sim.edu.sg

Figure 59: Consent form

Appendix C: Google Forms

Appendix C.1 Wellbeing Check Form

Link: <https://forms.gle/7Pj6jVmd9TOtrdoa8>

This appendix reproduces the thirteen-item Wellbeing Check Google Form, based on the Copenhagen Burnout Inventory, that participants completed alongside their recording. It covers the six personal and seven work-related burnout items and the reverse-scored energy item, scored using the CBI's own weighting. The collected responses are given in [Appendix D](#).

The image shows a screenshot of a Google Form titled "Wellbeing Check". The form has a purple header bar. Below the title, there is a paragraph: "Please answer these few questions about how you feel today, before using the app. Your answers are anonymous. There are no right or wrong answers." Below this, the email address "subathra27012004@gmail.com" is displayed with a "Switch account" link and a cloud icon. Below the email, there is a "Not shared" status with a lock icon. A red asterisk indicates a required question. The first question is "Participant number *", followed by a text input field labeled "Your answer". The second question is "How often do you feel tired? *", followed by five radio button options: "Always", "Often", "Sometimes", "Seldom", and "Never or almost never".

Wellbeing Check

Please answer these few questions about how you feel today, before using the app. Your answers are anonymous. There are no right or wrong answers.

subathra27012004@gmail.com [Switch account](#)

Not shared

* Indicates required question

Participant number *

Your answer

How often do you feel tired? *

☐ Always

☐ Often

☐ Sometimes

☐ Seldom

☐ Never or almost never

How often are you physically exhausted? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Seldom
- ☐ Never or almost never

How often are you emotionally exhausted? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Seldom
- ☐ Never or almost never

How often do you think: "I can't take it anymore"? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Seldom
- ☐ Never or almost never

How often do you feel worn out? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Seldom
- ☐ Never or almost never

How often do you feel weak and susceptible to illness? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Seldom
- ☐ Never or almost never

Is your work emotionally exhausting? *

- ☐ To a very high degree
- ☐ To a high degree
- ☐ Somewhat
- ☐ To a low degree
- ☐ To a very low degree

Do you feel burnt out because of your work? *

- ☐ To a very high degree
- ☐ To a high degree
- ☐ Somewhat
- ☐ To a low degree
- ☐ To a very low degree

Does your work frustrate you? *

- ☐ To a very high degree
- ☐ To a high degree
- ☐ Somewhat
- ☐ To a low degree
- ☐ To a very low degree

Do you feel worn out at the end of the working day? *

☐ Always

☐ Often

☐ Sometimes

☐ Seldom

☐ Never or almost never

Are you exhausted in the morning at the thought of another day at work? *

☐ Always

☐ Often

☐ Sometimes

☐ Seldom

☐ Never or almost never

Do you feel that every working hour is tiring for you? *

☐ Always

☐ Often

☐ Sometimes

☐ Seldom

☐ Never or almost never

Do you have enough energy for family and friends during leisure time? *

☐ Always

☐ Often

☐ Sometimes

☐ Seldom

☐ Never or almost never

Submit

Clear form

Figure 60: Wellbeing Check Form

Appendix C.2 App Feedback Form

Link: <https://forms.gle/1BAbZ6xSSUw5seSw6>

This appendix contains the App Feedback Google Form that participants completed after their check-in. It records the five-point ratings for ease of check-in, clarity of the score and recommendation, and whether the result matched how they felt, together with the free-text prompts for anything unclear and one suggested improvement. It supports the usability evaluation in Section 5.8.

The screenshot shows a Google Form titled "App Feedback". The form has a blue header bar. Below the title, there is a paragraph: "Now that you have tried the app, please share your thoughts. Your feedback helps improve the tool. Answers are anonymous." Below this, the user's email "subathra27012004@gmail.com" is displayed with a "Switch account" link and a cloud icon. Below the email, it says "Not shared" with a lock icon. A red asterisk indicates a required question. The first question is "Participant Number *", followed by a text input field labeled "Your answer". The second question is "How easy was it to complete a check-in? *", followed by the text "1 = Very Difficult and 5= Very Easy". Below this, there are five stars, each with a number above it: 1, 2, 3, 4, and 5. The stars are empty, indicating a rating has not been selected.

App Feedback

Now that you have tried the app, please share your thoughts. Your feedback helps improve the tool. Answers are anonymous.

subathra27012004@gmail.com [Switch account](#)

Not shared

* Indicates required question

Participant Number *

Your answer

How easy was it to complete a check-in? *

1 = Very Difficult and 5= Very Easy

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

How clear were your wellbeing score and the recommendation? *

1 = Very Unclear and 5= Very Clear

1

2

3

4

5

☆

☆

☆

☆

☆

Did the result roughly match how you feel today? *

☐ Yes

☐ Somewhat

☐ No

Was anything confusing or frustrating? *

Your answer

What is one thing you would improve? (Optional)

Your answer

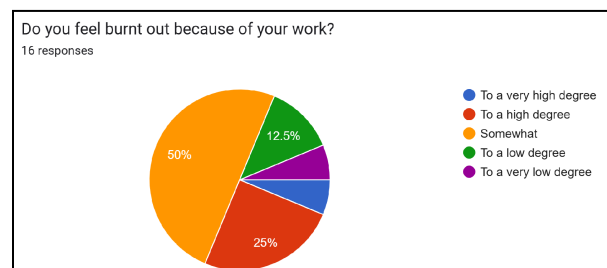
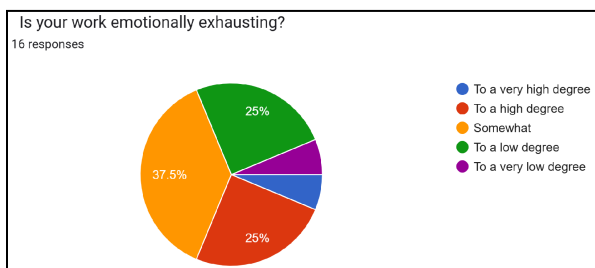
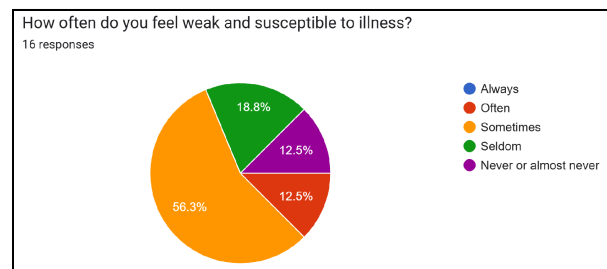
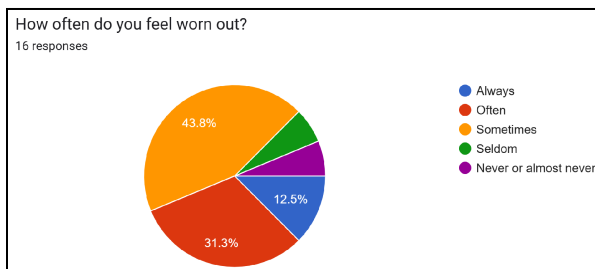
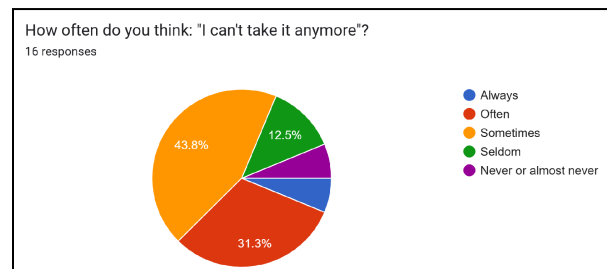
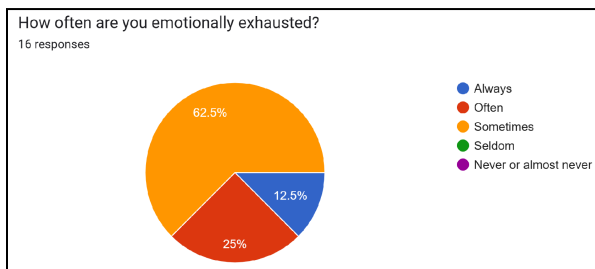
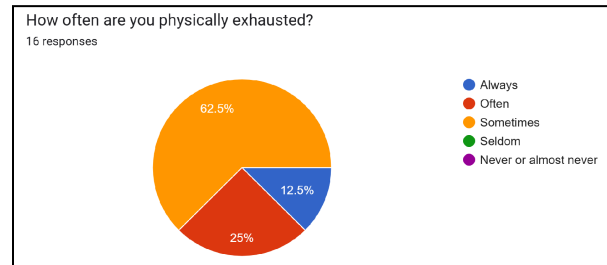
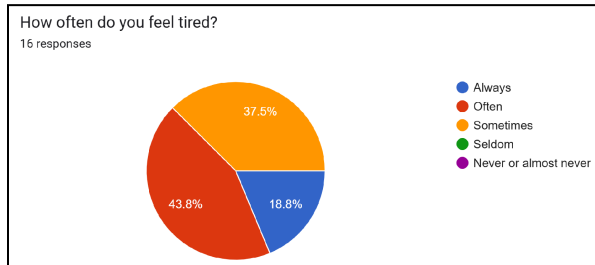
Submit

Clear form

Figure 61: App Feedback Form

Appendix D: CBI Questionnaire Results

This appendix presents the collected Wellbeing Check responses, giving each participant's personal, work and overall Copenhagen Burnout Inventory scores. This is the exported response data used for the validation against the burnout measure in Section 5.7, where the scores are correlated and band-matched against Solace's wellbeing scores.



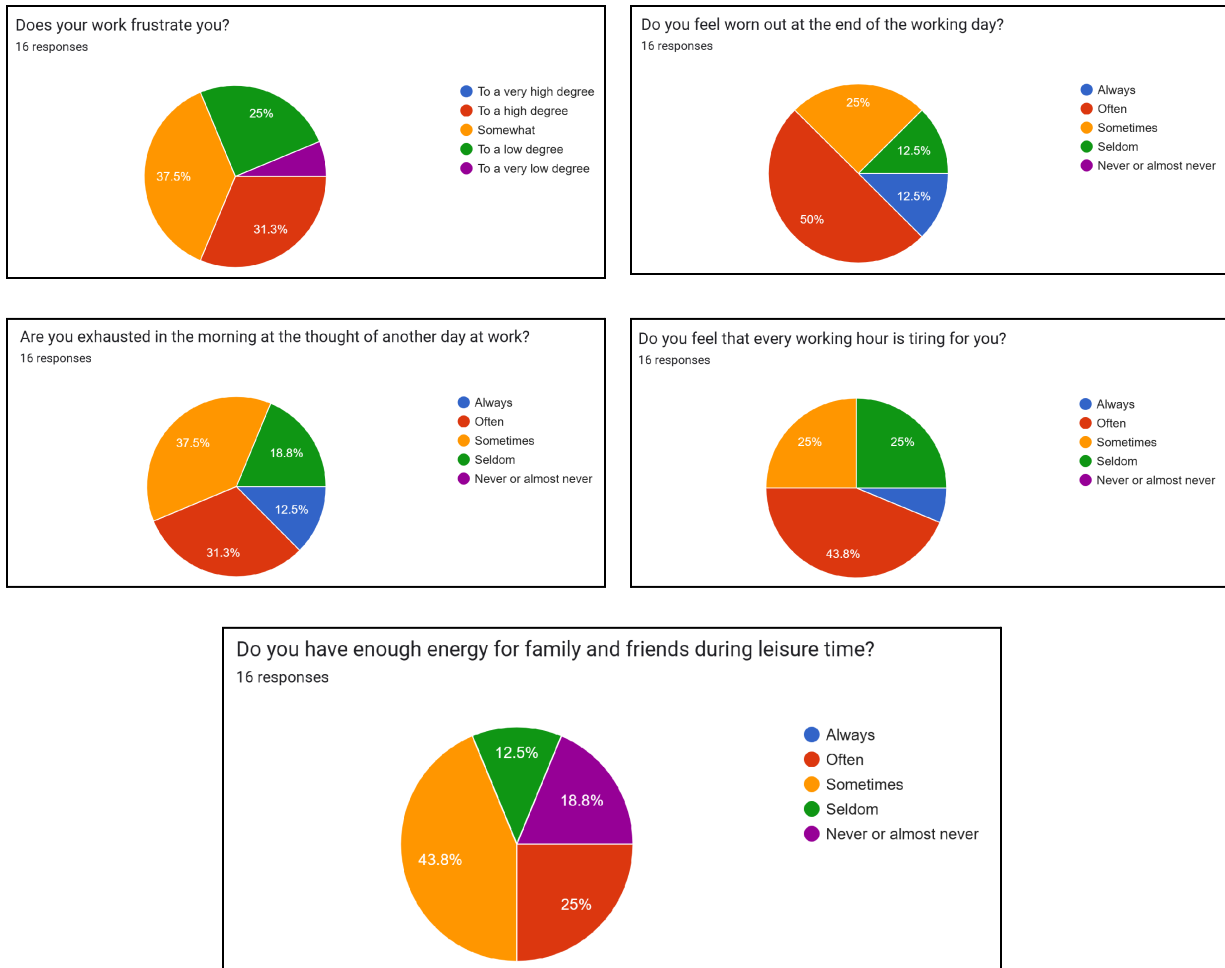


Figure 62: CBI Questionnaire Google Form Results

Appendix E: App Feedback Results

This appendix presents the collected App Feedback responses, giving the per-participant ratings and free-text comments summarised in Section 5.8.2. It includes the two design suggestions that were acted on, namely integrating the CBI questionnaire into the app and storing the original English transcript in the database for reviewers who do not read the check-in language.

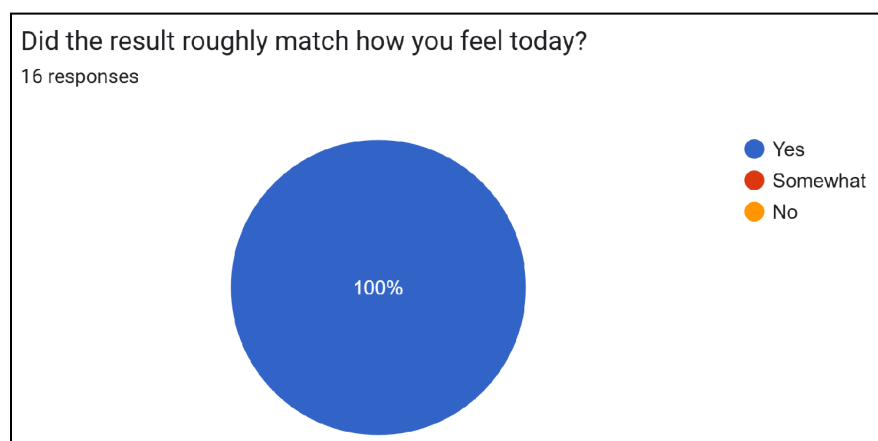
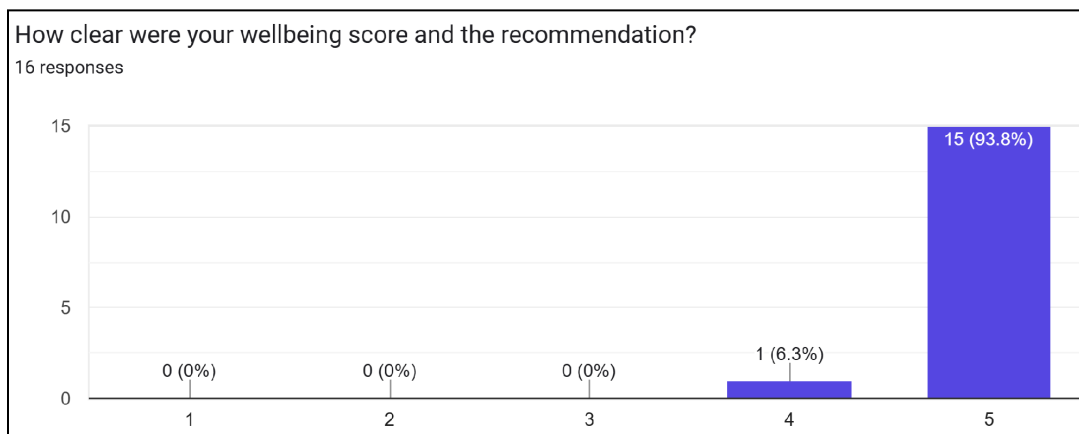
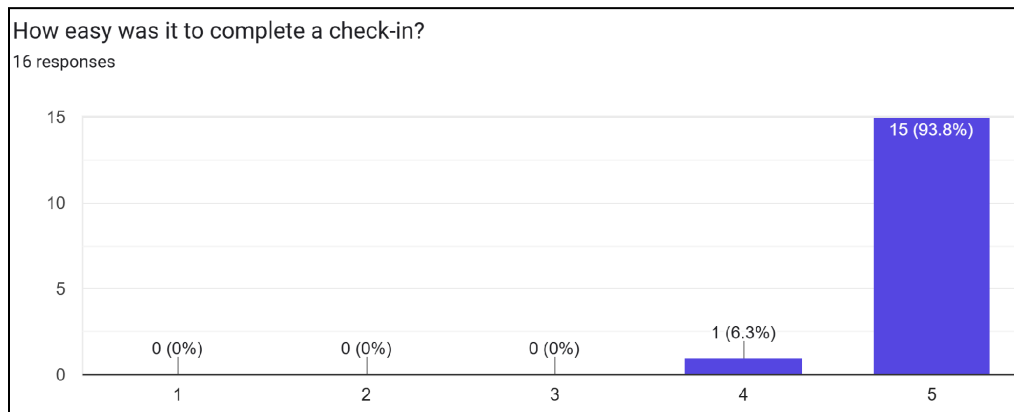


Figure 63: App Feedback Google Form Results

Participant Number	Was anything confusing or frustrating?	What is one thing you would improve? (Optional)
P08	No	Database user is an English person unable to read other languages, think about how you can improve this
P01	No	-
P03	Yes, the separate questionnaire	Improve by incorporating the questionnaire inside the application
P09	Explanation can be improved cause sometimes feel lost	-
P02	No	-
P016	No	-
P010	No	-
P012	No	Nil
P04	No	-
P05	Buttons' texts were small.	-
P06	No	Nil
P011	It was good overall	-
P013	No	Nil
P07	Nice	-
P015	No	-
P014	No it was good overall	Nil

Table 32: App Feedback Google Form Results for last two questions

CHAPTER 8: REFERENCES

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